

# Notes on Functional Analysis

Mengxi Wu  
University of Southern California

## Contents

|                                                           |    |
|-----------------------------------------------------------|----|
| Metric Spaces and Normed Vector Spaces . . . . .          | 2  |
| Main Theorems of Normed Vector Spaces . . . . .           | 7  |
| Hilbert Spaces . . . . .                                  | 14 |
| $L^p$ Spaces . . . . .                                    | 22 |
| Convolution and Smoothing . . . . .                       | 28 |
| Duals of Continuous Function Spaces . . . . .             | 35 |
| Representations of Positive Linear Functionals . . . . .  | 38 |
| Schwartz Space of Functions with Rapid Decrease . . . . . | 48 |
| Simple Properties of Fourier Transform . . . . .          | 54 |
| Inverse of $\mathcal{F}$ . . . . .                        | 57 |
| Plancherel's Equality . . . . .                           | 62 |
| Some Applications of the Fourier Transform . . . . .      | 64 |
| Tempered Distributions . . . . .                          | 66 |
| Tempered Distributions (continued) . . . . .              | 69 |
| Main Operations on $S' = S'(\mathbb{R}^d)$ . . . . .      | 70 |
| Some Applications of Tempered Distributions . . . . .     | 74 |
| Weak Convergence . . . . .                                | 76 |

# Metric Spaces and Normed Vector Spaces

## Metric Space

A metric on a set  $M$  is a function  $\rho : M \times M \rightarrow [0, \infty)$  such that

1.  $\rho(x, y) = \rho(y, x), \forall x, y \in M$ .
2.  $\rho(x, y) \leq \rho(x, z) + \rho(z, y), \forall x, y, z \in M$ .
3.  $\rho(x, y) = 0$  if and only if  $x = y$ .

## Example

Let  $M = \mathbb{R}^p$ . Then, for  $x = (x_1, \dots, x_p) \in \mathbb{R}^p$  and  $y = (y_1, \dots, y_p) \in \mathbb{R}^p$ ,

$$\rho_2(x, y) = \sqrt{(x_1 - y_1)^2 + \dots + (x_p - y_p)^2},$$
$$\rho_1(x, y) = |x_1 - y_1| + \dots + |x_p - y_p|, \quad \rho_\infty(x, y) = \max_{1 \leq i \leq p} |x_i - y_i|.$$

## Convergence

$x_n \in M$  converges to  $x \in M$  if  $\rho(x_n, x) \xrightarrow{n \rightarrow \infty} 0$ . We write  $\lim_n x_n = x$  or  $x_n \rightarrow x$  as  $n \rightarrow \infty$ .

*Note.* For the standard metrics on  $\mathbb{R}^p$  ( $\rho_1, \rho_2, \rho_\infty$ ),  $x^n = (x_1^n, \dots, x_p^n) \rightarrow x = (x_1, \dots, x_p)$  if and only if  $x_i^n \rightarrow x_i$  for all  $1 \leq i \leq p$ .

## Example

Let  $M = C([0, 1])$ , the space of continuous functions on  $[0, 1]$ . Then,

$$\rho_1(x, y) = \int_0^1 |x(t) - y(t)| dt, \quad \rho_\infty(x, y) = \sup_{0 \leq t \leq 1} |x(t) - y(t)|.$$

Let  $K = \rho_\infty(x, y)$ .

$$\rho_1(x, y) = \int_0^1 |x(t) - y(t)| dt \leq \int_0^1 K dt = K = \rho_\infty(x, y).$$

Note that  $\rho_1(x, y) \leq \rho_\infty(x, y)$ , so  $\rho_\infty$ -convergence implies  $\rho_1$ -convergence, but not necessarily vice versa.

## Cauchy Sequence

A sequence  $x_n \in M$  is Cauchy if  $\rho(x_n, x_m) \rightarrow 0$  as  $n, m \rightarrow \infty$ .

## Proposition 1

If  $x_n \rightarrow x$  in  $M$ , then  $(x_n)$  is Cauchy.

**Proof.**  $\rho(x_n, x_m) \leq \rho(x_n, x) + \rho(x_m, x) \rightarrow 0$  as  $n, m \rightarrow \infty$ .

## Completeness of Metric Space

$(M, \rho)$  is complete if every Cauchy sequence has a limit. That is, if  $\rho(x_n, x_m) \rightarrow 0$  as  $n, m \rightarrow \infty$ , then there exists  $x \in M$  such that  $x_n \rightarrow x$ .

### Example

- a.  $M = \mathbb{R}$  is complete, but  $\mathbb{Q} \subset \mathbb{R}$  is not, with the subspace metric.
- b.  $M = L^1([0, 1])$  with  $\rho_1(x, y) = \int_0^1 |x(t) - y(t)| dt$  is complete.
- c.  $C([0, 1]) \subset L^1([0, 1])$  with  $\rho_1(x, y)$  is not complete.

**Remark.** For example c, recall every  $x \in L^1$  can be approximated by a sequence  $x_n \in C([0, 1])$ .  $\int_0^1 |x_n(t) - x(t)| dt \rightarrow 0$ . Consider the specific sequence defined by the following step function:

$$x(t) = \begin{cases} 0 & \text{if } 0 \leq t < \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases} \in L^1([0, 1]), \quad x_n(t) = \begin{cases} 0 & \text{if } 0 \leq t < \frac{1}{2} \\ n(t - \frac{1}{2}) & \text{if } \frac{1}{2} \leq t < \frac{1}{2} + \frac{1}{n} \\ 1 & \text{if } \frac{1}{2} + \frac{1}{n} \leq t \leq 1 \end{cases}.$$

Then,

$$\int_0^1 |x_n(t) - x(t)| dt = \frac{1}{2n} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$x_n$  is Cauchy with respect to  $\rho_1$  but  $x \notin C([0, 1])$ .

### Dense

$B \subseteq M$  is dense if every  $x \in M$  can be approximated by  $x_n \in B$ :  $x_n \in B \rightarrow x \in M$ .

### Theorem 1

For any  $(M, \rho)$  there exists a complete  $(\tilde{M}, \tilde{\rho})$  such that  $M$  is isometric to a dense subset  $B \subseteq \tilde{M}$ . That is, there exists a bijection  $h : M \rightarrow B$  such that

$$\rho(x, y) = \tilde{\rho}(h(x), h(y)), \quad x, y \in M.$$

$(\tilde{M}, \tilde{\rho})$  is called the completion of  $(M, \rho)$ .

*Remark.* Isometric means distance-preserving. When we say that a map  $h : M \rightarrow B$  is an isometry, it means that the distance between any two points in  $M$  is exactly equal to the distance between their corresponding images in  $B$  under the mapping  $h$ . In Theorem 1 above,  $h$  and its inverse  $h^{-1} : B \rightarrow M$  are continuous.

### Example

- 1.  $\mathbb{R}$  is the completion of  $\mathbb{Q}$ .
- 2.  $L^1([0, 1])$  is the completion of  $C([0, 1])$  with respect to  $\rho_1(x, y) = \int_0^1 |x(t) - y(t)| dt$ .

## Equivalence relation

If  $(x_n), (y_n)$  are Cauchy sequences,  $(x_n) \sim (y_n)$  if  $\rho(x_n, y_n) \xrightarrow{n \rightarrow \infty} 0$ .

## Continuous Functions

Let  $f : (M, \rho) \rightarrow (Y, \nu)$  be a function.

1.  $f$  is continuous at  $x_0$  if  $x_n \rightarrow x_0$  implies  $f(x_n) \rightarrow f(x_0)$ . Equivalently, for any  $\epsilon > 0$ , there exists  $\delta > 0$  such that

$$\nu(f(x), f(x_0)) \leq \epsilon \text{ if } \rho(x_0, x) \leq \delta.$$

We write  $\lim_{x \rightarrow x_0} f(x) = f(x_0)$ .

2.  $f$  is continuous if  $f$  is continuous at all  $x \in M$ .

## Open Balls and Closed Ball

$$B_\delta(x) = \{y \in M \mid \rho(x, y) < \delta\}, \quad B_\delta[x] = \{y \in M \mid \rho(x, y) \leq \delta\}.$$

## Open Set

$A \subseteq M$  is open if  $\forall x \in A, \exists B_\delta(x) \subseteq A$  for some  $\delta > 0$ .

## Closed Set

$F \subseteq M$  is closed if  $F^c = M \setminus F$  is open.

## Theorem 2

Let  $\mathcal{T}$  be the collection of all open sets (all open subsets of  $M$ ). Note that  $M \in \mathcal{T}, \emptyset \in \mathcal{T}$ .  $\mathcal{T}$  is closed under arbitrary unions and finite intersections.

**Remark.**  $\emptyset$  and  $M$  are open and closed.  $\mathcal{T}$  is called the topology of  $M$ .

## Corollary 1

Let  $\mathcal{F}$  be the set of all closed sets.  $\mathcal{F}$  is closed under arbitrary intersections and finite unions.

## Neighborhood

$A \subseteq M$  is a neighborhood of  $x$  if  $x \in A$  and there exists  $B_\delta(x) \subseteq A$  with  $\delta > 0$ .

## Inner Point

$x \in A$  is an inner point of  $A$  if there exists  $B_\delta(x) \subseteq A$  with  $\delta > 0$ .

*Fact.* The set  $A^\circ$  of all inner points of  $A$  is open.

## Closure

Given  $A \subseteq M$ , its closure  $\overline{A}$  is the smallest closed set that contains  $A$ .

### Theorem 3

$\overline{A} = \{x \in M \mid \text{there exists } \{x_n\} \subseteq A \text{ with } \lim_{n \rightarrow \infty} x_n = x\}$ .

### Corollary 2

$F \subseteq M$  is closed iff  $\overline{F} = F$  iff  $\{x_n\} \subseteq F$  with  $\lim_{n \rightarrow \infty} x_n = x$  implies  $x \in F$ .

## Normed Vector Spaces (nvs)

A normed vector space  $E$  is a vector space over  $\mathbb{R}$  (or  $\mathbb{C}$ ) with a function, which we call a norm,  $\|\cdot\|_E: E \rightarrow [0, \infty)$  such that

1.  $\|\alpha x\|_E = |\alpha| \|x\|_E$ .
2.  $\|x + y\|_E \leq \|x\|_E + \|y\|_E$ .
3.  $\|x\|_E = 0$  iff  $x = 0$ .

*Note.*  $\|\cdot\|_E$  defines the distance  $\rho(x, y) = \|x - y\|_E$ ,  $x, y \in E$ .

## Triangle inequality

$$\|x - y\|_E = \|(x - z) + (z - y)\|_E \leq \|x - z\|_E + \|z - y\|_E.$$

## Example

1. Examples of norms on  $\mathbb{R}^d$  from which the previous metrics arise: for  $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ ,

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_d^2},$$

$$\|x\|_1 = |x_1| + \dots + |x_d|,$$

$$\|x\|_\infty = \max_i |x_i|.$$

2.  $E = C([0, 1])$  with  $\rho_1, \rho_\infty$ :

$$\|x\|_1 = \int_0^1 |x(t)| dt, \quad \|x\|_\infty = \sup_{0 \leq t \leq 1} |x(t)|.$$

## Completeness of Normed Vector Space

If a normed vector space  $E$  is complete, it is called Banach.

### Example

1.  $\mathbb{R}^d$  with  $\|x\|_1, \|x\|_2, \|x\|_\infty$  is Banach.
2.  $C[0, 1]$  is Banach with sup norm  $\|x\|_\infty$  but not Banach with  $\|x\|_1$ -norm.
3.  $L^1[0, 1]$  is Banach  $\rightarrow \|x\|_1 = \int_0^1 |x(t)| dt$ .  $C[0, 1]$  is dense in  $L^1[0, 1]$ .
4.  $\mathbb{Q}^d$  is not Banach.

### Linear Operators

Let  $E, F$  be normed vector spaces,  $T : E \rightarrow F$  be linear. The following claims are equivalent:

- (a)  $T$  is continuous at 0.
- (b)  $T$  is continuous on  $E$ .
- (c)  $T$  is bounded: There exists  $C > 0$  so that  $\|T(x)\|_F \leq C\|x\|_E, \forall x \in E$ .

*Notation.* We denote  $\mathcal{L}(E, F)$  the set of all bounded linear  $T : E \rightarrow F$

**Proof.** (b)  $\implies$  (a) is obvious. For (a)  $\implies$  (b), let  $x_n \rightarrow x$  in  $E$ . Then, by linearity,  $x_n - x \rightarrow 0$  in  $E$  and  $T(x_n - x) = T(x_n) - T(x) \rightarrow 0$  as  $n \rightarrow \infty$ . Thus, (a)  $\implies$  (b).

For (a)  $\implies$  (c), since  $T$  is continuous at 0, there exists  $\delta > 0$  such that

$$\|T(x)\|_F = \|T(x) - T(0)\|_F \leq 1 \text{ if } \|x\|_E = \|x - 0\|_E \leq \delta.$$

For  $y \in E$  and  $y \neq 0$ ,

$$\|T(y)\|_F = \left\| T\left(\frac{y}{\|y\|_E} \cdot \delta\right) \right\|_F \cdot \frac{\|y\|_E}{\delta} \leq 1 \cdot \frac{\|y\|_E}{\delta} = \frac{1}{\delta} \cdot \|y\|_E = C\|y\|_E.$$

where  $C = 1/\delta$ . For (c)  $\implies$  (a), if  $x_n \rightarrow 0$  then,

$$\|T(x_n)\|_F \leq C\|x_n\|_E \rightarrow 0 \text{ as } n \rightarrow \infty.$$

*Exercise.*  $T$  is bounded iff  $\sup \|T(x)\|_F < \infty$  for  $\|x\|_E \leq R, \forall R > 0$ .

### Operator Norm

Given  $T \in \mathcal{L}(E, F)$ , its operator norm is the number:

$$\|T\| = \sup_{\|x\|_E \leq 1} \|T(x)\|_F = \sup_{\|x\|_E = 1} \|T(x)\|_F.$$

*Note.*

- (i)  $\|T\| \leq C$  in (c).
- (ii)  $\|T\|$  is the smallest  $C > 0$  for which (c) holds.
- (iii) **Important:**  $\|T(\mathbf{x})\|_F \leq \|T\| \|\mathbf{x}\|_E$ .

*Exercise.*  $\mathcal{L}(E, F)$  with  $\|\cdot\|$  is nvs.

**Proof.** We provide the proof for triangle inequality proof. Let  $T, G \in \mathcal{L}(E, F)$ ,  $\|x\|_E \leq 1$ .

$$\sup_{\|x\|_E \leq 1} \|(T + G)(x)\|_F = \sup_{\|x\|_E \leq 1} \|T(x) + G(x)\|_F \leq \|T\| + \|G\|.$$

## Dual Space

Given nvs  $E$ , its **dual space** is  $E^* = \mathcal{L}(E, \mathbb{R})$  (or  $E^* = \mathcal{L}(E, \mathbb{C})$ ). Note that  $E^*$  is Banach. For  $\ell \in E^*$ ,  $x \in E$ , we denote

$$\ell(x) = \langle \ell, x \rangle_{E^*, E} = \langle \ell, x \rangle.$$

## Example

1.  $E = \mathbb{R}^d$ ,  $\ell \in E^* = \mathcal{L}(\mathbb{R}^d, \mathbb{R})$  is  $\ell(x) = a \cdot x = \langle a, x \rangle = \sum_{i=1}^d a_i x_i$  where  $a = (a_1, \dots, a_d) \in \mathbb{R}^d$ ,  $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ .
2. Let  $E, F$  be nvs. Then  $E \times F$  is nvs with  $\|(x, y)\|_2 = \sqrt{\|x\|_E^2 + \|y\|_F^2}$ . Also,  $\|(x, y)\|_1 = \|x\|_E + \|y\|_F$ , and  $\|(x, y)\|_\infty = \max\{\|x\|_E, \|y\|_F\}$ . Moreover, if  $E, F$  are Banach, then  $E \times F$  is Banach.

## Main Theorems of Normed Vector Spaces

### Hahn-Banach Theorem

Let  $E$  be a normed vector space and  $M \subseteq E$  be a vector subspace. Let  $f \in M^* = \mathcal{L}(M, \mathbb{R})$  (the dual space of  $M$ ). Then there exists  $\ell \in E^* = \mathcal{L}(E, \mathbb{R})$  extending  $f$  such that  $\|\ell\|_{E^*} = \|f\|_{M^*}$ .

**Remark.** The proof uses Zorn's Lemma, but Zorn's Lemma is not needed if  $E$  is separable (i.e., if  $E$  has a countable dense subset).

### Corollary 1

Let  $E$  be a normed vector space and  $F \subset E$  be a closed vector subspace such that  $F \neq E$ . Then, there exists a nonzero  $\ell \in E^*$  so that  $\ell|_F = 0$  (i.e.,  $\ell(y) = 0$  for all  $y \in F$ ). Moreover, if  $x_0 \notin F$ , then there exists  $\ell \in E^*$  so that  $\ell(x_0) = 1$  and  $\ell(y) = 0$  for any  $y \in F$ .

**Proof.** Let  $x_0 \notin F$ . We begin by constructing the subspace  $M$ :

$$M = \mathbb{R}x_0 + F = \{tx_0 + y : t \in \mathbb{R}, y \in F\}.$$

Before proceeding, we establish a crucial property: if  $tx_0 + y = 0$ , then  $t = 0$  and  $y = 0$ . To prove this, suppose  $tx_0 + y = 0$  with  $t \neq 0$ . Then we would have  $x_0 = -\frac{y}{t} \in F$  since  $F$  is a subspace, contradicting our assumption that  $x_0 \notin F$ . Therefore,  $t$  must be 0, and consequently  $y = 0$ .

We now define a linear functional  $f : M \rightarrow \mathbb{R}$  by  $f(tx_0 + y) = t$ . Note that this definition implies  $f(x_0) = 1$  and  $f(y) = 0$  for all  $y \in F$ . To show  $f$  is well-defined, suppose  $tx_0 + y = t'x_0 + y'$ . Then  $(t - t')x_0 + (y - y') = 0$ . By our previous observation, this implies  $t = t'$  and  $y = y'$ , so  $f(tx_0 + y) = f(t'x_0 + y')$ . The linearity of  $f$  follows naturally:

$$f(\alpha_1(t_1x_0 + y_1) + \alpha_2(t_2x_0 + y_2)) = \alpha_1t_1 + \alpha_2t_2 = \alpha_1f(t_1x_0 + y_1) + \alpha_2f(t_2x_0 + y_2).$$

To prove  $f$  is bounded, let  $d = \inf_{z \in F} \|x_0 - z\|_E$ . We claim that  $d > 0$ . Since  $F$  is closed and  $x_0 \notin F$ , if we had  $d = 0$ , there would exist a sequence  $\{z_n\} \subseteq F$  with  $z_n \rightarrow x_0$ , contradicting  $x_0 \notin F$ .

For any  $tx_0 + y \in M$ , we have

$$\|tx_0 + y\|_E = |t| \left\| x_0 + \frac{y}{t} \right\|_E = |t| \left\| x_0 - \left( -\frac{y}{t} \right) \right\|_E \geq |t|d.$$

Therefore, if  $\|tx_0 + y\|_E \leq 1$ , then  $|t| \leq \frac{1}{d}$ . This shows:

$$\|f\|_{M^*} = \sup_{\|tx_0 + y\|_E \leq 1} |f(tx_0 + y)| = \sup_{\|tx_0 + y\|_E \leq 1} |t| \leq \frac{1}{d} < \infty.$$

By the Hahn-Banach Theorem,  $f$  extends to a functional  $\ell \in E^*$ . This extended functional  $\ell$  satisfies  $\ell(x_0) = f(x_0) = 1$  and  $\ell(y) = f(y) = 0$  for all  $y \in F$ , completing our proof.

*Comment.* Corollary 1 is a standard tool to prove by contradiction that a vector subspace  $D \subseteq E$  is dense ( $\overline{D} = E$ ).

1. Assume the opposite,  $\overline{D} \neq E$ . Note that  $\overline{D}$  is a closed vector subspace.
2. By Corollary 1, there exists a nonzero  $\ell \in E^*$  such that  $\ell|_{\overline{D}} = 0$ .
3. But if other arguments show that  $\ell|_{\overline{D}} = 0 \implies \ell = 0$  on  $E$ , we reach a contradiction.

Note that if  $\ell|_D = 0$  and  $\ell$  is continuous, then for any sequence  $\{x_n\} \subseteq D$  with  $x_n \rightarrow x_0 \in \overline{D}$ , we have  $\ell(x_n) \rightarrow \ell(x_0) \implies \ell(x_0) = 0$ .

## Corollary 2

- (a) For each nonzero  $x \in E$ , there exists a nonzero  $\ell \in E^*$  such that

$$\ell(x) = \langle \ell, x \rangle = \|\ell\|_{E^*} \|x\|_E.$$

We say  $\ell$  and  $x$  are aligned.

- (b) For any  $x \in E$ ,

$$\|x\|_E = \sup_{\|\ell\|_{E^*} \leq 1} |\ell(x)|.$$

Recall that  $\|\ell\|_{E^*} = \sup_{\|x\|_E \leq 1} |\ell(x)|$ .

*Remark.* It is always the case that  $|\ell(x)| \leq \|\ell\|_{E^*} \|x\|_E$ . For (a):



1. If  $E = \mathbb{R}^d$  with  $\|\cdot\|_2$ , then  $E^* = \mathbb{R}^d$  with  $\|\cdot\|_2$ . Two vectors  $\ell = (a_1, \dots, a_d)$  and  $x = (x_1, \dots, x_d)$  are aligned if

$$\ell(x) = \langle a, x \rangle = a_1x_1 + \dots + a_dx_d = \|a\|_2\|x\|_2.$$

2. If  $E = \mathbb{R}^d$  with  $\|\cdot\|_1$ , then  $E^* = \mathbb{R}^d$  with  $\|\cdot\|_\infty$ , and we have:

$$\langle a, x \rangle = \|a\|_\infty\|x\|_1.$$

**Proof.** Assume the base field is  $\mathbb{R}$ .

- (a) Let  $x_0 \in E$  and  $x_0 \neq 0$ . Define

$$M = \mathbb{R}x_0 = \{tx_0 : t \in \mathbb{R}\}.$$

Define a linear map  $f : M \rightarrow \mathbb{R}$  by  $f(tx_0) = t\|x_0\|_E$ , and then:

$$\|f\|_{M^*} = \sup_{\|tx_0\|_E \leq 1} |f(tx_0)| = \sup_{|t|\|x_0\|_E \leq 1} |t|\|x_0\|_E = 1.$$

Note that  $f(x_0) = \|x_0\|_E = \|f\|_{M^*}\|x_0\|_E$ . Thus,  $f$  is linear and bounded, so  $f \in \mathcal{L}(M, \mathbb{R})$ . By the Hahn-Banach Theorem, there exists  $\ell \in E^*$  extending  $f$  with

$$\begin{aligned} \|\ell\|_{E^*} &= \|f\|_{M^*} = 1, \\ \implies \ell(x_0) &= f(x_0) = \|x_0\|_E = \|\ell\|_{E^*}\|x_0\|_E. \end{aligned}$$

- (b) Let  $x_0 \in E$  and  $\|\ell\|_{E^*} \leq 1$ . Then,

$$|\ell(x_0)| \leq \|\ell\|_{E^*}\|x_0\|_E \leq \|x_0\|_E \implies \sup_{\|\ell\|_{E^*} \leq 1} |\ell(x_0)| \leq \|x_0\|_E.$$

By part (a), there exists  $\ell \in E^*$  such that  $\|\ell\|_{E^*} = 1$  and  $\ell(x_0) = \|x_0\|_E$ .

## Norm of Adjoint Operator

Let  $E, F$  be normed vector spaces and  $T \in \mathcal{L}(E, F)$ . We define the adjoint  $T' : F^* \rightarrow E^*$  to be the linear map that assigns to each  $\ell \in F^*$  the functional  $T'\ell \in E^*$  defined by

$$(T'\ell)(x) := \ell(T(x)), \quad x \in E.$$

### Proposition 1

$T' \in \mathcal{L}(F^*, E^*)$  and  $\|T'\| = \|T\|$ .

**Proof.**

$$\begin{aligned} \|T'\| &= \sup_{\|\ell\|_{F^*} \leq 1} \|T'\ell\|_{E^*} = \sup_{\|\ell\|_{F^*} \leq 1} \sup_{\|x\|_E \leq 1} |(T'\ell)(x)| = \sup_{\|\ell\|_{F^*} \leq 1} \sup_{\|x\|_E \leq 1} |\ell(T(x))| \\ \implies \|T'\| &= \sup_{\|\ell\|_{F^*} \leq 1} \sup_{\|x\|_E \leq 1} |\ell(T(x))| = \sup_{\|x\|_E \leq 1} \sup_{\|\ell\|_{F^*} \leq 1} |\ell(T(x))|. \end{aligned}$$

By Corollary 2 (b),

$$\|T'\| = \sup_{\|x\|_E \leq 1} \sup_{\|\ell\|_{F^*} \leq 1} |\ell(T(x))| = \sup_{\|x\|_E \leq 1} \|T(x)\|_F = \|T\|.$$

## Embedding

Let  $E$  be a normed vector space and  $E^{**} = (E^*)^* = \mathcal{L}(E^*, \mathbb{R})$ . Every  $x_0 \in E$  defines naturally a functional  $J_{x_0}$  defined by

$$J_{x_0}(\ell) = \ell(x_0), \quad \ell \in E^*.$$

Note that  $|J_{x_0}(\ell)| = |\ell(x_0)| \leq \|\ell\|_{E^*} \|x_0\|_E$ .

## Proposition 2

The map  $J : E \rightarrow E^{**}$  that assigns to  $x_0 \in E$  the linear functional  $J_{x_0} \in E^{**}$  is an isometry:

$$\|x_0\|_E = \|J_{x_0}\|_{E^{**}}$$

i.e., the space  $E$  is isometrically embedded into  $E^{**}$  and we write  $E \subseteq E^{**}$ .

**Proof.** For  $x_0 \in E$ , by Corollary 2 (b),

$$\|x_0\|_E = \sup_{\|\ell\|_{E^*} \leq 1} |\ell(x_0)| = \sup_{\|\ell\|_{E^*} \leq 1} |J_{x_0}(\ell)| = \|J_{x_0}\|_{E^{**}}.$$

## Baire's Lemma

Let  $(M, \rho)$  be a complete metric space and  $M = \bigcup_n M_n$  with all  $M_n$  closed. Then, there exists  $n_0 \geq 1$  so that  $\overset{\circ}{M}_{n_0} \neq \emptyset$ , i.e., there exists  $B_{r_0}(x_0) \subseteq M_{n_0}$ . Recall  $\overset{\circ}{M}_{n_0} = \text{Int}(M_{n_0}) = \{x \in M_{n_0} : B_r(x) \subseteq M_{n_0} \text{ for some } r > 0\}$ .

**Proof.** We prove by contradiction. Assume  $\overset{\circ}{M}_n = \emptyset, \forall n \geq 1$ . Then for each  $n$ ,

$$O_n = M_n^c = M \setminus M_n \text{ is dense and open in } M.$$

Every  $B_r(x) \subseteq M$  contains  $y \in O_n$ . We will show that  $G = \bigcap_n O_n$  is dense, but

$$G = \bigcap_n O_n = \bigcap_n M_n^c = \left( \bigcup_n M_n \right)^c = M^c = \emptyset$$

and this is a contradiction.

To see  $G$  is dense, note  $G$  is dense if every  $B_{r_0}(x_0)$  contains  $x \in G$ . Fix  $B_{r_0}(x_0)$ . Since  $O_1$  is dense,

$$\exists \overline{B}_{r_1}(x_1) \subseteq B_{r_0}(x_0) \cap O_1, \text{ with } r_1 \leq \frac{r_0}{2}.$$

To get  $\overline{B}_{r_1}(x_1)$ , we can always find an open ball in a nonempty open set. Thus, we can take a closed ball to be the closure of an open ball with half the radius (or the closure of any other smaller ball).

Since  $O_2$  is dense,  $B_{r_1}(x_1) \cap O_2$  is nonempty and open.

$$\exists \overline{B}_{r_2}(x_2) \subseteq B_{r_1}(x_1) \cap O_2 \text{ with } r_2 \leq \frac{r_1}{2} \leq \frac{r_0}{2^2}.$$

Since  $O_{n+1}$  is dense,

$$\exists \overline{B}_{r_{n+1}}(x_{n+1}) \subseteq B_{r_n}(x_n) \cap O_{n+1} \text{ with } r_{n+1} \leq \frac{r_0}{2^{n+1}}.$$

We claim the sequence  $\{x_n\}$  is Cauchy. If  $j, j' \geq n$ , then  $x_j, x_{j'} \in B_{r_n}(x_n)$ , and

$$\rho(x_j, x_{j'}) \leq \rho(x_j, x_n) + \rho(x_{j'}, x_n) \leq 2 \cdot \frac{r_0}{2^n} \rightarrow 0 \text{ as } j, j' \rightarrow \infty.$$

Since  $M$  is complete,  $x = \lim_{n \rightarrow \infty} x_n$  exists. Note that

$$x \in \bigcap_{n \geq 1} \overline{B}_{r_n}(x_n) \subseteq \bigcap_{n \geq 0} B_{r_n}(x_n) \subseteq O_m, \quad \forall m \geq 1$$

because  $\overline{B}_{r_n}(x_n)$  is closed so the limit belongs to it. This means  $x \in \bigcap_{m \geq 1} O_m = G$ .

## Banach-Steinhaus Theorem

### Theorem 2

Let  $E, F$  be normed vector spaces and  $E$  be Banach. Let  $T_i \in \mathcal{L}(E, F)$ ,  $i \in I$ , and for each  $x \in E$ ,

$$\sup_{i \in I} \|T_i(x)\|_F < \infty \text{ (pointwise uniform boundedness).}$$

Then  $\sup_{i \in I} \|T_i\| < \infty$ , or equivalently, for every  $R > 0$ ,

$$\sup_{i \in I} \sup_{\|x\|_E \leq R} \|T_i(x)\|_F = \sup_{\|x\|_E \leq R} \sup_{i \in I} \|T_i(x)\|_F < \infty.$$

**Proof.** By assumption, for each  $n \geq 1$ , the set

$$M_n = \{x \in E : \sup_{i \in I} \|T_i(x)\|_F \leq n\} = \bigcap_{i \in I} \{x \in E : \|T_i(x)\|_F \leq n\}$$

is closed because each  $T_i$  is continuous. Since every  $x \in E$  is pointwise bounded, we have  $E = \bigcup_{n=1}^{\infty} M_n$ . By Baire's Lemma, there exists  $n_0 \geq 1$  such that  $\overset{\circ}{M}_{n_0} \neq \emptyset$ , meaning:

$$\exists \overline{B}_{r_0}(x_0) \subseteq \{x \in E : \sup_{i \in I} \|T_i(x)\|_F \leq n_0\}.$$

We will show that

$$\|T_i\| = \frac{1}{r_0} \sup_{\|y\|_E \leq r_0} \|T_i(y)\|_F \leq \frac{1}{r_0} C_0$$

where  $C_0$  is independent of  $i$ , by showing:

$$\sup_{\|y\|_E \leq r_0} \|T_i(y)\|_F \leq C_0.$$

Let  $y \in E$  with  $\|y\|_E \leq r_0$ . Then,

$$\|T_i(y)\|_F = \|T_i(y + x_0 - x_0)\|_F \leq \|T_i(x_0 + y)\|_F + \|T_i(x_0)\|_F \leq n_0 + \sup_{i \in I} \|T_i(x_0)\|_F = 2n_0$$

because  $x_0 + y \in \overline{B_{r_0}}(x_0)$  and  $\sup_{i \in I} \|T_i(x_0)\|_F \leq n_0$ . Thus, we can set  $C_0 = 2n_0$ .

**Remark.** Let  $T_i \in \mathcal{L}(\mathbb{R}^d, \mathbb{R}^d)$  be given by matrix components:

$$T_i = (a_{k\ell}^i), \quad 1 \leq k \leq d, 1 \leq \ell \leq d.$$

Then  $\sup_i \|T_i(x)\| < \infty$  for all  $x \in \mathbb{R}^d$  implies  $\sup_i |a_{k\ell}^i| < \infty$  for all  $k, \ell$ . Indeed, if  $e_k$  ( $k = 1, \dots, d$ ) is the standard basis of  $\mathbb{R}^d$ , then

$$\sup_i |a_{k\ell}^i| = \sup_i |e_k \cdot (T_i(e_\ell))| \leq \sup_i \|T_i(e_\ell)\| < \infty, \quad \ell = 1, \dots, d.$$

As  $\|e_k\| = 1$  and  $|\cos \alpha| \leq 1$ , we have:

$$e_k \cdot (T_i(e_\ell)) = \|e_k\| \|T_i(e_\ell)\| \cos \alpha.$$

### Open Mapping Theorem

Let  $M, Y$  be metric spaces and  $f : M \rightarrow Y$ . The following are equivalent:

- (a)  $f$  is continuous.
- (b)  $f^{-1}(U) \subseteq M$  is open for any open  $U \subseteq Y$ .
- (c)  $f^{-1}(F) \subseteq M$  is closed for any closed  $F \subseteq Y$ .

Let  $E, F$  be Banach spaces and  $T \in \mathcal{L}(E, F)$  be surjective (onto). If  $U \subseteq E$  is open, then  $T(U) \subseteq F$  is open.

### Theorem 3

Let  $E, F$  be Banach spaces and  $T \in \mathcal{L}(E, F)$  be surjective. Then there exists  $c > 0$  so that  $B_c^F \subseteq T(B_1^E)$ , where

$$B_c^F = \{y \in F : \|y\|_F < c\}, \quad B_1^E = \{x \in E : \|x\|_E < 1\}.$$

Here,  $B_r^H = \{x \in H : \|x\|_H < r\}$  denotes a general ball of radius  $r$  in space  $H$ .

**Proof.** Since  $T$  is onto,

$$E = \bigcup_{n=1}^{\infty} B_n^E, \quad F = \bigcup_{n=1}^{\infty} \overline{T(B_n^E)}.$$

We take the closure here to apply Baire's Lemma. By Baire's Lemma, there exists  $n_0 \geq 1$  such that  $B_{r_0}^F(x_0) \subseteq \overline{T(B_{n_0}^E)}$ . Since  $T$  is linear, we have  $y = T(v) \implies -y = T(-v)$ , meaning  $y \in T(B_{n_0}^E) \implies -y \in T(B_{n_0}^E)$ . Thus,  $B_{r_0}^F(-x_0) \subseteq \overline{T(B_{n_0}^E)}$ .

For  $y \in B_{r_0}^F$ ,

$$y = \frac{1}{2}(x_0 + y) + \frac{1}{2}(-x_0 + y) \in \overline{T(B_{n_0}^E)},$$

where  $x_0 + y \in B_{r_0}^F(x_0)$  and  $-x_0 + y \in B_{r_0}^F(-x_0)$ . By convexity,  $B_{r_0}^F \subseteq \overline{T(B_{n_0}^E)}$ . Now we claim:

$$B_{r_0/n_0}^F \subseteq \overline{T(B_1^E)} \subseteq T(B_2^E).$$

Let  $\delta = r_0/n_0$ . If this holds, then  $B_\delta^F \subseteq T(B_2^E) \implies B_{\delta/2}^F \subseteq T(B_1^E)$ .

We first show  $\overline{T(B_1^E)} \subseteq T(B_2^E)$ . Let  $z \in \overline{T(B_1^E)}$ . Then there exists  $x_1 \in B_1^E$  such that

$$z - T(x_1) \in B_{\delta/2}^F \subseteq \overline{T(B_{1/2}^E)}.$$

This is because  $z$  lies in the closure of the image of  $B_1^E$ , allowing us to approximate  $z$  closely by elements in the image. Similarly, there exists  $x_2 \in B_{1/2}^E$  such that:

$$z - T(x_1) - T(x_2) \in B_{\delta/2^2}^F \subseteq \overline{T(B_{1/2^2}^E)},$$

Inductively, there exists  $x_n \in B_{1/2^{n-1}}^E$  such that:

$$z - T(x_1 + \cdots + x_n) \in B_{\delta/2^n}^F \subseteq \overline{T(B_{1/2^n}^E)},$$

Note that:

$$\|z - T(x_1 + \cdots + x_n)\|_F \leq \frac{\delta}{2^n} \xrightarrow{n \rightarrow \infty} 0.$$

The series  $\sum_{k=1}^{\infty} x_k$  converges absolutely since:

$$\sum_{k=1}^{\infty} \|x_k\|_E \leq 1 + \sum_{k=1}^{\infty} \frac{1}{2^k} = 2.$$

Since  $E$  is Banach,  $x = \sum_{k=1}^{\infty} x_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n x_k$  exists, and:

$$\|x\|_E \leq \sum_{k=1}^{\infty} \|x_k\|_E < 2.$$

Therefore,

$$z = \lim_{n \rightarrow \infty} T(x_1 + \cdots + x_n) = T(x) \quad \text{and} \quad \|x\|_E < 2.$$

Hence,  $z \in T(B_2^E)$ , which proves  $\overline{T(B_1^E)} \subseteq T(B_2^E)$ .

### Corollary 3

Let  $E, F$  be Banach spaces and  $T \in \mathcal{L}(E, F)$  be onto. Then,  $T(U) \subseteq F$  is open for any open  $U \subseteq E$ .

**Proof.** Let  $U \subseteq E$  be open and  $y_0 \in T(U)$ . Then, there exists  $x_0 \in U$  and  $\delta > 0$  so that  $B_\delta^E(x_0) \subseteq U$  and  $y_0 = T(x_0)$ . By Theorem 3, there exists  $c > 0$  such that  $B_c^F \subseteq T(B_1^E)$ , implying  $B_{c\delta}^F \subseteq T(B_\delta^E)$ . We have:

$$y_0 + B_{c\delta}^F \subseteq T(x_0 + B_\delta^E) = T(B_\delta^E(x_0)) \subseteq T(U).$$

Therefore,  $T(U)$  is an open set.

## Inverse Mapping Theorem

Let  $E, F$  be Banach spaces and  $T \in \mathcal{L}(E, F)$  be bijective. Then,  $T^{-1} \in \mathcal{L}(F, E)$ .

**Proof.** Let  $U \subseteq E$  be open. Then  $(T^{-1})^{-1}(U) = T(U)$  is open by the Open Mapping Theorem. Thus,  $T^{-1}$  is continuous, so it is bounded.

## Closed Graph Theorem

Let  $E, F$  be Banach spaces and  $T : E \rightarrow F$  be linear. Assume its graph:

$$\Gamma = \{(x, T(x)) \in E \times F : x \in E\} \subseteq E \times F \text{ is closed.}$$

Then,  $T \in \mathcal{L}(E, F)$ .

**Proof.** Since  $\Gamma$  is a closed vector subspace of  $E \times F$ ,  $\Gamma$  is a Banach space as well. Both projections of  $\Gamma$  are continuous:

$$(x, T(x)) \in \Gamma \xrightarrow{\pi_1} x \in E,$$

$$(x, T(x)) \in \Gamma \xrightarrow{\pi_2} T(x) \in F.$$

Since  $\|\pi_1(x, T(x))\|_E = \|x\|_E \leq \sqrt{\|x\|_E^2 + \|T(x)\|_F^2} = \|(x, T(x))\|_{E \times F}$ ,  $\pi_1$  is bounded and clearly bijective. By the Inverse Mapping Theorem,  $\pi_1^{-1} : E \rightarrow \Gamma$  is continuous. Since  $T = \pi_2 \circ \pi_1^{-1}$  is the composition of two continuous functions,  $T$  is continuous and thus bounded.

*Comment.* Proving continuity of  $T : E \rightarrow F$ :

1. In a standard way, given  $x_n \rightarrow x$  in  $E$ :

(i) Prove  $\lim_n T(x_n) = y$  exists.

(ii) Prove  $y = T(x)$ .

2. Using the Closed Graph Theorem: Given  $x_n \rightarrow x$  in  $E$  and  $T(x_n) \rightarrow y$  in  $F$ , prove  $y = T(x)$ .

## Hilbert Spaces

### Inner Product

Let  $H$  be a complex vector space. An inner product on  $H$  is a  $\mathbb{C}$ -valued function denoted by  $\langle x, y \rangle$ , for  $x, y \in H$ , such that:

i.  $\langle x, x \rangle \geq 0 \ \forall x \in H$ ;  $\langle x, x \rangle = 0$  if and only if  $x = 0$

ii.  $\forall x, y, z \in H$ , and  $\alpha, \beta \in \mathbb{C}$ :

$$\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle,$$

$$\langle z, \alpha x + \beta y \rangle = \overline{\alpha} \langle z, x \rangle + \overline{\beta} \langle z, y \rangle,$$

$$\langle x, y \rangle = \overline{\langle y, x \rangle}.$$

## Examples

1.  $H = \mathbb{C}^d$ ,  $x = (x_1, \dots, x_d)$ ,  $y = (y_1, \dots, y_d) \in \mathbb{C}^d$ :

$$\langle x, y \rangle = \sum_{j=1}^d x_j \bar{y}_j.$$

Then,  $\langle x, x \rangle = \sum_{j=1}^d |x_j|^2$ .

2.  $\ell^2$  is the space of all sequences  $(x_1, x_2, \dots)$  such that

$$\sum_{n=1}^{\infty} |x_n|^2 < \infty,$$

with the inner product  $\langle x, y \rangle = \sum_{n=1}^{\infty} x_n \bar{y}_n$ .

3.  $\mathcal{C} = C([a, b])$  is the space of continuous complex-valued functions on  $[a, b]$  with

$$\langle f, g \rangle = \int_a^b f(t) \overline{g(t)} dt.$$

Then,  $\langle f, f \rangle = \int_a^b |f(t)|^2 dt$ .

4.  $L^2(X, \mu)$  is the space of complex-valued square-integrable functions (equivalence classes) on a measure space  $(X, \mathcal{X}, \mu)$  with

$$\langle f, g \rangle = \int f(x) \overline{g(x)} d\mu.$$

In fact, Examples 1 and 2 are just specific cases of Example 4 using the counting measure.

## Inner Product Space (Pre-Hilbert Space)

A complex vector space equipped with an inner product is called an inner product space or pre-Hilbert space.

### Orthogonality

Let  $H$  be a pre-Hilbert space. We say  $x \perp y$  in  $H$  if  $\langle x, y \rangle = 0$ . The induced norm is defined as:

$$\|x\|_H := \sqrt{\langle x, x \rangle}, \quad x \in H.$$

### Pythagorean Theorem

If  $x \perp y$ , then

$$\|x + y\|_H^2 = \|x\|_H^2 + \|y\|_H^2.$$

**Proof.** Let  $\langle x, y \rangle = \langle y, x \rangle = 0$ . Then,

$$\begin{aligned}\|x + y\|_H^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|_H^2 + \|y\|_H^2\end{aligned}$$

## Orthonormal System

A subset  $\{e_j : j = 1, \dots, N\}$  is an orthonormal system (ONS) if

$$\langle e_k, e_\ell \rangle = \begin{cases} 1 & \text{if } k = \ell \\ 0 & \text{if } k \neq \ell \end{cases}.$$

Note that  $\|e_k\|_H^2 = \langle e_k, e_k \rangle = 1$ .

## Theorem 2

Let  $\{e_j : j = 1, \dots, N\}$  be an ONS. Then for any  $x \in H$ :

- i.  $\sum_{j=1}^N \langle x, e_j \rangle e_j$  and  $x - \sum_{j=1}^N \langle x, e_j \rangle e_j$  are orthogonal.
- ii.  $\|x\|_H^2 = \|x - \sum_{j=1}^N \langle x, e_j \rangle e_j\|_H^2 + \|\sum_{j=1}^N \langle x, e_j \rangle e_j\|_H^2$ .
- iii. **Bessel's Inequality:**

$$\sum_{j=1}^N |\langle x, e_j \rangle|^2 \leq \|x\|_H^2.$$

**Proof.** We first prove (i) by showing that  $x - \sum_{j=1}^N \langle x, e_j \rangle e_j \perp e_k$  for all  $k = 1, \dots, N$ :

$$\langle x - \sum_{j=1}^N \langle x, e_j \rangle e_j, e_k \rangle = \langle x, e_k \rangle - \langle x, e_k \rangle \langle e_k, e_k \rangle = \langle x, e_k \rangle - \langle x, e_k \rangle = 0.$$

Next, (ii) follows directly from (i) and the Pythagorean Theorem, and (iii) clearly follows by dropping the non-negative first term from the right-hand side of (ii).

**Remark.** If  $\{e_j : j = 1, \dots, N\}$  is an ONS, then  $\{e_1, \dots, e_N\}$  is linearly independent. Indeed,

$$0 = \sum_{j=1}^N \alpha_j e_j \implies \alpha_k = 0 \quad \forall k$$

which is shown by taking the inner product of both sides with each  $e_k$ .



## Cauchy-Schwarz Inequality

For any  $x, y \in H$ ,

$$|\langle x, y \rangle| \leq \|x\|_H \|y\|_H.$$

**Proof.** Let  $y \neq 0$ . Consider the single-element ONS  $\{\frac{y}{\|y\|_H}\}$ . By Bessel's inequality:

$$\sum_{j=1}^1 |\langle x, e_j \rangle|^2 \leq \|x\|_H^2 \implies \left| \left\langle x, \frac{y}{\|y\|_H} \right\rangle \right|^2 \leq \|x\|_H^2 \implies |\langle x, y \rangle|^2 \leq \|x\|_H^2 \|y\|_H^2.$$

## Corollary 1

$\|x\|_H = \sqrt{\langle x, x \rangle}$ ,  $x \in H$ , is a well-defined norm.

**Proof.** We prove the triangle inequality. For any  $x, y \in H$ :

$$\begin{aligned} \|x + y\|_H^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &\leq \langle x, x \rangle + |\langle x, y \rangle| + |\langle y, x \rangle| + \langle y, y \rangle \\ &\leq \|x\|_H^2 + \|y\|_H^2 + 2\|x\|_H \|y\|_H \quad (\text{by Cauchy-Schwarz}) \\ &= (\|x\|_H + \|y\|_H)^2. \end{aligned}$$

## Hilbert Space

A pre-Hilbert space  $H$  is called a **Hilbert space** if it is complete with respect to the induced norm  $\|\cdot\|_H$ .

## Projection Theorem

Let  $H$  be a pre-Hilbert space. Given a vector subspace  $M \subseteq H$ , its orthogonal complement is defined as:

$$M^\perp = \{x \in H : \langle x, m \rangle = 0 \quad \forall m \in M\}.$$

We write  $x \perp M$  if  $x \in M^\perp$ . Note that  $M^\perp$  is always a closed vector subspace of  $H$  due to the continuity of the inner product.

## Theorem 3

Let  $H$  be a Hilbert space and  $M \subseteq H$  be a closed vector subspace.

i.  $\forall x \in H$ , there exists a unique  $m_0 \in M$  such that

$$d := \inf_{m \in M} \|x - m\|_H = \|x - m_0\|_H$$

Moreover,  $(x - m_0) \perp M$ . The vector  $m_0$  is called the **orthogonal projection** of  $x$  onto  $M$ , denoted by  $P_M x = m_0$ .

ii. Let  $m_1 \in M$  be such that  $(x - m_1) \perp M$ . Then  $m_1 = m_0$ .

**Remark.** Theorem 3 implies that  $\forall x \in H$ , there exists a unique decomposition  $x = m_0 + w$  where  $m_0 \in M$  and  $w \in M^\perp$ . Thus,  $H = M \oplus M^\perp$ .

**Proof of i.** Let  $x \in H$ . By the definition of infimum, there exists a sequence  $\{m_k\} \subseteq M$  such that  $\|x - m_k\|_H \rightarrow d$  as  $k \rightarrow \infty$ . We show that  $\{m_k\}$  is a Cauchy sequence using the Parallelogram Law:

$$\|m_k - m_\ell\|_H^2 = \|(x - m_\ell) - (x - m_k)\|_H^2 = 2\|x - m_\ell\|_H^2 + 2\|x - m_k\|_H^2 - \|2x - (m_\ell + m_k)\|_H^2.$$

Since  $M$  is a vector subspace,  $\frac{m_\ell + m_k}{2} \in M$ , which implies  $\|x - (\frac{m_\ell + m_k}{2})\|_H \geq d$ . Thus,

$$\|2x - (m_\ell + m_k)\|_H^2 = 4 \left\| x - \left( \frac{m_\ell + m_k}{2} \right) \right\|_H^2 \geq 4d^2.$$

Substituting this back gives:

$$\|m_k - m_\ell\|_H^2 \leq 2\|x - m_\ell\|_H^2 + 2\|x - m_k\|_H^2 - 4d^2 \xrightarrow{k, \ell \rightarrow \infty} 2d^2 + 2d^2 - 4d^2 = 0.$$

Since  $H$  is complete,  $m_0 := \lim_{k \rightarrow \infty} m_k$  exists, and by closure of  $M$ ,  $m_0 \in M$  with  $\|x - m_0\|_H = d$ .

For uniqueness, if  $\tilde{m}_0 \in M$  also achieves the distance  $d$ , applying the same inequality with  $m_k = m_0$  and  $m_\ell = \tilde{m}_0$  yields:

$$\|m_0 - \tilde{m}_0\|_H^2 \leq 2d^2 + 2d^2 - 4d^2 = 0 \implies m_0 = \tilde{m}_0.$$

To prove that  $(x - m_0) \perp M$ , let  $m \in M$  and  $t \in \mathbb{R}$ . Expanding the norm using inner product properties:

$$\|(x - m_0) - tm\|_H^2 = \|x - m_0\|_H^2 - 2t\operatorname{Re}\{\langle x - m_0, m \rangle\} + t^2\|m\|_H^2.$$

Since  $m_0$  minimizes the distance, we have  $d^2 \leq \|(x - m_0) - tm\|_H^2$ , which simplifies to:

$$0 \leq t^2\|m\|_H^2 - 2t\operatorname{Re}\{\langle x - m_0, m \rangle\} \quad \forall t \in \mathbb{R}.$$

For this quadratic function to remain non-negative for all  $t$ , the linear coefficient must vanish:

$$\operatorname{Re}\{\langle x - m_0, m \rangle\} = 0.$$

Replacing  $t$  by  $it$  in the original formulation yields  $\operatorname{Im}\{\langle x - m_0, m \rangle\} = 0$ . Since both parts are zero,  $\langle x - m_0, m \rangle = 0$ , proving orthogonality.

**Proof of ii.** Let  $m_1 \in M$  satisfy  $(x - m_1) \perp M$ . For any  $m \in M$ , since  $(m_1 - m) \in M$ , we have  $(x - m_1) \perp (m_1 - m)$ . By the Pythagorean Theorem:

$$\|x - m\|_H^2 = \|(x - m_1) + (m_1 - m)\|_H^2 = \|x - m_1\|_H^2 + \|m_1 - m\|_H^2 \geq \|x - m_1\|_H^2.$$

Taking the infimum over all  $m \in M$  implies  $\|x - m_1\|_H^2 = d^2$ . By the uniqueness proven in part (i),  $m_1 = m_0$ .

## Corollary 2

Let  $\{e_1, \dots, e_N\}$  be an ONS in a Hilbert space  $H$ , and let  $M = \text{span}\{e_1, \dots, e_N\}$ . Then for any  $x \in H$ :

$$P_M(x) = \sum_{j=1}^N \langle x, e_j \rangle e_j.$$

**Proof.** Theorem 2 established that  $(x - \sum_{j=1}^N \langle x, e_j \rangle e_j) \perp e_k$  for all  $k$ , which implies it is orthogonal to the entire span  $M$ . Applying Theorem 3 (ii) completes the proof.

## Hilbert Space Basis

Let  $H$  be a Hilbert space. An orthonormal system  $S \subseteq H$  is called a **complete orthonormal system (CONS)** or a **Hilbert basis** if for any  $z \in H$ :

$$\langle z, e_\alpha \rangle = 0 \quad \forall e_\alpha \in S \implies z = 0.$$

## Example

Let  $\mu$  be the counting measure on  $[0, 1]$ . Consider  $H = L^2([0, 1], \mu)$ , the space of functions  $f : [0, 1] \rightarrow \mathbb{R}$  such that

$$\sum_{t \in [0, 1]} [f(t)]^2 = \int_{[0, 1]} [f(t)]^2 d\mu < \infty.$$

The inner product is given by  $\langle f, g \rangle = \sum_{t \in [0, 1]} f(t)g(t)$ . For each  $\alpha \in [0, 1]$ , defining the standard indicator functions  $e_\alpha(t) = 1$  if  $\alpha = t$  and 0 otherwise forms a non-countable CONS where  $f(t) = \sum_{\alpha \in [0, 1]} \langle f, e_\alpha \rangle e_\alpha(t)$ .

## Theorem 4

Let  $H$  be a Hilbert space with a CONS  $\{e_\alpha : \alpha \in A\}$ . For any  $x \in H$ :

- i. The set  $\mathcal{A} := \{\alpha \in A : \langle x, e_\alpha \rangle \neq 0\}$  is at most countable:  $\mathcal{A} = \{\alpha_1, \alpha_2, \dots\}$ .
- ii.  $x = \sum_{\alpha \in A} \langle x, e_\alpha \rangle e_\alpha = \sum_{k=1}^{\infty} \langle x, e_{\alpha_k} \rangle e_{\alpha_k} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \langle x, e_{\alpha_k} \rangle e_{\alpha_k}$ .
- iii. **Parseval's Identity:**

$$\|x\|_H^2 = \sum_{\alpha \in A} |\langle x, e_\alpha \rangle|^2 = \sum_{k=1}^{\infty} |\langle x, e_{\alpha_k} \rangle|^2.$$

## Proof.

- i. For any finite subset  $\{\beta_1, \dots, \beta_n\} \subseteq A$ , Bessel's inequality states  $\sum_{k=1}^n |\langle x, e_{\beta_k} \rangle|^2 \leq \|x\|_H^2 < \infty$ . For each  $n \geq 1$ , define  $A_n = \{\alpha \in A : |\langle x, e_\alpha \rangle| > \frac{1}{n}\}$ . If any  $A_n$  were infinite, it would contain a countable sequence contradicting Bessel's bounded sum. Hence, each  $A_n$  is finite, making  $\mathcal{A} = \bigcup_{n=1}^{\infty} A_n$  a countable union of finite sets, which is countable.

- ii. Let  $\{\alpha_k\}_{k=1}^\infty$  be an enumeration of  $\mathcal{A}$  and define the partial sums  $s_n = \sum_{k=1}^n \langle x, e_{\alpha_k} \rangle e_{\alpha_k}$ . For  $m > n$ , by the Pythagorean theorem:

$$\|s_m - s_n\|_H^2 = \sum_{k=n+1}^m |\langle x, e_{\alpha_k} \rangle|^2.$$

As  $n, m \rightarrow \infty$ , this goes to 0 because the total series  $\sum_{k=1}^\infty |\langle x, e_{\alpha_k} \rangle|^2$  converges. Thus  $\{s_n\}$  is Cauchy, and since  $H$  is complete, it converges to some  $y \in H$ . To show  $y = x$ , we look at the inner product for any  $\alpha \in A$ :

$$\langle x - y, e_\alpha \rangle = \langle x, e_\alpha \rangle - \lim_{n \rightarrow \infty} \sum_{k=1}^n \langle x, e_{\alpha_k} \rangle \langle e_{\alpha_k}, e_\alpha \rangle.$$

If  $\alpha \notin \mathcal{A}$ , both terms are 0. If  $\alpha \in \mathcal{A}$ , then  $\alpha = \alpha_n$  for some  $n$ , yielding  $\langle x, e_{\alpha_n} \rangle - \langle x, e_{\alpha_n} \rangle = 0$ . Since  $\langle x - y, e_\alpha \rangle = 0$  for all  $\alpha$ , by the completeness of the system,  $x - y = 0 \implies x = y$ .

- iii. Applying Theorem 2 (ii) to the finite partial sums yields:

$$\|x\|_H^2 = \left\| x - \sum_{k=1}^n \langle x, e_{\alpha_k} \rangle e_{\alpha_k} \right\|_H^2 + \sum_{k=1}^n |\langle x, e_{\alpha_k} \rangle|^2.$$

Taking  $n \rightarrow \infty$ , the first norm term vanishes due to part (ii), leaving Parseval's Identity.

## Theorem 5

A Hilbert space  $H$  has a countable CONS if and only if  $H$  is separable (contains a countable dense subset).

**Proof.** ( $\implies$ ) Let  $\{e_n\}_{n \geq 1}$  be a countable CONS. The set of all finite linear combinations  $\sum_{k=1}^n r_k e_k$  where  $r_k \in \mathbb{Q} + i\mathbb{Q}$  forms a countable dense subset in  $H$ . ( $\impliedby$ ) Let  $\{y_1, y_2, \dots\}$  be a countable dense subset. We can filter out linearly dependent vectors to get a linearly independent sequence  $\{x_n\}$  spanning the same subspace. Applying the standard Gram-Schmidt procedure to  $\{x_n\}$  constructs a countable orthonormal system  $\{e_n\}$  whose span is dense in  $H$ , hence forming a CONS.

## Proposition 2

Let  $H$  be a pre-Hilbert space. For a fixed  $z \in H$ , define  $\ell_z(x) = \langle x, z \rangle$ . Then  $\ell_z \in H^*$  and  $\|\ell_z\|_{H^*} = \|z\|_H$ .

**Proof.** Boundedness follows from Cauchy-Schwarz:  $|\ell_z(x)| = |\langle x, z \rangle| \leq \|z\|_H \|x\|_H$ , showing  $\|\ell_z\|_{H^*} \leq \|z\|_H$ . Testing the functional on the unit vector  $x = \frac{z}{\|z\|_H}$  (for  $z \neq 0$ ) yields exactly  $\|z\|_H$ , establishing equality.

## Riesz Representation Theorem

Let  $H$  be a Hilbert space. For every bounded linear functional  $\ell \in H^*$ , there exists a unique  $z_\ell \in H$  such that

$$\ell(x) = \langle x, z_\ell \rangle, \quad \forall x \in H.$$

Moreover,  $\|\ell\|_{H^*} = \|z_\ell\|_H$ .

**Proof.** If  $\ell = 0$ , choosing  $z_\ell = 0$  works. Assume  $\ell \neq 0$ , and let  $M = \ker(\ell)$ . Since  $\ell$  is continuous,  $M$  is a closed subspace, and  $M \neq H$ . By the Projection Theorem,  $M^\perp \neq \{0\}$ . Pick a nonzero  $x_0 \in M^\perp$  scaled such that  $\ell(x_0) = 1$ . For any  $x \in H$ , the vector  $x - \ell(x)x_0$  belongs to  $M$  because  $\ell(x - \ell(x)x_0) = \ell(x) - \ell(x) = 0$ . Since  $x_0 \in M^\perp$ , we have:

$$\langle x - \ell(x)x_0, x_0 \rangle = 0 \implies \langle x, x_0 \rangle = \ell(x)\|x_0\|_H^2 \implies \ell(x) = \left\langle x, \frac{x_0}{\|x_0\|_H^2} \right\rangle.$$

Setting  $z_\ell = \frac{x_0}{\|x_0\|_H^2}$  proves existence. Uniqueness follows since if  $\langle x, z_1 \rangle = \langle x, z_2 \rangle$  for all  $x$ , then  $\langle x, z_1 - z_2 \rangle = 0 \implies z_1 - z_2 = 0$ .

## Adjoint Operator

Let  $H$  be a Hilbert space. For each  $T \in \mathcal{L}(H)$ , there exists a unique  $T^* \in \mathcal{L}(H)$  such that

$$\langle Tx, y \rangle = \langle x, T^*y \rangle \quad \forall x, y \in H.$$

Moreover,  $\|T^*\| = \|T\|$ .

**Proof.** For a fixed  $y \in H$ ,  $x \mapsto \langle Tx, y \rangle$  is a bounded linear functional on  $H$ . By the Riesz Representation Theorem, there exists a unique vector, which we denote as  $T^*y$ , such that  $\langle Tx, y \rangle = \langle x, T^*y \rangle$ . To verify that  $T^*$  is linear on a complex Hilbert space, consider  $y_1, y_2 \in H$  and  $\alpha, \beta \in \mathbb{C}$ . For any  $x \in H$ :

$$\begin{aligned} \langle x, T^*(\alpha y_1 + \beta y_2) \rangle &= \langle Tx, \alpha y_1 + \beta y_2 \rangle \\ &= \bar{\alpha} \langle Tx, y_1 \rangle + \bar{\beta} \langle Tx, y_2 \rangle \quad (\text{conjugate-linear in 2nd slot}) \\ &= \bar{\alpha} \langle x, T^*y_1 \rangle + \bar{\beta} \langle x, T^*y_2 \rangle \\ &= \langle x, \alpha T^*y_1 + \beta T^*y_2 \rangle \quad (\text{bringing scalars back inside}) \end{aligned}$$

Since this holds for all  $x \in H$ ,  $T^*(\alpha y_1 + \beta y_2) = \alpha T^*y_1 + \beta T^*y_2$ . The norm equality follows from:

$$\|T^*\| = \sup_{\|y\| \leq 1} \sup_{\|x\| \leq 1} |\langle x, T^*y \rangle| = \sup_{\|y\| \leq 1} \sup_{\|x\| \leq 1} |\langle Tx, y \rangle| = \sup_{\|x\| \leq 1} \|Tx\|_H = \|T\|.$$

## Self-Adjoint Operator

An operator  $T \in \mathcal{L}(H)$  is called **self-adjoint** if  $T^* = T$ .

### Lemma

Let  $H$  be a Hilbert space and  $T \in \mathcal{L}(H)$ . Then,

$$\|T^*T\| = \|T\|^2 = \|T^*\|^2.$$

**Proof.** For any  $x \in H$  with  $\|x\|_H \leq 1$ :

$$\|Tx\|_H^2 = \langle Tx, Tx \rangle = \langle x, T^*Tx \rangle \leq \|x\|_H \|T^*Tx\|_H \leq \|T^*T\|.$$

Taking the supremum over all such  $x$  yields  $\|T\|^2 \leq \|T^*T\|$ . Combined with the standard norm inequality  $\|T^*T\| \leq \|T^*\| \|T\| = \|T\|^2$ , we get equality.

### Remark

Let  $H$  be a Hilbert space. Then:

1.  $\mathcal{A} := \mathcal{L}(H)$  is a  $C^*$ -algebra.
2.  $\mathcal{A}$  is a Banach algebra satisfying  $\|ST\| \leq \|S\| \|T\|$ .
3. The adjoint operation  $T \mapsto T^*$  is an involution satisfying:
  - $(\alpha T + \beta S)^* = \bar{\alpha} T^* + \bar{\beta} S^*$ .
  - $(ST)^* = T^* S^*$ .
  - $\|T^*T\| = \|T\|^2$ .

### $L^p$ Spaces

Let  $(X, \mathcal{X}, \mu)$  be a measure space. For  $0 < p \leq \infty$ , we define the space  $L^p(X, \mathcal{X}, \mu)$  (abbreviated as  $L^p(\mu)$  or  $L^p$ ) as:

$$L^p = \left\{ f : X \rightarrow \mathbb{C} \text{ is measurable} : \int_X |f|^p d\mu < \infty \right\}.$$

We identify functions that are equal almost everywhere (a.e.) to form equivalence classes. Additionally, we define  $L^\infty$  to be the space of equivalence classes of measurable functions  $f : X \rightarrow \mathbb{C}$  that are essentially bounded:

$$|f| \leq C \text{ a.e. for some } C > 0.$$

**Remark.**  $L^p$  is a vector space. For instance, if  $p \in (0, \infty)$  and  $a, b > 0$ , the inequality  $(a + b)^p \leq 2^p(a^p + b^p)$  holds because:

$$(a + b)^p \leq (2 \max\{a, b\})^p = 2^p \max\{a^p, b^p\} \leq 2^p(a^p + b^p).$$

For  $f \in L^p$  where  $p \in (0, \infty)$ , we define the  $L^p$  norm (or quasi-norm if  $p < 1$ ) as:

$$\|f\|_{L^p} = \|f\|_p = \left( \int_X |f|^p d\mu \right)^{1/p}.$$

For  $p = \infty$ , the essential supremum norm is defined as:

$$\|f\|_{L^\infty} = \|f\|_\infty = \inf\{C \geq 0 : |f| \leq C \text{ a.e.}\}.$$

*Note.* By definition,  $|f| \leq \|f\|_\infty$  a.e.

### Young's Inequality

For  $a, b \geq 0$ ,  $p \in (1, \infty)$ , and  $\frac{1}{p} + \frac{1}{q} = 1$ ,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

**Proof.** Let  $a, b > 0$  (the case where  $a = 0$  or  $b = 0$  is trivial). Since the natural logarithm function  $\ln$  is concave, we can apply Jensen's inequality:

$$\ln\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \geq \frac{1}{p} \ln(a^p) + \frac{1}{q} \ln(b^q) = \ln(a) + \ln(b) = \ln(ab).$$

Taking the exponential (exp) of both sides yields the desired result.

### Hölder's Inequality

Let  $p \in [1, \infty]$  and let  $\frac{1}{p} + \frac{1}{q} = 1$  (with  $\frac{1}{\infty} = 0$ ). If  $f \in L^p$  and  $g \in L^q$ , then  $fg \in L^1$  and:

$$\int_X |fg| d\mu \leq \|f\|_p \|g\|_q.$$

**Proof.** For  $p = 1$  or  $p = \infty$ , the proof is straightforward and left as an exercise. Let  $p \in (1, \infty)$ ,  $f \neq 0$ , and  $g \neq 0$  a.e. By Young's inequality, for any parameter  $t > 0$ , we have:

$$\int_X |fg| d\mu = \int_X |(tf)(t^{-1}g)| d\mu \leq \frac{t^p}{p} \int_X |f|^p d\mu + \frac{t^{-q}}{q} \int_X |g|^q d\mu.$$

Let  $A = \int_X |f|^p d\mu$  and  $B = \int_X |g|^q d\mu$ . Define the auxiliary function  $L(t)$  for  $t > 0$ :

$$L(t) = \frac{t^p}{p} A + \frac{t^{-q}}{q} B.$$

To minimize  $L(t)$ , we find its derivative and set it to zero:

$$L'(t) = t^{p-1} A - t^{-q-1} B = 0 \implies t^{p+q} = \frac{B}{A} \implies t = \left(\frac{B}{A}\right)^{\frac{1}{pq}}.$$

Evaluating  $L(t)$  at this critical point gives:

$$\inf_{t>0} L(t) = \frac{1}{p} \left(\frac{B}{A}\right)^{1/q} A + \frac{1}{q} \left(\frac{B}{A}\right)^{-1/p} B = \frac{1}{p} A^{1/p} B^{1/q} + \frac{1}{q} A^{1/p} B^{1/q} = A^{1/p} B^{1/q}.$$

Substituting  $A^{1/p} = \|f\|_p$  and  $B^{1/q} = \|g\|_q$  completes the proof.

**Remark.** If  $f \in L^p$  and  $\frac{1}{p} + \frac{1}{q} = 1$ , then  $|f|^{p-1} \in L^q$ .

### Proposition 1 (Minkowski's Inequality)

For  $p \in [1, \infty]$ ,  $\|\cdot\|_p$  satisfies the triangle inequality and forms a norm on  $L^p$ .

**Proof.** For  $p = 1$  or  $p = \infty$ , the proof is left as an exercise. Let  $p \in (1, \infty)$  and  $f, g \in L^p$ . We assume  $f + g \neq 0$  dynamically. Splitting the integrand yields:

$$\int_X |f + g|^p d\mu = \int_X |f + g|^{p-1} |f + g| d\mu \leq \int_X |f + g|^{p-1} |f| d\mu + \int_X |f + g|^{p-1} |g| d\mu.$$

Applying Hölder's inequality to both integrals on the right-hand side using conjugate exponents  $p$  and  $q = \frac{p}{p-1}$ , we get:

$$\int_X |f + g|^p d\mu \leq \left( \int_X |f + g|^{(p-1)q} d\mu \right)^{1/q} \|f\|_p + \left( \int_X |f + g|^{(p-1)q} d\mu \right)^{1/q} \|g\|_p.$$

Since  $(p-1)q = p$  and  $1 - \frac{1}{q} = \frac{1}{p}$ , this simplifies directly to:

$$\int_X |f + g|^p d\mu \leq \left( \int_X |f + g|^p d\mu \right)^{1-\frac{1}{p}} (\|f\|_p + \|g\|_p).$$

Dividing both sides by  $\left( \int_X |f + g|^p d\mu \right)^{1-\frac{1}{p}}$  yields:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

**Remark.** Let  $f_n, f \in L^\infty$ . Then  $\|f_n - f\|_\infty \rightarrow 0$  if and only if there exist representatives  $\tilde{f}_n, \tilde{f}$  in their respective equivalence classes such that  $\sup_{x \in X} |\tilde{f}_n(x) - \tilde{f}(x)| \rightarrow 0$  as  $n \rightarrow \infty$ .

### Chebyshev's Inequality

Let  $p \in (0, \infty)$  and  $f \in L^p$ . Then, for any  $\epsilon > 0$ :

$$\mu(\{x \in X : |f(x)| > \epsilon\}) \leq \frac{\int_X |f|^p d\mu}{\epsilon^p}.$$

**Proof.** Let  $\epsilon > 0$ . By utilizing monotonic sets, we observe:

$$\int_X |f|^p d\mu \geq \int_{\{|f|>\epsilon\}} |f|^p d\mu \geq \epsilon^p \mu(\{x \in X : |f(x)| > \epsilon\}).$$

### Theorem 2

$L^p$  is a Banach space for any  $p \in [1, \infty]$ .

**Proof.** The case  $p = \infty$  is left as an exercise. Let  $p \in [1, \infty)$  and let  $(f_n)$  be a Cauchy sequence in  $L^p$ . By Chebyshev's Inequality, for any  $\epsilon > 0$ :

$$\mu(\{|f_n - f_m| > \epsilon\}) \leq \frac{\int_X |f_n - f_m|^p d\mu}{\epsilon^p} \xrightarrow{n,m \rightarrow \infty} 0.$$



Hence,  $(f_n)$  is Cauchy in measure. This implies there exists a measurable function  $f$  and a subsequence  $(f_{n_k})$  such that  $f_{n_k} \rightarrow f$  a.e. as  $k \rightarrow \infty$ .

Fix  $\epsilon > 0$ . Choose  $N$  such that  $\int_X |f_n - f_m|^p d\mu \leq \epsilon$  for all  $n, m \geq N$ . Letting  $m = n_l \rightarrow \infty$ , Fatou's Lemma implies:

$$\int_X |f_n - f|^p d\mu = \int_X \lim_{l \rightarrow \infty} |f_n - f_{n_l}|^p d\mu \leq \liminf_{l \rightarrow \infty} \int_X |f_n - f_{n_l}|^p d\mu \leq \epsilon$$

for all  $n \geq N$ . Thus,  $f - f_n \in L^p$ , which means  $f = (f - f_n) + f_n \in L^p$ , and  $f_n \rightarrow f$  in  $L^p$ .

### Comparison of $L^p$ Spaces

In general, spaces are not nested: neither  $L^p \subseteq L^q$  nor  $L^q \subseteq L^p$  if  $p \neq q$  on arbitrary measure spaces.

**Proof.** Let  $X = (0, \infty)$  equipped with Lebesgue measure  $\mu$ . Consider the function:

$$f(x) = x^{-\alpha} \chi_{(0,1)}(x) + x^{-\beta} \chi_{[1,\infty)}(x) \quad (\alpha, \beta > 0).$$

Calculating the  $L^r$  integral gives  $\int_X |f(x)|^r d\mu = \int_0^1 x^{-\alpha r} dx + \int_1^\infty x^{-\beta r} dx$ . This converges if and only if  $\alpha < \frac{1}{r} < \beta$ . Thus, if we select exponents such that  $\frac{1}{q} < \alpha < \frac{1}{p} < \beta$ , then  $f \in L^p$  but  $f \notin L^q$ . Conversely, choosing  $\alpha < \frac{1}{q} < \beta < \frac{1}{p}$  yields  $f \in L^q$  but  $f \notin L^p$ .

### Proposition 3

Let  $\mu(X) < \infty$  and  $0 < p < q \leq \infty$ . Then  $L^q \subseteq L^p$  and:

$$\|f\|_p \leq \|f\|_q \mu(X)^{\frac{1}{p} - \frac{1}{q}}.$$

**Proof.** Applying Hölder's inequality with conjugate exponents  $\frac{q}{p}$  and  $\frac{q}{q-p}$  yields:

$$\int_X |f|^p \cdot 1 d\mu \leq \left( \int_X |f|^q d\mu \right)^{\frac{p}{q}} \left( \int_X 1 d\mu \right)^{1 - \frac{p}{q}} = \|f\|_q^p \mu(X)^{1 - \frac{p}{q}}.$$

Taking the  $p$ -th root of both sides proves the assertion.

### Proposition 4

Let  $0 < p < q < r \leq \infty$ . Then  $L^q \subseteq L^p + L^r$ .

**Proof.** Let  $f \in L^q$ . We decompose  $f = f \chi_{\{|f| \leq 1\}} + f \chi_{\{|f| > 1\}} =: k + h$ . Then  $|k| \leq |f|$  and  $|k| \leq 1$ , so  $|k|^r \leq |f|^q \in L^1$ , implying  $k \in L^r$ . Similarly,  $|h| \leq |f|$  and on the support of  $h$ ,  $|f| > 1$ , meaning  $|h|^p \leq |f|^q \in L^1$ , implying  $h \in L^p$ .

### Proposition 5

Let  $0 < p < q < r \leq \infty$ . Then  $L^p \cap L^r \subseteq L^q$  and  $\|f\|_q \leq \|f\|_p^t \|f\|_r^{1-t}$  where  $\frac{1}{q} = \frac{t}{p} + \frac{1-t}{r}$ .

**Proof.** Let  $f \in L^p \cap L^r$ .

- i. If  $r = \infty$ , then  $\int_X |f|^q d\mu = \int_X |f|^{q-p} |f|^p d\mu \leq \|f\|_\infty^{q-p} \|f\|_p^p$ . Taking the  $q$ -th root gives the result with  $t = p/q$ .
- ii. If  $r < \infty$ , rewrite  $|f|^q = |f|^{tq} |f|^{(1-t)q}$ . Apply Hölder's inequality with conjugate exponents  $\frac{p}{tq}$  and  $\frac{r}{(1-t)q}$ . The conditions require  $\frac{tq}{p} + \frac{(1-t)q}{r} = 1$ , which aligns perfectly with the definition of  $t$ .

### Proposition 6 (Sequence Spaces)

If  $X = \mathbb{N}$  and  $\mu$  is the counting measure, we denote  $\ell^p = L^p(\mathbb{N}, \mu)$ . If  $0 < p < q \leq \infty$ , then  $\ell^p \subseteq \ell^q$  and  $\|x\|_q \leq \|x\|_p$ .

**Proof.** Let  $x \in \ell^p$ . For  $q = \infty$ ,  $\|x\|_\infty^p = \sup_n |x_n|^p \leq \sum_n |x_n|^p = \|x\|_p^p \implies \|x\|_\infty \leq \|x\|_p$ . For  $q < \infty$ , by interpolation (Proposition 5),  $\|x\|_q \leq \|x\|_p^{p/q} \|x\|_\infty^{1-p/q} \leq \|x\|_p$ .

### Simple Function Spaces

Define  $\Sigma = \{f \in \mathcal{E} : f \text{ vanishes outside a set of finite measure}\}$ , where  $\mathcal{E}$  is the space of all complex-valued simple functions:

$$\mathcal{E} = \left\{ f = \sum_{j=1}^n z_j \chi_{E_j} : n \geq 1, z_j \in \mathbb{C}, E_j \in \mathcal{X} \text{ are disjoint} \right\}.$$

**Remark.**  $\mathcal{E} \subseteq L^\infty$  and  $\Sigma \subseteq L^p$  for any  $p \in (0, \infty]$ .

### Theorem 2.10 [Folland]

Let  $(X, \mathcal{X})$  be a measurable space. If  $f : X \rightarrow \mathbb{C}$  is measurable, there exists a sequence  $\{f_n\} \subseteq \mathcal{E}$  such that  $0 \leq |f_1| \leq |f_2| \leq \dots \leq |f|$ ,  $f_n \rightarrow f$  pointwise, and  $f_n \rightarrow f$  uniformly on any set where  $f$  is bounded.

### Proposition 7

- i.  $\mathcal{E}$  is dense in  $L^\infty$ .
- ii.  $\Sigma$  is dense in  $L^p$  if  $p \in [1, \infty)$ .

**Proof.** Apply Theorem 2.10. For (i), since  $f \in L^\infty$  is bounded outside a null set,  $f_n \rightarrow f$  uniformly a.e., meaning  $\|f_n - f\|_\infty \rightarrow 0$ . For (ii), if  $f \in L^p$ ,  $|f_n - f|^p \leq 2^{p+1} |f|^p \in L^1$ . By the Dominated Convergence Theorem,  $\int_X |f_n - f|^p d\mu \rightarrow 0$ .

## Generalized Dominated Convergence Theorem

Let  $\{f_n\}$  be a sequence of measurable functions such that  $f_n \rightarrow f$  a.e. Suppose there exists a sequence of nonnegative integrable functions  $\{g_n\}$  such that  $|f_n| \leq g_n$  a.e.,  $g_n \rightarrow g$  a.e., and  $\int_X g_n d\mu \rightarrow \int_X g d\mu < \infty$ . Then  $f_n \rightarrow f$  in  $L^1$ .

**Proof.** Consider the sequence of nonnegative functions  $h_n = g_n + g - |f_n - f|$ . Since  $|f_n - f| \leq g_n + g$  a.e.,  $h_n \geq 0$  a.e. Applying Fatou's Lemma to  $\{h_n\}$  yields:

$$\int_X \liminf_{n \rightarrow \infty} h_n d\mu \leq \liminf_{n \rightarrow \infty} \int_X h_n d\mu.$$

Since  $f_n \rightarrow f$  and  $g_n \rightarrow g$  a.e.,  $\liminf h_n = 2g$  a.e. Thus:

$$\int_X 2g d\mu \leq \int_X g d\mu + \int_X g d\mu - \limsup_{n \rightarrow \infty} \int_X |f_n - f| d\mu.$$

Subtracting  $2 \int_X g d\mu < \infty$  from both sides results in  $\limsup_{n \rightarrow \infty} \int_X |f_n - f| d\mu \leq 0$ , which ensures  $\lim_{n \rightarrow \infty} \|f_n - f\|_1 = 0$ .

## The Dual of $L^p$

Let  $p \in (1, \infty]$  and  $q \in [1, \infty)$  with  $\frac{1}{p} + \frac{1}{q} = 1$ . For any  $h \in L^q$ , define the linear functional  $T_h$  on  $L^p$  by:

$$T_h(f) = \int_X fh d\mu, \quad f \in L^p.$$

Then  $T_h \in (L^p)^*$  and  $\|T_h\|_{(L^p)^*} = \|h\|_q = \sup\{|\int_X fh d\mu| : \|f\|_p \leq 1\}$ .

## Theorem 4 (Riesz Representation Theorem for $L^p$ )

Let  $p \in (1, \infty)$ . For each bounded linear functional  $T \in (L^p)^*$ , there exists a unique  $h \in L^q$  such that  $T(f) = \int_X hf d\mu$  for all  $f \in L^p$ , and  $\|T\|_{(L^p)^*} = \|h\|_q$ . If  $\mu$  is  $\sigma$ -finite, this also holds for  $p = 1$  and  $q = \infty$ .

## Weak $L^p$

Let  $p \in (0, \infty)$ . A measurable function  $f$  belongs to **weak**  $L^p$  if its distribution function  $\lambda_f(\alpha) = \mu(\{x \in X : |f(x)| > \alpha\})$  satisfies:

$$[f]_p := \left( \sup_{\alpha > 0} \alpha^p \lambda_f(\alpha) \right)^{1/p} < \infty.$$

1. By Chebyshev's Inequality,  $L^p \subseteq \text{weak } L^p$  and  $[f]_p \leq \|f\|_p$ .
2. Using layer-cake integration by parts, if  $f \in L^p$ , its norm can be computed via:

$$\|f\|_p^p = \int_X |f|^p d\mu = p \int_0^\infty \alpha^{p-1} \lambda_f(\alpha) d\alpha.$$

## Convolution and Smoothing

Let  $X = \mathbb{R}^d$ ,  $\mathcal{X} = \mathcal{L}$ , and  $\mu = m \implies d\mu = dm = dx$ .

### Lemma 1

Let  $f, g$  be measurable,  $x_0 \in \mathbb{R}^d$ . Then,

$$f(x_0 - \cdot)g \in L^1 \iff g(x_0 - \cdot)f \in L^1 \quad (1)$$

and if Eq.(1) holds,

$$\int f(x_0 - y)g(y) dy = \int g(x_0 - y)f(y) dy.$$

All functions involved in Eq.(1) are measurable including  $f(x_0 - \cdot)$  and  $g(x_0 - \cdot)$ .

**Proof.** Let  $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$  be bijective with  $G, G^{-1} \in C^1$  and  $x = G(y) = (g_1(y), \dots, g_d(y))$ . Note that  $C^1$  denotes the class of continuously differentiable functions. Then, by the formula of changing variables,

$$\int f(x) dx = \int f(G(y)) |\det \nabla G(y)| dy.$$

$$\nabla G(y) = \begin{bmatrix} \frac{\partial g_1(y)}{\partial y_1} & \dots & \frac{\partial g_1(y)}{\partial y_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_d(y)}{\partial y_1} & \dots & \frac{\partial g_d(y)}{\partial y_d} \end{bmatrix}.$$

In our case, we have  $G(y) = x_0 - y = z$ . Then,

$$\nabla G(y) = \begin{bmatrix} -1 & 0 & \dots & 0 \\ 0 & -1 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -1 \end{bmatrix} = -I_d \implies |\det \nabla G(y)| = |(-1)^d| = 1.$$

$$\implies \int f(x_0 - y)g(y) dy = \int f(G(y))g(x_0 - G(y)) |\det \nabla G(y)| dy = \int g(x_0 - z)f(z) dz.$$

### Convolution

Let  $f, g$  be measurable and  $f(x - \cdot)g \in L^1$  a.e. with respect to  $x$ . The following function is well-defined:

$$(f * g)(x) = \int f(x - y)g(y) dy = \int g(x - y)f(y) dy = (g * f)(x).$$

$f * g$  is called the convolution of  $f$  and  $g$ .

## Well-defined Convolution

Let  $f, g$  be measurable. If

$$\int |f(x-y)||g(y)| dy < \infty \quad \text{a.e. with respect to } x,$$

then we say

$$(f * g)(x) = \int f(x-y)g(y) dy,$$

is well-defined. We say  $(f * g)(x)$  is defined for the  $x$  such that

$$\int |f(x-y)||g(y)| dy < \infty.$$

## Support

Given  $f : \mathbb{R}^d \rightarrow \mathbb{C}$ , the support of  $f$ , denoted  $\text{supp}(f)$ , is the closure of  $\{x : f(x) \neq 0\}$ .  $C_c(\mathbb{R}^d)$  is the space of continuous functions with compact support.

## Lemma 2

If  $f, g \in C_c(\mathbb{R}^d)$ , then  $f * g \in C_c(\mathbb{R}^d)$ . Moreover,

$$\text{supp}(f * g) \subseteq \overline{\text{supp}(f) + \text{supp}(g)},$$

where  $A + B = \{x + y : x \in A, y \in B\}$ .

**Proof.** Since  $f, g \in C_c(\mathbb{R}^d)$ , both  $f, g$  are bounded. Therefore,

$$\int |f(x-y)||g(y)| dy < \infty \text{ for any } x \in \mathbb{R}^d.$$

Hence,  $f * g$  is defined for any  $x \in \mathbb{R}^d$ . Now, let  $\{x_n\}$  be a sequence that converges to  $x_0 \in \mathbb{R}^d$ . Define  $h_n(y) = [f(x_n - y) - f(x_0 - y)]g(y)$  and  $h(y) = [f(x_0 - y) - f(x_0 - y)]g(y) = 0$ . Next, we apply the Dominated Convergence Theorem to prove  $f * g$  is continuous. DCT needs:

1.  $h_n \rightarrow h$  a.e.
2. There exists a nonnegative  $\phi \in L^1$  with  $|h_n| \leq \phi$  a.e. for all  $n$ .

Since  $f \in C_c(\mathbb{R}^d)$ ,  $f$  is continuous. Therefore, for each fixed  $y$ , we have  $f(x_n - y) \rightarrow f(x_0 - y)$  as  $x_n \rightarrow x_0$ . This gives us pointwise convergence:  $h_n(y) \rightarrow 0 = h(y)$  for each  $y$ . We need an integrable function that dominates  $|h_n(y)|$ . Note that since  $f$  has compact support and is continuous, it's bounded:  $|f(z)| \leq M$  for some constant  $M$ . Then,

$$|h_n(y)| = |[f(x_n - y) - f(x_0 - y)]g(y)| \leq 2M|g(y)|.$$

Since  $g \in C_c(\mathbb{R}^d)$ ,  $g$  is bounded, so  $\phi = 2M|g(y)| \in L^1$ . By DCT,

$$\lim_{n \rightarrow \infty} \left( \int f(x_n - y)g(y) dy - \int f(x_0 - y)g(y) dy \right) = \lim_{n \rightarrow \infty} \int h_n(y) dy = \int \lim_{n \rightarrow \infty} h_n(y) dy = 0.$$

Therefore,  $f * g$  is continuous.

Now, we claim  $\{x : (f * g)(x) \neq 0\} \subseteq \text{supp}(f) + \text{supp}(g)$ .

$$\begin{aligned}
(f * g)(x) \neq 0 &\implies \text{There exists } y \text{ such that } f(x - y)g(y) \neq 0, \\
&\implies x - y \in \text{supp}(f), \quad y \in \text{supp}(g), \\
&\implies x = (x - y) + y \in \text{supp}(f) + \text{supp}(g), \\
&\implies \{x : (f * g)(x) \neq 0\} \subseteq \text{supp}(f) + \text{supp}(g).
\end{aligned}$$

Taking the closure of both sides,

$$\text{supp}(f * g) = \overline{\{x : (f * g)(x) \neq 0\}} \subseteq \overline{\text{supp}(f) + \text{supp}(g)}.$$

Since both  $\text{supp}(f)$  and  $\text{supp}(g)$  are compact sets, we use a theorem from topology: The Minkowski sum of two compact sets is compact. Thus,  $\text{supp}(f) + \text{supp}(g)$  is compact. Since it is compact, it is closed in  $\mathbb{R}^d$ . Therefore,

$$\overline{\text{supp}(f) + \text{supp}(g)} = \text{supp}(f) + \text{supp}(g).$$

This gives us  $\text{supp}(f * g) \subseteq \text{supp}(f) + \text{supp}(g)$ . Since  $\text{supp}(f * g)$  is a closed subset of the compact set  $\text{supp}(f) + \text{supp}(g)$ , it is compact. Hence,  $f * g$  is a continuous function with compact support, so  $f * g \in C_c(\mathbb{R}^d)$ .

**Remark.**

1. If a function has compact support, it must be bounded, and here's why: A set in  $\mathbb{R}^d$  is compact if and only if it is closed and bounded. So when we say a function  $f$  has compact support, it means that the closure of the set  $\{x : f(x) \neq 0\}$  is both closed and bounded. For a continuous function  $f$  with compact support, we can apply the extreme value theorem, which states that a continuous function on a compact set attains both its maximum and minimum values. Specifically:

- Since  $f$  is continuous and its support is compact
- By the extreme value theorem,  $f$  must attain a maximum value on its support
- Outside its support,  $f$  is zero by definition
- Therefore,  $f$  is bounded by the maximum absolute value it achieves on its support

This means that there exists some constant  $M > 0$  such that  $|f(x)| \leq M$  for all  $x \in \mathbb{R}^d$ .

2. Lebesgue measure on  $\mathbb{R}^d$  is indeed sigma-finite. We can express  $\mathbb{R}^d$  as a countable union of bounded sets:

$$\mathbb{R}^d = \bigcup_{n=1}^{\infty} B_n.$$

where  $B_n = \{x \in \mathbb{R}^d : |x| \leq n\}$  is the closed ball of radius  $n$  centered at the origin. Each  $B_n$  has finite Lebesgue measure. Specifically:

$$\lambda(B_n) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} \cdot n^d < \infty.$$

This decomposition into a countable union of finite-measure sets is exactly the definition of a sigma-finite measure space.

### Proposition 1

Let  $p \in [1, \infty]$ ,  $\frac{1}{p} + \frac{1}{q} = 1$ ,  $f \in L^p$ ,  $h \in L^q$ . Then,  $(f * h)(x)$  is defined for all  $x \in \mathbb{R}^d$  and

$$\sup_x |(f * h)(x)| \leq \|f\|_p \|h\|_q.$$

Moreover,  $f * h$  is continuous. Also,  $\lim_{|x| \rightarrow \infty} (f * h)(x) = 0$  if  $p \in (1, \infty)$ .

**Proof.** For each  $x \in \mathbb{R}^d$ , by Hölder's inequality,

$$\begin{aligned} \int |f(x-y)| |h(y)| dy &\leq \|f(x-\cdot)\|_p \|h\|_q = \|f\|_p \|h\|_q < \infty, \\ \implies |(f * h)(x)| &= \left| \int f(x-y) h(y) dy \right| \leq \int |f(x-y)| |h(y)| dy \leq \|f\|_p \|h\|_q, \\ \implies \sup_x |(f * h)(x)| &\leq \|f\|_p \|h\|_q. \end{aligned}$$

### Continuity

- (i) Let  $p \in (1, \infty)$ . There exist  $f_n, h_n \in C_c(\mathbb{R}^d)$  such that  $\|f_n - f\|_p \rightarrow 0$  and  $\|h_n - h\|_q \rightarrow 0$  as  $n \rightarrow \infty$ . Recall by Lemma 2,  $f_n * h_n \in C_c(\mathbb{R}^d)$ , and

$$\begin{aligned} \sup_x |(f_n * h_n)(x) - (f * h)(x)| &\leq \sup_x |((f_n - f) * h_n)(x)| + \sup_x |(f * (h_n - h))(x)| \\ &\leq \|f_n - f\|_p \|h_n\|_q + \|f\|_p \|h_n - h\|_q \\ &\leq \left( \sup_k \|h_k\|_q \right) \|f_n - f\|_p + \|f\|_p \|h_n - h\|_q \xrightarrow{n \rightarrow \infty} 0. \end{aligned}$$

Thus,  $f_n * h_n \rightarrow f * h$  uniformly. The uniform convergence of a sequence of continuous functions preserves continuity, so  $f * h$  must be continuous. Also,

$$\lim_{|x| \rightarrow \infty} (f_n * h_n)(x) = 0 \implies \lim_{|x| \rightarrow \infty} (f * h)(x) = 0.$$

- (ii) Let  $p = 1$ ,  $f \in L^1$ ,  $h \in L^\infty$ . There exists  $f_n \in C_c(\mathbb{R}^d)$  such that  $\|f_n - f\|_1 \xrightarrow{n \rightarrow \infty} 0$ . Then,

$$\sup_x |(f_n * h)(x) - (f * h)(x)| = \sup_x |((f_n - f) * h)(x)| \leq \|f_n - f\|_1 \|h\|_\infty \xrightarrow{n \rightarrow \infty} 0.$$

Thus,  $f_n * h$  converges uniformly to  $f * h$  as  $n \rightarrow \infty$ . So we just need to show that  $(f_n * h)(x) = \int f_n(x-y) h(y) dy$  is continuous in  $x$ . By the DCT and continuity of  $f_n$ :

$$\lim_{x \rightarrow x_0} ((f_n * h)(x) - (f_n * h)(x_0)) = \int \lim_{x \rightarrow x_0} [f_n(x-y) - f_n(x_0-y)] h(y) dy = 0.$$

The uniform convergence of a sequence of continuous functions preserves continuity, so  $f * h$  must be continuous.

**Remark.** The property  $\lim_{|x| \rightarrow \infty} (f_n * h_n)(x) = 0$  for functions  $f_n * h_n \in C_c(\mathbb{R}^d)$  follows directly from the definition of compact support. By definition, a function  $g \in C_c(\mathbb{R}^d)$  has compact support, meaning that the set  $\text{supp}(g) = \overline{\{x \in \mathbb{R}^d : g(x) \neq 0\}}$  is compact. Since  $\text{supp}(g)$  is compact in  $\mathbb{R}^d$ , it must be bounded. This means there exists some radius  $R > 0$  such that:

$$\text{supp}(g) \subseteq B(0, R) = \{x \in \mathbb{R}^d : |x| \leq R\}.$$

In other words, for any  $|x| > R$ , we have  $x \notin \text{supp}(g)$ , which means  $g(x) = 0$ . Applying this to  $f_n * h_n \in C_c(\mathbb{R}^d)$ , there exists some  $R_n > 0$  such that for all  $|x| > R_n$ , we have  $(f_n * h_n)(x) = 0$ . Therefore,  $\lim_{|x| \rightarrow \infty} (f_n * h_n)(x) = 0$  is immediate because the function is exactly zero outside some bounded region.

## Proposition 2

Let  $p \in [1, \infty)$ ,  $f \in L^p$ ,  $h \in L^1$ . Then,  $f * h$  is well-defined and  $f * h \in L^p$  with

$$\|f * h\|_p \leq \|f\|_p \|h\|_1.$$

**Proof.** By Minkowski's inequality for integrals,

$$\begin{aligned} \left( \int \left| \int f(x-y)h(y) dy \right|^p dx \right)^{1/p} &\leq \int \left( \int |f(x-y)|^p dx \right)^{1/p} |h(y)| dy \\ &= \|f\|_p \int |h(y)| dy \\ &= \|f\|_p \|h\|_1 < \infty. \end{aligned}$$

Hence,  $\int |f(x-y)||h(y)| dy < \infty$  a.e. with respect to  $x$  and  $\|f * h\|_p \leq \|f\|_p \|h\|_1$ .

## Continuity of Shifts

For  $f$  on  $\mathbb{R}^d$  and  $z \in \mathbb{R}^d$ ,

$$\tau_z f(x) = f(x+z), \quad x \in \mathbb{R}^d.$$

Note that  $\|\tau_z f\|_p = \|f\|_p$ ,  $p \in (0, \infty]$ . Also note that  $\tau_0 f = f$ .

## Lemma 3

Let  $p \in [1, \infty)$ ,  $f \in L^p$ . Then,

1.  $\|\tau_z f - f\|_p \xrightarrow{z \rightarrow 0} 0$  or  $\int |f(x-z) - f(x)|^p dx \xrightarrow{z \rightarrow 0} 0$
2.  $\|\tau_z f - \tau_{z_0} f\|_p = \|\tau_{z-z_0} f - f\|_p \xrightarrow{z \rightarrow z_0} 0$

The second part follows directly from  $\|\tau_z f - f\|_p \xrightarrow{z \rightarrow 0} 0$ .

**Proof.** Let  $g \in C_c(\mathbb{R}^d)$ . Then, by DCT,

$$\int |g(x+z) - g(x)|^p dx \xrightarrow{z \rightarrow 0} 0.$$



Since if  $|z| \leq 1$ ,  $\text{supp}(g) \subseteq B_R(0)$ ,

$$|g(x+z) - g(x)| \leq |g(x+z)| + |g(x)| \leq \chi_{B_{R+1}}(x) \cdot 2\|g\|_\infty \in L^p.$$

Let  $f \in L^p$  with  $p \in [1, \infty)$ . For any  $g \in \mathcal{C}_c(\mathbb{R}^d)$ ,

$$\begin{aligned} \|\tau_z f - f\|_p &\leq \|\tau_z f - \tau_z g\|_p + \|\tau_z g - g\|_p + \|g - f\|_p \\ &= \|f - g\|_p + \|\tau_z g - g\|_p + \|g - f\|_p \\ &= 2\|f - g\|_p + \|\tau_z g - g\|_p. \end{aligned}$$

Since  $g$  is continuous,

$$\limsup_{z \rightarrow 0} \|\tau_z f - f\|_p \leq 2\|f - g\|_p + \limsup_{z \rightarrow 0} \|\tau_z g - g\|_p = 2\|f - g\|_p.$$

Functions in  $\mathcal{C}_c(\mathbb{R}^d)$  are dense in  $L^p$ . The density property states that for any function  $f \in L^p$  and any  $\epsilon > 0$ , there exists a function  $g \in \mathcal{C}_c(\mathbb{R}^d)$  such that  $\|f - g\|_p < \epsilon$ . Using this density property, for any  $\epsilon > 0$ , we can find a  $g \in \mathcal{C}_c(\mathbb{R}^d)$  with  $\|f - g\|_p < \epsilon/2$ . This means  $2\|f - g\|_p < \epsilon$ . Therefore,

$$\limsup_{z \rightarrow 0} \|\tau_z f - f\|_p \leq 2\|f - g\|_p < \epsilon.$$

Since  $\epsilon$  is arbitrary,

$$\limsup_{z \rightarrow 0} \|\tau_z f - f\|_p = 0.$$

Since  $\limsup_{z \rightarrow 0} \|\tau_z f - f\|_p \geq \liminf_{z \rightarrow 0} \|\tau_z f - f\|_p \geq 0$ , we conclude  $\lim_{z \rightarrow 0} \|\tau_z f - f\|_p = 0$ .

## Mollifiers

Let  $\tilde{w}(x) := \ell(|x|_2^2)$ ,  $x \in \mathbb{R}^d$  where  $|x| := \sqrt{x_1^2 + \dots + x_d^2}$ . Then,

$$\tilde{w} \in \mathcal{C}_c^\infty(\mathbb{R}^d), \quad \text{supp}(\tilde{w}) = B_1(0).$$

$\mathcal{C}_c^\infty(\mathbb{R}^d)$  is the space of infinitely differentiable functions with compact support. Let  $w(x) = c\tilde{w}(x)$ ,  $x \in \mathbb{R}^d$ , with  $c > 0$  chosen so that  $\int w \, dx = 1$ . For  $\varepsilon > 0$ , set

$$w_\varepsilon(x) := \varepsilon^{-d} w\left(\frac{x}{\varepsilon}\right), \quad x \in \mathbb{R}^d.$$

**Note.**  $w_\varepsilon \in \mathcal{C}_c^\infty(\mathbb{R}^d)$ ,  $\text{supp}(w_\varepsilon) = B_\varepsilon(0)$  with  $w_\varepsilon > 0$ , and  $\int w_\varepsilon(x) \, dx = 1$  for any  $\varepsilon > 0$ .

**Remark.**  $w$  is a probability density function of a random variable  $\xi \in \mathbb{R}^d$ ,  $|\xi| \leq 1$  almost surely.  $w_\varepsilon$  is a probability density function of  $\varepsilon\xi$ .

$$\mathbb{E}[f(\varepsilon\xi)] = \int f(\varepsilon x) w(x) \, dx, \quad y = \varepsilon x \implies \int f(y) w_\varepsilon(y) \, dy.$$

## Smoothing

Let  $p \in [1, \infty)$ ,  $f \in L^p$ . Then,

$$f_\varepsilon(x) = (f * w_\varepsilon)(x) = \int f(x-y)w_\varepsilon(y) dy = \mathbb{E}[f(x - \varepsilon\xi)], \quad x \in \mathbb{R}^d.$$

1.  $f_\varepsilon \in L^p$ ,  $\|f_\varepsilon\|_p \leq \|f\|_p$ ,  $\|f_\varepsilon - f\|_p \xrightarrow{\varepsilon \rightarrow 0} 0$ .
2.  $f_\varepsilon \in C_b^\infty$ , where  $C_b^\infty$  is the space of all infinitely differentiable bounded functions with all derivatives bounded.
3. Let  $\gamma = (\gamma_1, \dots, \gamma_d)$ ,  $|\gamma| = \gamma_1 + \dots + \gamma_d$ , and  $D^\gamma = \frac{\partial^{|\gamma|}}{\partial x_1^{\gamma_1} \dots \partial x_d^{\gamma_d}}$ . Then,

$$D^\gamma f_\varepsilon(x) = \int D^\gamma w_\varepsilon(x-y)f(y) dy = (D^\gamma w_\varepsilon) * f(x), \quad x \in \mathbb{R}^d.$$

### **Proof.**

1. By Proposition 2,

$$\|f_\varepsilon\|_p = \|f * w_\varepsilon\|_p \leq \|f\|_p \cdot \|w_\varepsilon\|_1 = \|f\|_p.$$

**Convergence.** First, since  $\int w_\varepsilon(y) dy = 1$ ,

$$|(f * w_\varepsilon)(x) - f(x)| = \left| \int [f(x-y) - f(x)]w_\varepsilon(y) dy \right| \leq \int |\tau_{-y}f(x) - f(x)|w_\varepsilon(y) dy.$$

By Minkowski's inequality for integrals,

$$\|f * w_\varepsilon - f\|_p \leq \left\| \int |\tau_{-y}f - f|w_\varepsilon(y) dy \right\|_p \leq \int \|\tau_{-y}f - f\|_p w_\varepsilon(y) dy \leq \int \sup_{|y| \leq \varepsilon} \|\tau_{-y}f - f\|_p w_\varepsilon(y) dy.$$

By Lemma 3,

$$\begin{aligned} \|f * w_\varepsilon - f\|_p &\leq \int \sup_{|y| \leq \varepsilon} \|\tau_{-y}f - f\|_p w_\varepsilon(y) dy = \sup_{|y| \leq \varepsilon} \|\tau_y f - f\|_p \xrightarrow{\varepsilon \rightarrow 0} 0, \\ &\implies \|f_\varepsilon - f\|_p \xrightarrow{\varepsilon \rightarrow 0} 0. \end{aligned}$$

2. Apply Theorem 2.27 and induction.
3. Note that  $D^\alpha w_\varepsilon \in C_c^\infty(\mathbb{R}^d) \subset L^q$ .  $D^\alpha w_\varepsilon$  belongs to  $L^q$  because:

- (a) First, note that  $w_\varepsilon$  is defined as  $\varepsilon^{-d}w(x/\varepsilon)$ , where  $w(x) = C \cdot \ell(|x|/\lambda)$ , and  $\ell$  is the standard bump function that is infinitely differentiable and has compact support.
- (b) Since  $w$  inherits the properties of  $\ell$ , it is also infinitely differentiable with compact support, meaning  $w \in C_c^\infty(\mathbb{R}^d)$ .

- (c) When we take any partial derivative  $D^\alpha$  of  $w_\varepsilon$ , the result is still a smooth function with compact support. This is because differentiation preserves smoothness, and the rescaling by  $\varepsilon$  doesn't change the compact support property - it just scales it.
- (d) Any function in  $C_c^\infty(\mathbb{R}^d)$  automatically belongs to every  $L^q$  space for  $q \in [1, \infty]$  because: (1) It's bounded since it's continuous on a compact set. (2) It has finite support (zero outside a compact set). Therefore, the integral  $\int |D^\alpha w_\varepsilon(x)|^q dx < \infty$  for any  $q \in [1, \infty)$  and  $\sup_x |D^\alpha w_\varepsilon(x)| < \infty$ .

By Proposition 1,

$$\sup_x |D^\alpha f_\varepsilon(x)| = \sup_x |(D^\alpha w_\varepsilon) * f(x)| \leq \|f\|_p \|D^\alpha w_\varepsilon\|_q.$$

## Duals of Continuous Function Spaces

Let  $X$  be a compact metric space (CM), and  $C(X)$  be the space of all continuous real-valued functions on  $X$  with finite norm:

$$\|f\|_u = \sup_{x \in X} |f(x)| = \max_{x \in X} |f(x)|.$$

We always consider the Borel  $\sigma$ -algebra  $\mathcal{B}(X)$  on  $X$ . Note that  $X$  is compact if any sequence  $x_n$  in  $X$  has a converging subsequence.

### Proposition 1

Let  $\mu$  be a finite signed measure on  $X$ . Define:

$$\ell_\mu(f) = \int f d\mu, \quad f \in C(X).$$

Then,  $\ell_\mu \in [C(X)]^*$  and

$$\|\ell_\mu\|_{[C(X)]^*} = \sup_{|f|_{C(X)} \leq 1} |\ell_\mu(f)| = \|\ell_\mu\| \leq \|\mu\| = |\mu|(X) < \infty,$$

where  $|\mu| = \mu^+ + \mu^-$  is the total variation of  $\mu$ .

**Proof.** For  $f \in C(X)$ ,

$$|\ell_\mu(f)| = \left| \int f d\mu \right| \leq \int |f| d|\mu| \leq \|f\|_u |\mu|(X).$$

Hence  $\ell_\mu \in [C(X)]^*$ ,  $\|\ell_\mu\| \leq \|\mu\|$ .

### Tietze-Urysohn Theorem

Let  $(E, \rho)$  be a metric space,  $A \subseteq E$  be closed, and  $f : A \rightarrow \mathbb{R}$  be bounded and continuous. Then,  $f$  has a continuous extension  $g : E \rightarrow \mathbb{R}$  such that:

$$\sup_{x \in E} g(x) = \sup_{x \in A} f(x), \quad \inf_{x \in E} g(x) = \inf_{x \in A} f(x),$$

$$g(x) = \begin{cases} f(x), & x \in A \\ \inf_{y \in A} \frac{f(y)\rho(x,y)}{\rho(x,A)}, & x \notin A \end{cases}.$$

where  $\rho(x, A) = \inf_{y \in A} \rho(x, y)$ . The ratio  $\frac{\rho(x,A)}{\rho(x,y)} \geq 1$ . Here is the reason: For a fixed  $x$ ,

- When  $y \in A$  is close to  $x$ , this ratio approaches 1.
- When  $y \in A$  is far from  $x$ , this ratio is smaller than 1.

$$\inf_{y \in A} f(y) \leq \inf_{y \in A} \frac{f(y)\rho(x,y)}{\rho(x,A)}.$$

### Corollary 0

Let  $A, B \subseteq E$  be closed and disjoint subsets of a metric space  $(E, \rho)$ . Then, there exists a continuous  $g : E \rightarrow [0, 1]$  such that

$$g|_A = 1, \quad g|_B = 0.$$

**Proof.** Applying Tietze-Urysohn Theorem to the closed set  $A \cup B$ ,

$$f(x) = \begin{cases} 1, & x \in A \\ 0, & x \in B \end{cases}.$$

### Lemma 1

Let  $X$  be a compact metric space,  $K \subseteq U \subseteq X$  where  $K$  is compact and  $U$  is open. Then, there exists open  $V \subseteq X$  such that  $K \subseteq V \subseteq \overline{V} \subseteq U$  with  $\overline{V}$  compact.

**Proof.** For all  $x \in K$ , there exists an open ball  $V_x$  and closed ball  $N_x$ , both centered at  $x$ , such that

$$V_x \subseteq N_x \subseteq U, \quad K \subseteq \bigcup_{x \in K} V_x.$$

Since  $K$  is compact, every open cover of  $K$  has a finite subcover. Then, there exists  $x_1, \dots, x_m \in K$  so that

$$K \subseteq \bigcup_{j=1}^m V_{x_j} \subseteq \bigcup_{j=1}^m N_{x_j} \subseteq U.$$

Then, we set

$$V = \bigcup_{j=1}^m V_{x_j}.$$

The  $\overline{V_{x_j}}$  is the smallest closed set containing  $V_{x_j}$ . Since  $N_{x_j}$  is closed and  $V_{x_j} \subseteq N_{x_j}$ , it follows that:

$$\overline{V_{x_j}} \subseteq N_{x_j}.$$

In general, the closure of a union satisfies  $\overline{A \cup B} = \overline{A} \cup \overline{B}$ . Therefore, for  $V = \bigcup_{j=1}^m V_{x_j}$ ,

$$\overline{V} = \overline{\bigcup_{j=1}^m V_{x_j}} = \bigcup_{j=1}^m \overline{V_{x_j}},$$

Since each  $\overline{V_{x_j}} \subseteq N_{x_j}$ ,

$$\overline{V} \subseteq \bigcup_{j=1}^m N_{x_j} \implies V \subseteq \overline{V} \subseteq \bigcup_{j=1}^m N_{x_j} \subseteq U.$$

$\bigcup_{j=1}^m N_{x_j}$  is compact because each closed ball  $N_{x_j}$  is bounded and closed. The finite union of compact sets is compact. Since  $\overline{V}$  is a closed subset of the compact set  $\bigcup_{j=1}^m N_{x_j}$ , it must also be compact.

**Notation.** Let  $U \subseteq X$  be open,  $f \in C(X)$ .

$$f \prec U \text{ if } 0 \leq f \leq 1 \text{ and } \text{supp}(f) \subseteq U.$$

### Proposition 1

Let  $X$  be a compact metric space,  $K \subseteq U \subseteq X$  where  $K$  is compact and  $U$  is open. Then, there exists  $f \prec U$  such that  $f|_K = 1$ .

**Proof.** By Lemma 1, there exists open  $V$  such that

$$K \subseteq V \subseteq \overline{V} \subseteq U.$$

By Corollary 0, there exists  $f \in C(X)$  such that

$$f|_K = 1, \quad f|_{V^c} = 0.$$

Note  $\text{supp}(f) \subseteq \overline{V} \subseteq U$ . By Corollary 0,  $0 \leq f \leq 1$  ( $g : E \rightarrow [0, 1]$ ).

### Corollary 1

Let  $X$  be a compact metric space,  $K \subseteq \bigcup_{j=1}^n U_j$  where  $K$  is compact and  $U_j$  are open for all  $j = 1, \dots, n$ . Then, there exists  $f_j \prec U_j$ ,  $j = 1, \dots, n$ , such that

$$\left( \sum_{j=1}^n f_j \right) \Big|_K = 1.$$

**Proof.** For any  $x \in K$ , there exists an open ball  $V_x$  and closed ball  $N_x$ , both centered at  $x$  such that:

$$V_x \subseteq N_x \subseteq U_j \text{ for some } j \in \{1, \dots, n\}$$

Hence,  $K \subseteq \bigcup_{x \in K} V_x$  and there exists  $x_1, \dots, x_m$  so that

$$K \subseteq \bigcup_{k=1}^m V_{x_k} \subseteq \bigcup_{k=1}^m N_{x_k}$$

Let  $K_j$  be the union of all  $N_{x_k}$  s.t.  $N_{x_k} \subseteq U_j$ . Then  $K \subseteq \bigcup_{j=1}^n K_j$  and all  $K_j$  are compact with  $K_j \subseteq U_j$ .

By Proposition 1, there exists  $g_j \prec U_j, g_j|_{K_j} = 1$ . Let  $U = \bigcup_{j=1}^n \{x : g_j(x) > 0\}$ . Then,  $\bigcup_{j=1}^n K_j \subseteq U$  and open since  $g_j \in C(X)$ . Specifically, the set  $\{x : g_j(x) > 0\}$  is open as the preimage of the open set  $(0, \infty)$  under the continuous function  $g_j$ .

By Proposition 1, there exists  $f \prec U$  such that  $f|_{\bigcup_{j=1}^n K_j} = 1$ . Set:

$$g_{n+1} = 1 - f = \begin{cases} 0 & \text{on } \bigcup_{j=1}^n K_j \\ 1 & \text{on } U^c \end{cases}$$

On  $K$ , since  $K \subseteq \bigcup_{j=1}^n K_j$ , for any  $x \in K$ , at least one  $g_j(x) = 1$ , so  $f(x) \geq 1$ . On  $U^c$ , all  $g_j = 0$ , so  $f = 0$ . Therefore, on  $\bigcup_{j=1}^n K_j$ ,  $f = 1$ , so  $g_{n+1} = 0$ .

$$f_j = \frac{g_j}{\sum_{i=1}^{n+1} g_i}, \quad j = 1, \dots, n$$

Since  $g_{n+1} \geq 0$ , we must have  $g_{n+1} = 0$  on  $\bigcup_{j=1}^n K_j$ . On  $U^c$ ,  $f = 0$ , so  $g_{n+1} = 1$ .

Since  $g_{n+1} = 0$  on  $K$ ,

$$\begin{aligned} \sum_{i=1}^{n+1} g_i = \sum_{j=1}^n g_j \text{ on } K &\implies f_j = \frac{g_j}{\sum_{k=1}^n g_k} \text{ on } K \implies \sum_{j=1}^n f_j = \frac{\sum_{j=1}^n g_j}{\sum_{k=1}^n g_k} = 1 \text{ on } K. \\ &\implies \left( \sum_{j=1}^n f_j \right) \Big|_K = 1. \end{aligned}$$

Since  $g_j \prec U_j$ ,  $\text{supp}(g_j) \subseteq U_j$ . The denominator  $\sum_{i=1}^{n+1} g_i$  is positive everywhere, so dividing by it does not change the support. Thus,  $\text{supp}(f_j) = \text{supp}(g_j) \subseteq U_j$ , meaning  $f_j \prec U_j$ .

## Representations of Positive Linear Functionals

A linear functional  $\ell : C(X) \rightarrow \mathbb{R}$  is **positive** if  $\ell(f) \geq 0$  if  $f \geq 0$ .

**Remark.** If  $f \geq g$ , then  $\ell(f) \geq \ell(g)$  for positive  $\ell$ . In fact,  $\ell$  is positive iff this holds.

### Proposition 2

If  $\ell$  is positive, then  $\ell \in [C(X)]^*$ .

**Proof.** Let  $f \in C(X)$ . Then,

$$-|f|_u \leq f \leq |f|_u \implies -|f|_u \ell(1) \leq \ell(f) \leq |f|_u \ell(1) \implies |\ell(f)| \leq \ell(1) |f|_u.$$

### Definition

Let  $\mu$  be a Borel measure on a metric space  $X$ . Let  $A \in \mathcal{B}$  (Borel  $\sigma$ -algebra). We say  $\mu$  is:

(i) **Outer regular** on  $A$  if

$$\mu(A) = \inf\{\mu(U) : U \supseteq A, U \text{ is open}\}$$

(ii) **Inner regular** on  $A$  if

$$\mu(A) = \sup\{\mu(K) : K \subseteq A, K \text{ is compact}\}$$

(iii) **Regular** if  $\mu$  is both inner and outer regular for all  $E \in \mathcal{B}$ .

**Theorem 1**

Let  $\ell : C(X) \rightarrow \mathbb{R}$  be a positive linear functional. Then, there exists a unique finite Borel regular measure  $\mu$  on  $X$  such that:

$$\ell(f) = \int f d\mu, \quad f \in C(X).$$

(i) For open  $U \subseteq X$ ,

$$\mu(U) = \sup\{\ell(f) : f \prec U\} \leq \ell(1) = \mu(X)$$

$$\ell(1) = \int_X 1 d\mu = \mu(X).$$

(ii) For compact  $K \subseteq X$ ,

$$\mu(K) = \inf\{\ell(f) : f \geq \chi_K, f \in C(X)\}.$$

*Note.* For open  $U \subseteq X$ ,

$$\chi_U(x) = \sup\{f(x) : f \prec U\}, \quad x \in X.$$

**Comment.** (i) implies (ii): if  $K$  is compact,

$$\begin{aligned} \mu(K) &= \mu(X) - \mu(K^c) \\ &= \ell(1) - \sup\{\ell(f) : f \prec K^c\} \\ &= \ell(1) - \sup\{\ell(f) : 0 \leq f \leq 1, \text{supp}(f) \subseteq K^c\} \\ &= \inf\{\ell(1 - f) : 0 \leq f \leq 1, \text{supp}(f) \subseteq K^c\} \\ &\geq \inf\{\ell(g) : g \geq \chi_K, g \in C(X)\}. \end{aligned}$$

For  $f$  with  $\text{supp}(f) \subseteq K^c$ ,  $f$  is zero on  $K$  since  $K$  and  $\text{supp}(f)$  are disjoint. Thus,  $1 - f \geq 1$  on  $K$  because  $f = 0$  on  $K$ , and  $1 - f \geq 0$  everywhere since  $f \leq 1$ . Therefore,  $1 - f \geq \chi_K$ , where  $\chi_K$  is the indicator function of  $K$ .

**Proof.** *Uniqueness.* Let  $\mu$  be a finite regular measure and  $\ell(f) = \int f d\mu$ . Let  $U \subseteq X$  be open. By the note above,

$$\mu(U) = \int_U 1 d\mu = \sup\{\ell(f) : f \prec U\}.$$

Let  $K \subseteq U$  be compact. By Proposition 1, there exists  $f \prec U$  such that  $f|_K = 1$ . Then,

$$\begin{aligned}\mu(K) &\leq \int_X f d\mu = \ell(f) \\ \implies \mu(U) &= \sup\{\mu(K') : K' \subseteq U, K' \text{ compact}\} \leq \sup\{\ell(f) : f \prec U\} \\ \implies \mu(U) &= \sup\{\ell(f) : f \prec U\} \text{ which is (i)}\end{aligned}$$

Why does it imply uniqueness? Suppose there were two measures  $\mu_1$  and  $\mu_2$  representing  $\ell$ :

$$\ell(f) = \int f d\mu_1 = \int f d\mu_2 \quad \forall f \in C(X).$$

For any open set  $U$ , both  $\mu_1(U)$  and  $\mu_2(U)$  must satisfy:

$$\mu_i(U) = \sup\{\ell(f) : f \prec U\}, \quad i = 1, 2.$$

Since the right-hand side depends only on  $\ell$ , we must have:

$$\mu_1(U) = \mu_2(U).$$

Thus,  $\mu_1$  and  $\mu_2$  agree on all open sets. The measure  $\mu$  is outer regular, meaning for any Borel set  $E \subseteq X$ ,

$$\mu(E) = \inf\{\mu(U) : U \supseteq E, U \text{ open}\}.$$

Since  $\mu_1$  and  $\mu_2$  agree on open sets, they must assign the same measure to any Borel set  $E$ :

$$\mu_1(E) = \inf\{\mu_1(U) : U \supseteq E\} = \inf\{\mu_2(U) : U \supseteq E\} = \mu_2(E).$$

Hence,  $\mu_1 = \mu_2$  on the entire Borel  $\sigma$ -algebra. In short, open sets generate the topology of  $X$ , and their measures are uniquely specified by  $\ell$ . Since  $\mu$  is outer regular, its values on arbitrary sets are determined by its values on open sets.

*Existence.*

$$\begin{aligned}\mu(U) &:= \sup\{\ell(f) : f \prec U\} \text{ for } U \subseteq X \text{ open.} \\ \mu^*(E) &:= \inf\{\mu(U) : U \supseteq E, U \text{ open}\} \text{ for } E \subseteq X.\end{aligned}$$

**Claim 1.**  $\mu^*$  is an outer measure.

An outer measure must satisfy:

1.  $\mu^*(\emptyset) = 0$ .
2.  $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$ .
3.  $\mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} \mu^*(E_n)$ .



$\mu^*(\emptyset) = 0$  because  $\emptyset$  is covered by itself, and  $\mu(\emptyset) = 0$ . If  $A \subseteq B$ , any open cover of  $B$  also covers  $A$ , so the infimum for  $A$  is over a larger set of covers, so  $\mu^*(A) \leq \mu^*(B)$ . Thus, it is enough to prove that

$$\mu^* \left( \bigcup_{n=1}^{\infty} U_n \right) \leq \sum_{n=1}^{\infty} \mu^*(U_n) \text{ for } U_n \text{ open.}$$

The key observation is that open sets can be used to approximate arbitrary sets when defining the outer measure. There exists  $f \prec \bigcup_{n=1}^{\infty} U_n = U$  such that  $\ell(f) > \mu \left( \bigcup_{n=1}^{\infty} U_n \right) - \epsilon$  for a given  $\epsilon > 0$  since  $\mu(U) = \sup\{\ell(f) : f \prec U\}$ . Since every open cover of a compact set has a finite subcover, there exists  $N$  such that  $\text{supp}(f) \subseteq \bigcup_{n=1}^N U_n$ . By Corollary 1, there exists  $f_n \prec U_n$  such that

$$\left( \sum_{n=1}^N f_n \right) \Big|_{\text{supp}(f)} = 1, \quad f = \sum_{n=1}^N (f_n f),$$

$$\mu \left( \bigcup_{n=1}^{\infty} U_n \right) \leq \ell(f) + \epsilon = \left[ \sum_{n=1}^N \ell(f f_n) \right] + \epsilon \leq \left[ \sum_{n=1}^N \mu(U_n) \right] + \epsilon.$$

Hence  $\mu^*$  is an outer measure.

**Claim 2.**  $\mu$  has a finite regular Borel extension.

By Caratheodory's theorem,  $\mu^*$  is a measure on the  $\sigma$ -algebra  $\mathcal{F}$  of all  $A \subseteq X$  such that

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \setminus A) \text{ for all } E \subseteq X \quad (2)$$

We now show  $\mathcal{F}$  contains all the open sets of  $X$ , i.e.,  $U \in \mathcal{F}$  if  $U \subseteq X$  is open. If this is the case,  $\mathcal{F}$  contains  $\mathcal{B}$  and  $\mu$  can be extended to a measure on  $\mathcal{B}$ .

Let  $U \subseteq X$  be open. Equation 2 holds if  $E$  is open. In this case,  $E \cap U$  is open so there exists  $f \prec E \cap U$  such that  $\ell(f) > \mu(E \cap U) - \frac{\epsilon}{2}$  where  $\epsilon > 0$ . There exists  $g \prec E \setminus \text{supp}(f)$  such that

$$\ell(g) > \mu(E \setminus \text{supp}(f)) - \frac{\epsilon}{2}.$$

Then,  $f + g \prec E$  since  $\text{supp}(f), \text{supp}(g) \subseteq E$ . Then,

$$\begin{aligned} \mu(E) &\geq \ell(f + g) = \ell(f) + \ell(g) \\ &> \mu(E \cap U) - \frac{\epsilon}{2} + \mu(E \setminus \text{supp}(f)) - \frac{\epsilon}{2} \\ &\geq \mu(E \cap U) + \mu^*(E \setminus U) - \epsilon. \end{aligned}$$

Now, we prove Equation 2 holds for arbitrary  $E \subseteq X$ . For all  $\epsilon > 0$ , there exists open  $\tilde{E} \supset E$  such that

$$\begin{aligned} \mu^*(E) &\geq \mu(\tilde{E}) - \epsilon \\ \implies \mu^*(E) &\geq \mu(\tilde{E}) - \epsilon \geq \mu^*(\tilde{E} \cap U) + \mu^*(\tilde{E} \setminus U) - \epsilon \geq \mu^*(E \cap U) + \mu^*(E \setminus U) - \epsilon. \end{aligned}$$

Since  $\mu(U) \leq \ell(1) < \infty$  for all open  $U \subseteq X$ ,  $\mu$  is finite.

We define for any subset  $E \subseteq X$  by:

$$\mu^*(E) = \inf\{\mu(U) : U \supseteq E, U \text{ open}\}.$$

This ensures  $\mu^*$  is outer regular. Why Outer Regularity on the Borel  $\sigma$ -Algebra Implies Inner Regularity (in a Compact Space)? Let  $A \in \mathcal{B}$  be a Borel set. Since  $\mu$  is outer regular, for the complement  $A^c$ , there exists an open set  $U \supseteq A^c$  such that:

$$\mu(U) \leq \mu(A^c) + \epsilon \text{ for any } \epsilon > 0.$$

Since  $U \supseteq A^c$ , its complement  $K = U^c$  satisfies  $K \subseteq A$ . In a compact space, closed subsets are compact, so  $K$  is compact. From  $\mu(X) = \mu(A) + \mu(A^c)$  and  $\mu(U) \leq \mu(A^c) + \epsilon$ ,

$$\mu(K) = \mu(U^c) = \mu(X) - \mu(U) \geq \mu(X) - (\mu(A^c) + \epsilon) = \mu(A) - \epsilon.$$

Since  $K \subseteq A$ , this shows:

$$\mu(A) \leq \mu(K) + \epsilon \leq \sup\{\mu(K') : K' \subseteq A, K' \text{ compact}\} + \epsilon.$$

Taking  $\epsilon \rightarrow 0$  gives:

$$\mu(A) \leq \sup\{\mu(K) : K \subseteq A, K \text{ compact}\}.$$

The reverse inequality ( $\geq$ ) is trivial since  $\mu(K) \leq \mu(A)$  for  $K \subseteq A$ . Thus,

$$\mu(A) = \sup\{\mu(K) : K \subseteq A, K \text{ compact}\}.$$

We conclude  $\mu$  is inner regular. Since  $\mu$  is both outer and inner regular,  $\mu$  is regular.

**Claim 3.**  $\int f d\mu = \ell(f), f \in C(X)$ .

Let  $0 \leq f \leq 1, f \in C(X)$ . Let  $N > 1$ . Then,

$$\begin{aligned} f &= \sum_{n=1}^N f \chi_{\frac{n-1}{N} \leq f < \frac{n}{N}} \\ &= \sum_{n=1}^N \left[ \left( f - \frac{n-1}{N} \right) \chi_{\frac{n-1}{N} \leq f < \frac{n}{N}} + \frac{n-1}{N} \chi_{\frac{n-1}{N} \leq f < \frac{n}{N}} \right] \\ \implies f &= \sum_{n=1}^N f \chi_{\frac{n-1}{N} \leq f < \frac{n}{N}} = \sum_{n=1}^N \left[ \left( f - \frac{n-1}{N} \right) \chi_{\frac{n-1}{N} \leq f < \frac{n}{N}} + \frac{1}{N} \chi_{\frac{n}{N} \leq f} \right] = \sum_{n=1}^N f_n \end{aligned}$$

where  $f_n \in C(X)$ .

$$\begin{aligned} &\sum_{n=1}^N \frac{n-1}{N} (\chi_{\frac{n-1}{N} \leq f} - \chi_{\frac{n}{N} \leq f}) \\ &= \frac{1}{N} \sum_{n=1}^N (n-1) (\chi_{\frac{n-1}{N} \leq f} - \chi_{\frac{n}{N} \leq f}) \\ &= \frac{1}{N} [1 \cdot (\chi_{\frac{1}{N} \leq f} - \chi_{\frac{2}{N} \leq f}) + 2 \cdot (\chi_{\frac{2}{N} \leq f} - \chi_{\frac{3}{N} \leq f}) + \dots + (N-1) \cdot (\chi_{\frac{N-1}{N} \leq f} - \chi_{\frac{N}{N} \leq f})] \\ &= \frac{1}{N} [\chi_{\frac{1}{N} \leq f} - \chi_{\frac{2}{N} \leq f} + 2\chi_{\frac{2}{N} \leq f} - 2\chi_{\frac{3}{N} \leq f} + \dots - (N-1)\chi_{\frac{N}{N} \leq f}] \\ &= \frac{1}{N} [\chi_{\frac{1}{N} \leq f} + \chi_{\frac{2}{N} \leq f} + \chi_{\frac{3}{N} \leq f} + \dots + \chi_{\frac{N-1}{N} \leq f} - (N-1) \cdot \chi_{\frac{N}{N} \leq f}]. \end{aligned}$$

Since  $(N-1) \cdot \chi_{\frac{N}{N} \leq f} = 0$ ,

$$\sum_{n=1}^N \frac{n-1}{N} \chi_{\frac{n-1}{N} \leq f < \frac{n}{N}} = \sum_{n=1}^N \frac{n-1}{N} (\chi_{\frac{n-1}{N} \leq f} - \chi_{\frac{n}{N} \leq f}) = \frac{1}{N} \sum_{n=1}^N \chi_{\frac{n}{N} \leq f}.$$

For each  $n$ ,

$$f_n = \left(f - \frac{n-1}{N}\right) \chi_{\frac{n-1}{N} \leq f \leq \frac{n}{N}} + \frac{1}{N} \chi_{\frac{n}{N} \leq f} \implies \frac{1}{N} \chi_{\frac{n}{N} \leq f} \leq f_n \leq \frac{1}{N} \chi_{\frac{n-1}{N} \leq f}$$

By (ii),

$$\begin{aligned} \mu\left(\frac{n}{N} \leq f\right) &\leq N \ell(f_n) \leq \mu\left(\frac{n-1}{N} \leq f\right) \\ \implies \frac{1}{N} \sum_{n=1}^N \mu\left(\frac{n}{N} \leq f\right) &\leq \sum_{n=1}^N \ell(f_n) \leq \frac{1}{N} \sum_{n=1}^N \mu\left(\frac{n-1}{N} \leq f\right) \end{aligned}$$

Focus on the right-hand side (RHS):

$$\ell(f) \leq \sum_{n=1}^N \frac{1}{N} \mu\left(\frac{n-1}{N} \leq f\right).$$

We can express  $\mu\left(\frac{n-1}{N} \leq f\right)$  as:

$$\mu\left(\frac{n-1}{N} \leq f\right) = \sum_{k=n}^N \mu\left(\frac{k-1}{N} \leq f < \frac{k}{N}\right) + \mu(f \geq 1).$$

Since  $f \leq 1$ ,

$$\mu\left(\frac{n-1}{N} \leq f\right) = \sum_{k=n}^N \mu\left(\frac{k-1}{N} \leq f < \frac{k}{N}\right).$$

Substitute back into the RHS:

$$\ell(f) \leq \sum_{n=1}^N \frac{1}{N} \sum_{k=n}^N \mu\left(\frac{k-1}{N} \leq f < \frac{k}{N}\right).$$

Interchange the order of summation:

$$\begin{aligned} \sum_{n=1}^N \sum_{k=n}^N \frac{1}{N} \mu\left(\frac{k-1}{N} \leq f < \frac{k}{N}\right) &= \sum_{k=1}^N \sum_{n=1}^k \frac{1}{N} \mu\left(\frac{k-1}{N} \leq f < \frac{k}{N}\right), \\ \implies \sum_{n=1}^k \frac{1}{N} \mu\left(\frac{k-1}{N} \leq f < \frac{k}{N}\right) &= \frac{k}{N} \mu\left(\frac{k-1}{N} \leq f < \frac{k}{N}\right), \\ \implies \ell(f) &\leq \sum_{k=1}^N \frac{k}{N} \mu\left(\frac{k-1}{N} \leq f < \frac{k}{N}\right), \end{aligned}$$

Write  $\frac{k}{N} = \frac{k-1}{N} + \frac{1}{N}$ :

$$\begin{aligned}\ell(f) &\leq \sum_{k=1}^N \left( \frac{k-1}{N} + \frac{1}{N} \right) \mu \left( \frac{k-1}{N} \leq f < \frac{k}{N} \right), \\ \Rightarrow \ell(f) &\leq \sum_{k=1}^N \frac{k-1}{N} \mu \left( \frac{k-1}{N} \leq f < \frac{k}{N} \right) + \frac{1}{N} \sum_{k=1}^N \mu \left( \frac{k-1}{N} \leq f < \frac{k}{N} \right),\end{aligned}$$

The second sum is:

$$\frac{1}{N} \sum_{k=1}^N \mu \left( \frac{k-1}{N} \leq f < \frac{k}{N} \right) = \frac{1}{N} \mu(f > 0).$$

because the sets  $\frac{k-1}{N} \leq f < \frac{k}{N}$  partition  $\{f > 0\}$ . Then,

$$\sum_{n=1}^N \frac{n-1}{N} \mu \left( \frac{n-1}{N} \leq f < \frac{n}{N} \right) \leq \ell(f) \leq \sum_{n=1}^N \frac{n-1}{N} \mu \left( \frac{n-1}{N} \leq f < \frac{n}{N} \right) + \frac{1}{N} \mu(f > 0)$$

The left-hand side is the lower Riemann sum and the right-hand side is the upper Riemann sum. As we integrate (Riemann sum) and let  $N \rightarrow \infty$ ,

$$\int f d\mu = \ell(f) = \int f d\mu$$

Therefore,  $\int f d\mu = \ell(f)$  for all  $f \in C(X)$ .

## Corollary 2

Let  $X$  be a compact metric space. Then, any finite Borel measure  $\gamma$  is regular.

**Proof.** Let  $\ell(f) = \int f d\gamma$ ,  $f \in C(X)$ . Then,  $\ell$  is positive. By Theorem 1, there exists a regular finite  $\mu$  so that:

$$\ell(f) = \int f d\mu = \int f d\gamma \text{ for any } f \in C(X).$$

Let  $K \subset X$  be compact and define:

$$F_n = \{x \in X : \rho(x, K) \geq \frac{1}{n}\} \text{ for all } n \geq 1.$$

We show that the complement  $F_n^c = \{x \in X : \rho(x, K) < 1/n\}$  is open: Take any  $x \in F_n^c$ , meaning  $\rho(x, K) = d < 1/n$ . By definition of infimum, there exists  $y \in K$  such that  $\rho(x, y) < (1/n + d)/2$ . Now, consider the open ball  $B(x, r)$  where  $r = (1/n - d)/2 > 0$ . For any  $z \in B(x, r)$ , by the triangle inequality:

$$\rho(z, K) \leq \rho(z, y) \leq \rho(z, x) + \rho(x, y) < r + \frac{1/n + d}{2} = \frac{1/n - d}{2} + \frac{1/n + d}{2} = 1/n.$$

Thus,  $\rho(z, K) < 1/n$ , so  $z \in F_n^c$ . Since every  $x \in F_n^c$  has an open neighborhood entirely contained in  $F_n^c$ ,  $F_n^c$  is open, and hence  $F_n$  is closed.

By Corollary 0, there exists continuous  $f_n : X \rightarrow [0, 1]$  such that  $f_n|_K = 1$ ,  $f_n|_{F_n} = 0$ . Note that  $f_n(x) \rightarrow \chi_K(x)$  for all  $x \in X$  and:

$$\int f_n d\mu = \int f_n d\gamma \implies \mu(K) = \gamma(K)$$

Therefore,  $\gamma$  is regular, and in fact  $\gamma = \mu$ , since  $\mu$  is regular. For open  $U$ ,

$$\gamma(U) = \gamma(X) - \gamma(U^c) = \mu(X) - \mu(U^c) = \mu(U)$$

Let  $A \in \mathcal{B}$ ,  $\epsilon > 0$ . Since  $\mu$  is regular, there exists  $U_\epsilon \supseteq A \supseteq K_\epsilon$  so that:

$$\mu(U_\epsilon \setminus K_\epsilon) < \epsilon \implies \gamma(U_\epsilon) \leq \gamma(K_\epsilon) + \epsilon \leq \gamma(A) + \epsilon.$$

Hence,  $\gamma$  is outer regular on  $A$ . Similarly,

$$\mu(K_\epsilon) \geq \mu(A) - \epsilon \implies \gamma(K_\epsilon) \geq \gamma(A) - \epsilon.$$

Thus,  $\gamma$  is inner regular. Then,  $\gamma$  is regular and  $\gamma = \mu$  since  $\gamma$  and  $\mu$  are equal on compact and open sets.

## Lemma 2

Let  $\ell \in [C(X)]^*$ . Then,  $\ell = \ell_1 - \ell_2$  with  $\ell_1, \ell_2$  positive.

**Proof.** Define  $P = \{f \in C(X) : f \geq 0\}$ . For any  $f \in P$ ,

$$\ell_1(f) := \sup\{\ell(g) : 0 \leq g \leq f, g \in C(X)\}.$$

Note that  $\ell_1(f) \geq 0$  since we can take  $g = 0$  in the supremum. We first show that  $\ell_1$  is linear on  $P$ . For any  $f, f' \in P$  and  $c \geq 0$ , we need to verify:

1. *Additivity.*  $\ell_1(f + f') = \ell_1(f) + \ell_1(f')$ .

For any  $0 \leq g \leq f$  and  $0 \leq g' \leq f'$ , we have  $0 \leq g + g' \leq f + f'$ , and thus:

$$\ell(g) + \ell(g') = \ell(g + g') \leq \ell_1(f + f').$$

Taking suprema over  $g$  and  $g'$  gives  $\ell_1(f) + \ell_1(f') \leq \ell_1(f + f')$ . For the reverse inequality, given any  $0 \leq h \leq f + f'$ , we can write  $h = u + u'$  where  $0 \leq u \leq f$  and  $0 \leq u' \leq f'$  by taking  $u = \min(h, f)$  and  $u' = h - u$ . Then,

$$\ell(h) = \ell(u) + \ell(u') \leq \ell_1(f) + \ell_1(f').$$

Taking supremum over  $h$  gives  $\ell_1(f + f') \leq \ell_1(f) + \ell_1(f')$ .

2. *Homogeneity.*  $\ell_1(cf) = c\ell_1(f)$ .

$$\ell_1(cf) = \sup\{\ell(g) : 0 \leq g \leq cf\} = \sup\{\ell(ch) : 0 \leq h \leq f\} = c \sup\{\ell(h) : 0 \leq h \leq f\}.$$

$$\implies \ell_1(cf) = c\ell_1(f)$$

We now extend  $\ell_1$  to all of  $C(X)$ . Any  $f \in C(X)$  can be written as  $f = f_1 - f_2$  where  $f_1, f_2 \in P$ , using positive and negative parts. Define:

$$\ell_1(f) := \ell_1(f_1) - \ell_1(f_2).$$

If  $f = f_1 - f_2 = f'_1 - f'_2$ , then  $f_1 + f'_2 = f'_1 + f_2$ , so by additivity on  $P$ :

$$\begin{aligned} \ell_1(f_1) + \ell_1(f'_2) &= \ell_1(f'_1) + \ell_1(f_2) \\ \implies \ell_1(f_1) - \ell_1(f_2) &= \ell_1(f'_1) - \ell_1(f'_2) \end{aligned}$$

Thus,  $\ell_1(f) = \ell_1(f_1) - \ell_1(f_2)$  is well-defined. Finally, define  $\ell_2 := \ell_1 - \ell$ . To see that  $\ell_2$  is positive, note that for any  $f \geq 0$ :

$$\ell_2(f) = \ell_1(f) - \ell(f) = \sup\{\ell(g) : 0 \leq g \leq f\} - \ell(f) \geq \ell(f) - \ell(f) = 0.$$

Thus,  $\ell_1$  and  $\ell_2$  are both positive linear functionals with  $\ell = \ell_1 - \ell_2$ .

#### Proposition 4

Let  $X$  be a compact metric space,  $\mu$  be a finite regular Borel measure,  $p \in [1, \infty)$ . Then,  $C(X)$  is dense in  $L^p$ .

**Proof.**  $\Sigma = \mathcal{E}$  as  $\mu(X) < \infty$ . Since  $\Sigma = \mathcal{E}$  is dense in  $L^p$ , it is enough to approximate  $\chi_E$  in  $L^p$  by continuous functions. Let  $E \in \mathcal{B}$ ,  $\epsilon > 0$ . Since  $\mu$  is regular, for any Borel set  $E$  and  $\epsilon > 0$ , there exist a compact set  $K_\epsilon \subseteq E$  and an open set  $U_\epsilon \supseteq E$ , such that:

$$\mu(U_\epsilon) < \mu(E) + \frac{\epsilon}{2}, \quad \mu(K_\epsilon) > \mu(E) - \frac{\epsilon}{2}.$$

Subtracting these inequalities gives:

$$\mu(U_\epsilon \setminus K_\epsilon) = \mu(U_\epsilon) - \mu(K_\epsilon) < \epsilon.$$

Since  $K_\epsilon \subseteq E \subseteq U_\epsilon$ , by Corollary 0, we can construct a continuous function  $f$  such that:  $f = 1$  on  $K_\epsilon$ ,  $f = 0$  outside  $U_\epsilon$ , and  $0 \leq f \leq 1$  everywhere. The difference  $|\chi_E - f|$  is nonzero only on  $U_\epsilon \setminus K_\epsilon$ , because:

- On  $K_\epsilon$ ,  $\chi_E = 1$  and  $f = 1$ , so  $|\chi_E - f| = 0$ .
- Outside  $U_\epsilon$ ,  $\chi_E = 0$  and  $f = 0$ , so  $|\chi_E - f| = 0$ .
- On  $U_\epsilon \setminus K_\epsilon$ ,  $|\chi_E - f| \leq 1$  since  $0 \leq f \leq 1$  and  $\chi_E \in \{0, 1\}$ .

Therefore:

$$\int_X |\chi_E - f|^p d\mu \leq \int_{U_\epsilon \setminus K_\epsilon} 1 d\mu = \mu(U_\epsilon \setminus K_\epsilon) < \epsilon.$$

Since simple functions are dense in  $L^p$ , and we can approximate any simple function by continuous functions, we can approximate any  $L^p$ -function by continuous functions. Thus,  $C(X)$  is dense in  $L^p(X, \mu)$ .

### Lusin's theorem

Let  $X$  be a compact metric space and  $\mu$  be finite. Let  $f$  be a bounded measurable function on  $X$  ( $|f|_\infty \leq C$ ). Then, for all  $\epsilon > 0$ , there exists  $g \in C(X)$  such that  $\mu(\{f \neq g\}) \leq \epsilon$ . Moreover,  $|g|_\infty \leq \sup_x |f(x)|$ , and  $0 \leq g \leq \sup_x |f(x)|$  if  $f \geq 0$ .

**Proof.** By Proposition 4, since  $C(X)$  is dense in  $L^1(X, \mu)$ , there exists a sequence  $\{f_n\} \subset C(X)$  such that  $f_n \rightarrow f$  in  $L^1$ , and hence  $f_n \rightarrow f$  almost everywhere. Since  $f_n \rightarrow f$  almost everywhere and  $\mu(X) < \infty$ , Egoroff's Theorem guarantees that for any  $\epsilon > 0$ , there exists a measurable set  $A \subset X$  such that:

$$\mu(X \setminus A) \leq \frac{\epsilon}{2},$$

and  $f_n \rightarrow f$  uniformly on  $A$ . Since  $\mu$  is regular from Corollary 2, for the set  $A$ , there exists a compact subset  $K \subseteq A$  such that:

$$\mu(A \setminus K) \leq \frac{\epsilon}{2}.$$

$$\implies \mu(X \setminus K) = \mu(X \setminus A) + \mu(A \setminus K) \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Since  $f_n \rightarrow f$  uniformly on  $A$ , and  $K \subseteq A$ , the restriction  $f|_K$  is continuous as a uniform limit of continuous functions. By the Tietze-Urysohn Theorem, since  $K$  is closed in  $X$  and  $X$  is a compact metric space, there exists a continuous function  $g \in C(X)$  such that:

$$g|_K = f|_K, \quad \sup_X g = \sup_K f, \quad \inf_X g = \inf_K f.$$

By construction,  $f = g$  on  $K$  and  $\mu(X \setminus K) \leq \epsilon$ . Therefore:

$$\{x \in X : f(x) \neq g(x)\} \subseteq X \setminus K, \implies \mu(\{f \neq g\}) \leq \mu(X \setminus K) \leq \epsilon.$$

Since  $g$  is an extension of  $f|_K$ , and  $f$  is bounded by  $C$ ,

$$|g(x)| \leq \sup_K |f| \leq \sup_X |f| = C.$$

If  $f \geq 0$ , then  $g \geq 0$  since  $\inf_X g \geq \inf_K f \geq 0$ .

### Proposition 5

Let  $X$  be a compact metric space,  $\mu$  be a finite signed measure. Let  $\ell_\mu(f) = \int f d\mu$ ,  $f \in C(X)$ . Then,  $\ell_\mu \in [C(X)]^*$  and  $\|\ell_\mu\| = |\ell_\mu|_{[C(X)]^*} = |\mu|(X) = \|\mu\|$ .

**Proof.** By Proposition 1, since  $\mu$  is a finite signed measure,  $\ell_\mu \in [C(X)]^*$  and  $\|\ell_\mu\| \leq \|\mu\|$ . Let  $D^+, D^-$  be the disjoint sets of Hahn's decomposition such that  $d|\mu| = \chi_{D^+} d\mu - \chi_{D^-} d\mu$  and  $d\mu = (\chi_{D^+} - \chi_{D^-}) d|\mu|$ . Note that  $|\rho| = \rho^2 = 1$  where  $\rho = \chi_{D^+} - \chi_{D^-}$ . Let  $\epsilon > 0$ . By

Lusin's theorem, there exists  $g \in C(X)$  such that  $|\mu|(\{\rho \neq g\}) \leq \epsilon$  and  $|g|_\infty \leq 1$ . Then:

$$\begin{aligned}
\ell_\mu(g) &= \int g d\mu \\
&= \int g\rho d|\mu| \\
&= \int_{g=\rho} g\rho d|\mu| + \int_{g \neq \rho} g\rho d|\mu| \\
&= \int_X \rho^2 d|\mu| - \int_{g \neq \rho} \rho^2 d|\mu| + \int_{g \neq \rho} g\rho d|\mu| \\
&\geq |\mu|(X) - \epsilon - \epsilon \\
&= |\mu|(X) - 2\epsilon.
\end{aligned}$$

Since:

$$0 \leq \int_{g \neq \rho} \rho^2 d|\mu| = |\mu|(\{g \neq \rho\}) \leq \epsilon,$$

$$\left| \int_{g \neq \rho} g\rho d|\mu| \right| \leq \int_{g \neq \rho} |g\rho| d|\mu| \leq \int_{g \neq \rho} d|\mu| = |\mu|(\{g \neq \rho\}) \leq \epsilon \implies \int_{g \neq \rho} g\rho d|\mu| \geq -\epsilon.$$

Therefore,

$$\|\mu\| \geq |\ell_\mu(g)| \geq \ell_\mu(g) \geq |\mu|(X) - 2\epsilon$$

Taking  $\epsilon \rightarrow 0$ , we have  $\|\ell_\mu\| = \|\mu\|$ .

**Notation.** Let  $M(X)$  be the space of finite signed measures with  $\|\mu\| = |\mu|(X)$  (total variation norm).

### Theorem 3

Let  $X$  be a compact metric space. Then, for all  $\ell \in [C(X)]^*$ , there exists a unique  $\mu \in M(X)$  such that  $\ell(f) = \int f d\mu$ ,  $f \in C(X)$ . Moreover,  $\|\ell\| = \|\mu\|$ .

**Proof.** Let  $\ell \in [C(X)]^*$ . By Lemma 2,  $\ell = \ell_1 - \ell_2$  with  $\ell_1, \ell_2$  positive. By Theorem 1, there exists finite regular measures  $\mu_1, \mu_2$  such that for all  $f \in C(X)$ :

$$\begin{aligned}
\ell_1(f) &= \int f d\mu_1, \quad \ell_2(f) = \int f d\mu_2 \\
\implies \ell(f) &= \int f d(\mu_1 - \mu_2) = \int f d\mu, \quad f \in C(X)
\end{aligned}$$

Then, we could prove  $\mu$  is a finite regular signed measure and is unique following a similar argument as Corollary 2 proof. Moreover, by Proposition 5,  $\|\ell\| = \|\mu\|$ .

## Schwartz Space of Functions with Rapid Decrease

### From Fourier Series to Fourier Transform

1. Let  $T > 0$ . Then,  $\{\frac{1}{\sqrt{2\pi}}e^{\frac{in\pi}{T}x}, n = 0, \pm 1, \dots\}$  is  $2T$ -periodic.

This forms a CONS in  $L^2([-T, T])$  (complex-valued functions).



2. If  $h \in L^2([-T, T])$ , then

$$h(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi}{T}x} \text{ in } L^2([-T, T]) \text{ with } c_n = \frac{1}{2T} \int_{-T}^T h(x) e^{-\frac{in\pi}{T}x} dx.$$

Also,  $\int_{-T}^T |h(x)|^2 dx = 2T \sum_{n=-\infty}^{\infty} |c_n|^2.$

**Proof.** First, let's compute the  $L^2$  norm squared of  $h$ :

$$\int_{-T}^T |h(x)|^2 dx = \int_{-T}^T h(x) \overline{h(x)} dx$$

Substitute the Fourier series for  $h(x)$  and  $\overline{h(x)}$ :

$$\int_{-T}^T \left( \sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi}{T}x} \right) \left( \sum_{m=-\infty}^{\infty} \overline{c_m} e^{-\frac{im\pi}{T}x} \right) dx$$

Rearranging, we get:

$$\int_{-T}^T \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} c_n \overline{c_m} e^{\frac{i(n-m)\pi}{T}x} dx$$

By Fubini's theorem,

$$\sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} c_n \overline{c_m} \int_{-T}^T e^{\frac{i(n-m)\pi}{T}x} dx$$

The exponential functions form an orthogonal system. When  $n \neq m$ ,

$$\int_{-T}^T e^{\frac{i(n-m)\pi}{T}x} dx = \left[ \frac{T}{i\pi(n-m)} e^{\frac{i(n-m)\pi}{T}x} \right]_{-T}^T = 0$$

When  $n = m$ ,

$$\int_{-T}^T e^0 dx = \int_{-T}^T 1 dx = 2T$$

Therefore,

$$\sum_{n=-\infty}^{\infty} c_n \overline{c_n} \cdot 2T = 2T \sum_{n=-\infty}^{\infty} |c_n|^2$$

3. Let  $f \in C_c^\infty(\mathbb{R})$ ,  $f(x) = 0$  if  $|x| > R > 0$ . Then, for  $T > R + 1$ ,  $|x| < T$ ,

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{1}{2T} \int_{-T}^T f(y) e^{-\frac{in\pi}{T}y} dy e^{\frac{in\pi}{T}x} = \sum_{n=-\infty}^{\infty} \frac{1}{2T} \int_{-\infty}^{\infty} f(y) e^{-i\frac{n}{2T}2\pi y} dy e^{i\frac{n}{2T}2\pi x}.$$

Let  $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(y) e^{-i2\pi\xi y} dy$ ,  $\xi \in \mathbb{R}$ . Note that  $\hat{f} \in C_c^\infty(\mathbb{R})$ . Hence,

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{1}{2T} \hat{f}\left(\frac{n}{2T}\right) e^{i\frac{n}{2T}2\pi x}, \quad |x| < T, \text{ for all } T > R + 1$$

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-T}^T |f(x)|^2 dx = \frac{1}{2T} \sum_{n=-\infty}^{\infty} \left| \hat{f}\left(\frac{n}{2T}\right) \right|^2 \text{ for } T > R + 1.$$

For any  $x \in \mathbb{R}$ ,

$$f(x) = \lim_{T \rightarrow \infty} \sum_{n=-\infty}^{\infty} \frac{1}{2T} \hat{f}\left(\frac{n}{2T}\right) e^{i2\pi \frac{n}{2T} x} = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{i2\pi \xi x} d\xi.$$

Also,

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \frac{1}{2T} \sum_{n=-\infty}^{\infty} \left| \hat{f}\left(\frac{n}{2T}\right) \right|^2 = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 d\xi \text{ for all } T > R + 1.$$

For finite  $T > R + 1$ , we established:

$$\int_{-T}^T |f(x)|^2 dx = \frac{1}{2T} \sum_{n=-\infty}^{\infty} \left| \hat{f}\left(\frac{n}{2T}\right) \right|^2$$

Since  $f$  has compact support within  $[-R, R]$ , when  $T > R + 1$ ,

$$\begin{aligned} \int_{-\infty}^{\infty} |f(x)|^2 dx &= \int_{-T}^T |f(x)|^2 dx \\ \sum_{n=-\infty}^{\infty} \left| \hat{f}\left(\frac{n}{2T}\right) \right|^2 \cdot \frac{1}{2T} &= \sum_{n=-\infty}^{\infty} |\hat{f}(\xi_n)|^2 \cdot \Delta\xi \end{aligned}$$

This is precisely the form of a Riemann sum for the function  $|\hat{f}(\xi)|^2$  over the entire real line:

$$\sum_{n=-\infty}^{\infty} |\hat{f}(\xi_n)|^2 \cdot \Delta\xi$$

As  $T \rightarrow \infty$ , the spacing  $\Delta\xi = \frac{1}{2T} \rightarrow 0$ , making the partition finer. Under appropriate conditions (which  $\hat{f}$  satisfies since it comes from a compactly supported  $f \in C_c^\infty(\mathbb{R})$ ), the Riemann sum converges to the integral:

$$\lim_{T \rightarrow \infty} \sum_{n=-\infty}^{\infty} |\hat{f}(\xi_n)|^2 \cdot \Delta\xi = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 d\xi$$

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{i2\pi \xi x} d\xi, \quad x \in \mathbb{R} \text{ and } \int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 d\xi$$

**Fourier transform** of  $f \in C_c^\infty(\mathbb{R})$ :

$$\hat{f}(\xi) = (\mathcal{F}f)(\xi) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi \xi x} dx, \quad \xi \in \mathbb{R}$$

**The inverse Fourier transform** of  $g \in C_c^\infty(\mathbb{R})$ :

$$(\mathcal{F}^{-1}g)(x) = \int_{-\infty}^{\infty} g(\xi) e^{i2\pi\xi x} d\xi, \quad x \in \mathbb{R}.$$

We also have:

$$\mathcal{F}^{-1}\mathcal{F}f = f.$$

Note that  $(\mathcal{F}^{-1}f)(x) = \hat{f}(-x)$ , for any  $x$ . Also,  $\mathcal{F}$  is "unitary". It turns out  $F(C_c^\infty(\mathbb{R})) \subseteq S(\mathbb{R})$ , the space of functions of rapid decrease. Moreover,  $F : S(\mathbb{R}) \rightarrow S(\mathbb{R})$  is a bijection.

### Space of functions of Rapid Decrease

$$S(\mathbb{R}) = \{f \in C^\infty(\mathbb{R}) : \sup_x (1 + |x|)^N |f^{(n)}(x)| < \infty \text{ for any } N, n \in \mathbb{Z} \text{ and } N, n \geq 0\}.$$

### Functions of Rapid Decrease

Let  $f, g : \mathbb{R}^d \rightarrow \mathbb{C}$ . For  $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d$ , where  $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ ,

$$|\alpha| = \alpha_1 + \dots + \alpha_d,$$

$$\partial^\alpha = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}, \text{ total derivative of order } |\alpha|,$$

$$\alpha! = \alpha_1! \dots \alpha_d!, \quad 0! = 1.$$

### Product Rule

$$\partial^\alpha(fg) = \sum_{\beta \leq \alpha} \frac{\alpha!}{\beta!(\alpha - \beta)!} \partial^\beta f \partial^{\alpha - \beta} g,$$

where  $\beta \leq \alpha \iff \beta_k \leq \alpha_k$  for any  $k$ .

$$x^\alpha := x_1^{\alpha_1} \dots x_d^{\alpha_d}, \quad \text{where } \alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d,$$

$$x = (x_1, \dots, x_d) \in \mathbb{R}^d.$$

### Definition of Schwartz Space

For  $\alpha \in \mathbb{N}_0^d$  and  $N \in \mathbb{N}_0$ ,

$$|f|_{N,\alpha} = \sup_{x \in \mathbb{R}^d} (1 + |x|)^N |\partial^\alpha f(x)|.$$

The Schwartz space of functions of rapid decrease is defined as:

$$S = S(\mathbb{R}^d) = \{f \in C^\infty(\mathbb{R}^d) : |f|_{N,\alpha} < \infty \text{ for any } N, \alpha\}.$$

## Properties

1. If  $|f|_{N,\alpha} < \infty$  if and only if  $(1 + |x|)^N |f(x)| \leq C_{\alpha,N}$  for all  $x \in \mathbb{R}^d$ , then

$$|\partial^\alpha f(x)| \leq \frac{C_{\alpha,N}}{(1 + |x|)^N} \text{ for any } x \in \mathbb{R}^d.$$

2. If  $f \in S$ , then

$$|\partial^\alpha f(x)| \leq \frac{C_{\alpha,N}}{(1 + |x|)^N} \text{ for any } x \in \mathbb{R}^d.$$

3.  $C_c^\infty(\mathbb{R}^d) \subseteq S(\mathbb{R}^d)$ . Example:  $e^{-|x|^2} \in S(\mathbb{R}^d)$  but  $e^{-|x|} \notin C_c^\infty(\mathbb{R}^d)$  because it is non-zero everywhere.

## Seminorms and Convergence

1.  $|f|_{N,\alpha}$  with  $f \in S$  is a seminorm.

$$|f + g|_{N,\alpha} \leq |f|_{N,\alpha} + |g|_{N,\alpha},$$

$$|af|_{N,\alpha} = |a| |f|_{N,\alpha},$$

$$|f|_{N,\alpha} = 0 \text{ not implies } f = 0.$$

2. Let  $f_k, f \in S$ . We say  $f_k \rightarrow f$  in  $S$  if

$$|f_k - f|_{N,\alpha} \rightarrow 0 \text{ for any } N, \alpha.$$

3. For  $N \in \mathbb{N}_0$  and  $f \in S$ ,

$$|f|_{(N)} = \max_{|\alpha| \leq N} |f|_{N,\alpha} = \max_{|\alpha| \leq N} \sup_x (1 + |x|)^N |\partial^\alpha f(x)|$$

$$|f|_{(N)} \text{ are norms on } S, \text{ increasing in } N, |f|_{(N)} \leq |f|_{(N+1)}.$$

4. For  $f, g \in S$ ,

$$\rho(f, g) = \sum_{N=1}^{\infty} 2^{-N} \min(|f - g|_{(N)}, 1)$$

## Properties of $S(\mathbb{R}^d)$

1. If  $f \in S$  then  $\partial^\alpha f \in S$ .
2. If  $f, g \in S$  then  $fg \in S$ .
3. If  $f \in S$  then  $Pf \in S$  for any polynomial  $P$ .
4.  $f \in S \Rightarrow f \in L^p(\mathbb{R}^d)$  for any  $p > 0$ .

**Proof.** For  $p = \infty$  this is easy. For  $0 < p < \infty$ ,

$$\int_{\mathbb{R}^d} |f|^p dx = \int_{\mathbb{R}^d} (|f|(1 + |x|)^{\frac{d+1}{p}})^p \frac{dx}{(1 + |x|)^{d+1}} \leq |f|_{\lceil \frac{d+1}{p} \rceil, 0}^p \int_{\mathbb{R}^d} \frac{dx}{(1 + |x|)^{d+1}} < \infty.$$

5. If  $f, g \in S$  then  $f * g \in S$ .

**Proof.** By definition of convolution,

$$\partial^\alpha(f * g)(x) = \int_{\mathbb{R}^d} \partial^\alpha f(x - y)g(y) dy,$$

$$1 + |x| \leq 1 + |x - y| + |y| \leq (1 + |x - y|)(1 + |y|).$$

$$\begin{aligned} (1 + |x|)^N |\partial^\alpha(f * g)(x)| &\leq \int (1 + |x - y|)^N |\partial^\alpha f(x - y)| (1 + |y|)^N |g(y)| dy \\ &\leq |f|_{N,\alpha} \int (1 + |y|)^{N+d+1} |g(y)| \frac{dy}{(1 + |y|)^{d+1}} \\ &\leq |f|_{N,\alpha} |g|_{N+d+1,0} \int \frac{dy}{(1 + |y|)^{d+1}} < \infty \end{aligned}$$

### Continuous Linear Functionals on $S$

Let  $T : S(\mathbb{R}^d) \rightarrow \mathbb{C}$  be linear. Then,  $T$  is continuous with respect to the metric  $\rho$  on  $S$  if and only if

There exists  $C > 0$  and  $N \in \mathbb{N}_0$  such that  $|T(f)| \leq C|f|_{(N)}$  for all  $f \in S$ .

**Proof.** ( $\Rightarrow$ ) Let  $T$  be continuous. Since  $T$  is continuous at zero,  $|T(f)| = |T(f) - T(0)| \leq 1$  if  $\rho(f, 0) \leq 2\delta$  for some  $\delta > 0$ .

$$\begin{aligned} \rho(f, 0) &= \sum_{N=1}^{\infty} 2^{-N} \min\{1, |f|_{(N)}\} \\ &\leq \sum_{N=1}^{N_0} 2^{-N} |f|_{(N_0)} + \sum_{N=N_0+1}^{\infty} 2^{-N} \\ &\leq |f|_{(N_0)} + 2^{-N_0} \end{aligned}$$

Choose  $N_0$  so that  $2^{-N_0} \leq \delta$ . Let  $g \in S$  and  $g \neq 0$ . Then,  $f = \frac{\delta}{|g|_{(N_0)}} g$  has  $|f|_{(N_0)} = \delta$ .

$$\Rightarrow |f|_{(N_0)} \leq \delta \Rightarrow \rho(f, 0) \leq 2\delta \Rightarrow |T(f)| \leq 1$$

$$\Rightarrow \sum_{N=1}^{\infty} 2^{-N} \min\{1, |f|_{(N)}\} \leq |f|_{(N_0)} + 2^{-N_0} \leq 2\delta \text{ for all } N_0 \in \mathbb{N},$$

$$|T(f)| = |T(g)| \cdot \frac{\delta}{|g|_{(N_0)}} \leq 1 \Rightarrow |T(g)| \leq \frac{1}{\delta} |g|_{(N_0)}.$$

( $\Leftarrow$ ) Let  $|T(f)| \leq C|f|_{(N_0)}$ ,  $f \in S$ . Let  $f_n \rightarrow f$  in  $S$ . Then, when  $|f_n - f|_{(N_0)} \rightarrow 0$ ,

$$|T(f_n) - T(f)| = |T(f_n - f)| \leq C|f_n - f|_{(N_0)} \rightarrow 0.$$

### Corollary 0

A linear functional  $T : S \rightarrow \mathbb{C}$  is continuous if and only if there exist  $C > 0$ ,  $\alpha_1, \dots, \alpha_m \in \mathbb{N}_0^d$ , and  $N_1, \dots, N_m \in \mathbb{N}_0$  such that:

$$|T(f)| \leq C \sum_{k=1}^m |f|_{N_k, \alpha_k}.$$

### Fourier Transform on $S(\mathbb{R}^d)$

(i) Given  $f \in L^1(\mathbb{R}^d)$ , its Fourier transform is:

$$\mathcal{F}f(\xi) = \hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-i2\pi x \cdot \xi} dx, \quad \xi \in \mathbb{R}^d.$$

where  $x \cdot \xi = x_1 \xi_1 + \dots + x_d \xi_d$ .

(ii) Given  $g \in L^1(\mathbb{R}^d)$ , its inverse Fourier transform is:

$$\mathcal{F}^{-1}g(x) = \int_{\mathbb{R}^d} g(\xi) e^{i2\pi x \cdot \xi} d\xi, \quad x \in \mathbb{R}^d.$$

### Note.

- (a)  $\mathcal{F}f(\xi)$  and  $\mathcal{F}^{-1}g(x)$  are bounded, continuous functions.
- (b)  $\mathcal{F}$  and  $\mathcal{F}^{-1}$  are linear operators.
- (c)  $|\mathcal{F}f(\xi)| \leq \int_{\mathbb{R}^d} |f| dx = \|f\|_{L^1}$ .
- (d)  $|\mathcal{F}^{-1}g(x)| \leq \|g\|_{L^1}$ .
- (e)  $\mathcal{F}^{-1}g(x) = \mathcal{F}g(-x)$ ,  $x \in \mathbb{R}^d$ .
- (f)  $\widehat{\partial^\alpha f} = (i2\pi\xi)^\alpha \hat{f}$ .

## Simple Properties of Fourier Transform

### Property 1

If  $f \in S$ , then for  $\alpha \in \mathbb{N}_0^d$ ,

$$\widehat{\partial^\alpha f}(\xi) = (i2\pi\xi)^\alpha \hat{f}(\xi) = (i2\pi)^{|\alpha|} \xi_1^{\alpha_1} \dots \xi_d^{\alpha_d} \hat{f}(\xi)$$

where  $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_d) \in \mathbb{N}_0^d$ ,  $x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \dots x_d^{\alpha_d}$ , and  $x = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$ .

**Proof.** For  $d = 1$  and  $\alpha = 1$ ,

$$\widehat{f^{(1)}}(\xi) = i2\pi\xi \hat{f}(\xi), \quad \xi \in \mathbb{R}.$$

By definition,

$$\begin{aligned}
\widehat{f^{(1)}}(\xi) &= \int_{-\infty}^{\infty} f'(x) e^{-i2\pi x \xi} dx = \lim_{n \rightarrow \infty} \int_{-n}^n f'(x) e^{-i2\pi x \xi} dx \\
&= \lim_{n \rightarrow \infty} [f(x) e^{-i2\pi x \xi}]_{-n}^n - (-i2\pi \xi) \int_{-n}^n f(x) e^{-i2\pi x \xi} dx \\
&= 0 + i2\pi \xi \hat{f}(\xi) = i2\pi \xi \hat{f}(\xi)
\end{aligned}$$

### Example 1

Let  $\Delta u(x) = \frac{\partial^2}{\partial x_1^2} u(x) + \cdots + \frac{\partial^2}{\partial x_d^2} u(x)$ , where  $u \in S$ . Find  $\widehat{\Delta u}(\xi)$ .

$\frac{\partial^2}{\partial x_1^2} = \partial^\alpha$  with  $\alpha = [2, 0, 0, \dots, 0]$ . By property 1,

$$\begin{aligned}
\widehat{\Delta u}(\xi) &= \widehat{\frac{\partial^2}{\partial x_1^2} u(\xi)} + \cdots + \widehat{\frac{\partial^2}{\partial x_d^2} u(\xi)} \\
&= (i2\pi \xi_1)^2 \hat{u}(\xi) + \cdots + (i2\pi \xi_d)^2 \hat{u}(\xi) \\
&= \hat{u}(\xi) (-4\pi^2 (\xi_1^2 + \xi_2^2 + \cdots + \xi_d^2)) \\
&= -4\pi^2 |\xi|^2 \hat{u}(\xi)
\end{aligned}$$

### Property 2

1. Let  $f \in S$ ,  $y \in \mathbb{R}^d$ . Then:

$$\widehat{f(\cdot + y)}(\xi) = e^{i2\pi \xi \cdot y} \hat{f}(\xi)$$

**Proof.**

$$\begin{aligned}
\widehat{f(\cdot + y)}(\xi) &= \int_{\mathbb{R}^d} f(x + y) e^{-i2\pi x \cdot \xi} dx \\
&= \int_{\mathbb{R}^d} f(z) e^{-i2\pi (z - y) \cdot \xi} dz \quad (\text{where } z = x + y \text{ and } dz = dx) \\
&= e^{i2\pi y \cdot \xi} \hat{f}(\xi)
\end{aligned}$$

2. Let  $f \in S$ ,  $\delta \in \mathbb{R}$  and  $\epsilon \neq 0$ . Then,

$$\widehat{f(\delta \cdot)}(\xi) = |\delta|^{-d} \hat{f}\left(\frac{\xi}{\delta}\right)$$

**Proof.**

$$\begin{aligned}
\widehat{f(\delta \cdot)}(\xi) &= \int_{\mathbb{R}^d} f(\delta x) e^{-i2\pi x \cdot \xi} dx \\
&= \int_{\mathbb{R}^d} f(z) e^{-i2\pi \frac{z \cdot \xi}{\delta}} \delta^{-d} dz \quad (\text{where } z = \delta x \text{ and } dz = \delta^d dx) \\
&= \delta^{-d} \hat{f}\left(\frac{\xi}{\delta}\right)
\end{aligned}$$

## Example 2

Let  $f \in S$ ,  $\int f dx \neq 0$ . Show that:

$$\lim_{\delta \rightarrow 0} |\widehat{f(\delta \cdot)}(\xi)| = \begin{cases} 0, & \xi \neq 0 \\ +\infty, & \xi = 0 \end{cases}.$$

Assume  $\delta \neq 0$ .

$$|\widehat{f(\delta \cdot)}(\xi)| \stackrel{\delta \neq 0}{=} |\delta|^{-d} \left| \hat{f}\left(\frac{\xi}{\delta}\right) \right|$$

Since  $\hat{f} \in S$ , it satisfies:

$$|\hat{f}(\xi)| \leq \frac{C}{(1 + |\xi|)^{d+1}},$$

where  $C = |\hat{f}|_{d+1,0}$  is a seminorm. When  $\xi \neq 0$ ,

$$\begin{aligned} |\widehat{f(\delta \cdot)}(\xi)| &= \frac{1}{|\xi|^d} \cdot \left( \frac{|\xi|}{|\delta|} \right)^d \cdot |\hat{f}(\xi)| \\ &\leq \frac{1}{|\xi|^d} \cdot \left( \frac{|\xi|}{|\delta|} \right)^d \cdot \frac{|\hat{f}|_{d+1,0}}{(1 + |\xi|/|\delta|)^{d+1}} \rightarrow 0 \quad \text{as } \delta \rightarrow 0 \end{aligned}$$

When  $\xi = 0$ ,

$$|\widehat{f(\delta \cdot)}(0)| = |\delta|^{-d} |\hat{f}(0)| \rightarrow \infty \quad \text{as } \delta \rightarrow 0$$

Note that  $\hat{f}(0) = \int f(x) dx \neq 0$ , so  $|\hat{f}(0)| \neq 0$ . On the other hand,  $\lim_{\delta \rightarrow 0} f(\delta x) = f(0)$  for  $f \in S$ . Note the opposite tendencies of  $f(\delta x)$  and  $\widehat{f(\delta \cdot)}(\xi)$  as  $\delta \rightarrow 0$ .

## Property 3

If  $f \in S$ , then  $\hat{f} \in C^\infty$  with all derivatives bounded, and:

$$\partial^\beta \hat{f}(\xi) = \hat{g}(\xi) \quad \text{with} \quad g(x) = (-i2\pi x)^\beta f(x) \in S, \quad \beta \in \mathbb{N}_0^d.$$

**Proof.** (See Folland 2-27)

$$\partial^\beta \hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) \frac{\partial^\beta}{\partial \xi} e^{-i2\pi x \cdot \xi} dx = \int_{\mathbb{R}^d} \underbrace{f(x)(-i2\pi x)^\beta}_{g(x)} e^{-i2\pi x \cdot \xi} dx = \hat{g}(\xi)$$

## Proposition 1

The Fourier transform  $F$  maps  $S$  to  $S$ . We write  $F : S \rightarrow S$ .

**Proof.** Let  $f \in S$ . By Property 3 above,  $\hat{f} \in C^\infty$ . Then,  $\hat{f} \in S$  if and only if  $\sup_{\xi \in \mathbb{R}^d} |\xi^\alpha \partial^\beta \hat{f}(\xi)| < \infty$  for any  $\alpha, \beta \in \mathbb{N}_0^d$ .

$$\begin{aligned} \xi^\alpha \partial^\beta \hat{f}(\xi) &= (i2\pi)^{-|\alpha|} (i2\pi \xi)^\alpha \partial^\beta \hat{f}(\xi) \\ &= (i2\pi)^{-|\alpha|} (i2\pi \xi)^\alpha \hat{g}(\xi) \quad \text{with } g(x) = (-i2\pi x)^\beta f(x) \in S \end{aligned}$$

Hence,

$$\sup_{\xi \in \mathbb{R}^d} |\xi^\alpha \partial^\beta \hat{f}(\xi)| = (2\pi)^{-|\alpha|} \sup_{\xi \in \mathbb{R}^d} |\widehat{\partial^\alpha g}(\xi)| \leq (2\pi)^{-|\alpha|} |\partial^\alpha g|_{L_1} < \infty$$



**Proposition 2**

Let  $f, g \in S$ . Then,  $\widehat{f * g}(\xi) = \hat{f}(\xi)\hat{g}(\xi)$ .

**Proof.** By Fubini's Theorem,

$$\begin{aligned} & \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x-y)g(y) dy e^{-i2\pi x \cdot \xi} dx \\ &= \int_{\mathbb{R}^d} \left( \int_{\mathbb{R}^d} f(x-y) e^{-i2\pi(x-y) \cdot \xi} dx \right) \cdot g(y) e^{-i2\pi y \cdot \xi} dy, \quad \text{set } z = x-y \text{ and } dz = dx \\ &= \int_{\mathbb{R}^d} (\hat{f}(\xi)) \cdot g(y) e^{-i2\pi y \cdot \xi} dy = \hat{f}(\xi) \cdot \hat{g}(\xi) \end{aligned}$$

We can swap integrals because  $\iint |f(x-y)| \cdot |g(y)| dy dx = |f|_{L^1} \cdot |g|_{L^1} < \infty$ .

**Proposition 3**

Let  $f, g \in S$ . Then,

$$\begin{aligned} \int_{\mathbb{R}^d} \hat{f}(z)g(z) dz &= \int_{\mathbb{R}^d} f(y)\hat{g}(y) dy \\ \int_{\mathbb{R}^d} \mathcal{F}^{-1}(f)(z)g(z) dz &= \int_{\mathbb{R}^d} f(y)\mathcal{F}^{-1}(g)(y) dy \end{aligned}$$

**Proof.** The proof will use Fubini's Theorem. Let  $f, g \in S$ . Then,

$$\begin{aligned} \int_{\mathbb{R}^d} \hat{f}(z)g(z) dz &= \int_{\mathbb{R}^d} \left( \int_{\mathbb{R}^d} f(y) e^{-i2\pi y \cdot z} dy \right) g(z) dz \\ &= \int_{\mathbb{R}^d} \left( \int_{\mathbb{R}^d} g(z) e^{-i2\pi y \cdot z} dz \right) f(y) dy \\ &= \int_{\mathbb{R}^d} \hat{g}(y)f(y) dy \end{aligned}$$

We can use Fubini's Theorem because:

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |f(y)| \cdot |g(z)| \cdot |e^{i2\pi y \cdot z}| dy dz = |f|_{L^1} \cdot |g|_{L^1} < \infty$$

since  $f, g \in L^1$  since  $f, g \in S$ . Analogous argument proves the  $\mathcal{F}^{-1}$  case.

**Inverse of  $\mathcal{F}$** **Theorem 1**

Let  $f \in S$ . Then,  $\mathcal{F}^{-1}\mathcal{F}(f)(x) = \mathcal{F}\mathcal{F}^{-1}(f)(x)$  for  $x \in \mathbb{R}^d$ . In particular,

$$\mathcal{F}^{-1}(\hat{f})(x) = \int_{\mathbb{R}^d} \hat{f}(\xi) e^{i2\pi x \cdot \xi} d\xi = f(x) \text{ for } x \in \mathbb{R}^d.$$

This means both  $\mathcal{F}, \mathcal{F}^{-1} : S \rightarrow S$  are bijections.

**Proof.** Let  $f \in S$ . We first prove that inversion holds at  $x = 0$ .

$$f(0) = \int_{\mathbb{R}^d} \hat{f}(\xi) d\xi.$$

For  $\delta > 0$  and  $k(x) = e^{-\pi|x|^2}$ ,

$$\int_{\mathbb{R}^d} f(\delta y) \hat{k}(y) dy \longrightarrow f(0) \text{ as } \delta \rightarrow 0 \text{ by dominant convergence theorem.}$$

By Proposition 3,

$$\begin{aligned} \int_{\mathbb{R}^d} f(\delta y) \hat{k}(y) dy &= \int \widehat{f(\delta \cdot)}(\xi) k(\xi) d\xi \\ &= \int \delta^{-d} \hat{f}\left(\frac{\xi}{\delta}\right) k(\xi) d\xi \text{ (by Property 2(b))} \\ &= \int \delta^{-d} \hat{f}(z) k(\delta z) \delta^d dz \text{ where } z = \frac{\xi}{\delta} \text{ and } d\xi = \delta^d dz \\ &= \int \hat{f}(z) k(\delta z) dz \end{aligned}$$

By dominant convergence theorem, when  $\delta \rightarrow 0$ ,  $\int \hat{f}(z) k(\delta z) dz \rightarrow \int \hat{f}(z) k(0) dz = \int \hat{f}(z) dz$  as  $k(0) = 1$  and  $\int \hat{f}(z) dz = f(0)$ . In general case, let  $y \in \mathbb{R}^d$  and consider  $f(x + y)$  for  $x \in \mathbb{R}^d$ . Note that  $f(y) = f(0 + y)$ . Using the result that inversion holds at 0,

$$\begin{aligned} f(y) &= f(0 + y) = \int \mathcal{F}(f(\cdot + y))(\xi) d\xi \\ &= \int \hat{f}(\xi) e^{i2\pi\xi \cdot y} d\xi \text{ (by property 2(a))} \\ &= \mathcal{F}^{-1}(\hat{f})(y) \end{aligned}$$

We have shown  $\mathcal{F}^{-1}\mathcal{F}(f) = f$  so we now show  $\mathcal{F}\mathcal{F}^{-1}(f) = f$ . Recall that  $\mathcal{F}(g)(x) = \mathcal{F}^{-1}(g)(-x)$ . Now  $\mathcal{F}\mathcal{F}^{-1}(f)(x) = \mathcal{F}^{-1}(\mathcal{F}^{-1}(f))(-x)$  and:

$$\mathcal{F}^{-1}f(x) = \int f(\xi) e^{i2\pi x \cdot \xi} d\xi = \int f(-z) e^{-i2\pi z \cdot x} dz = \mathcal{F}\tilde{f}(x)$$

By the change of variable,  $\xi = -z$  and  $\tilde{f}(x) = f(-x)$  for  $x \in \mathbb{R}^d$ . Therefore,

$$\mathcal{F}(\mathcal{F}^{-1}(f))(x) = \mathcal{F}^{-1}(\mathcal{F}^{-1}(f))(-x) = \mathcal{F}^{-1}(\mathcal{F}(\tilde{f}))(-x) = \tilde{f}(-x) = f(x).$$

Thus, we prove that  $\mathcal{F}\mathcal{F}^{-1}(f)(x) = f(x) = \mathcal{F}^{-1}\mathcal{F}(f)(x)$ .

**Remark.**

1. First Use of DCT in the Proof.

$$\int_{\mathbb{R}^d} f(\delta y) \hat{k}(y) dy \longrightarrow f(0) \text{ as } \delta \rightarrow 0.$$

- *Pointwise Convergence.* Since  $f$  is continuous (because  $f \in S$ , the Schwartz space),  $f(\delta y) \rightarrow f(0)$  as  $\delta \rightarrow 0$  for each  $y$ .
- *Dominating Function.* Since  $\hat{f} \in S$ , it is rapidly decreasing, meaning  $|\hat{f}(\xi)| \leq \frac{C}{(1+|\xi|)^{d+1}}$  for some  $C > 0$ . Meanwhile,  $|k(\delta\xi)| \leq 1$  (since  $k$  is a Gaussian). Thus,  $|\hat{f}(\xi)k(\delta\xi)| \leq |\hat{f}(\xi)|$ , and  $\hat{f}(\xi)$  is integrable (because  $\hat{f} \in S$ ).

DCT allows us to move the limit inside:

$$\lim_{\delta \rightarrow 0} \int f(\delta y) \hat{k}(y) dy = \int \lim_{\delta \rightarrow 0} f(\delta y) \hat{k}(y) dy = \int f(0) \hat{k}(y) dy = f(0).$$

## 2. Second Use of DCT in the Proof.

$$\int \hat{f}(\xi) k(\delta\xi) d\xi \longrightarrow \int \hat{f}(\xi) d\xi \quad \text{as } \delta \rightarrow 0.$$

- *Pointwise Convergence.*  $k(\delta\xi) = e^{-\pi\delta^2|\xi|^2} \rightarrow 1$  as  $\delta \rightarrow 0$  for each  $\xi$ .
- *Dominating Function.* Since  $\hat{f} \in S$ , it is rapidly decreasing, meaning  $|\hat{f}(\xi)| \leq \frac{C}{(1+|\xi|)^{d+1}}$  for some  $C > 0$ . Meanwhile,  $|k(\delta\xi)| \leq 1$  since  $k$  is a Gaussian. Thus,  $|\hat{f}(\xi)k(\delta\xi)| \leq |\hat{f}(\xi)|$ , and  $\hat{f}(\xi)$  is integrable because  $\hat{f} \in S$ .

DCT allows us to move the limit inside:

$$\lim_{\delta \rightarrow 0} \int \hat{f}(\xi) k(\delta\xi) d\xi = \int \hat{f}(\xi) \lim_{\delta \rightarrow 0} k(\delta\xi) d\xi = \int \hat{f}(\xi) d\xi.$$

## 3. Why the Schwartz Space $S$ is Important?

- Functions in  $S$  decay rapidly: Both  $f$  and  $\hat{f}$  decrease faster than any polynomial, ensuring integrability.
- Gaussians are nice dominating functions:  $k(x) = e^{-\pi|x|^2}$  is smooth, integrable, and dominates the integrals involved.

If  $f$  were not in  $S$ , we might not have a suitable dominating function.

## Plancherel's Equality

For  $f \in S$ ,  $\int |f(x)|^2 dx = \int |\hat{f}(\xi)|^2 d\xi$ , so  $|f|_{L^2} = |\hat{f}|_{L^2}$  where  $L^2 = L^2(\mathbb{R}^d)$ . *Idea of the proof:* Use  $f(0) = \int \hat{f}(\xi) d\xi$  and the Fourier transform of the convolution: Let  $(f * g)(x) = \int f(x-y)g(y) dy$  and  $\mathcal{F}(f * g)(\xi) = \hat{f}(\xi)\hat{g}(\xi)$ . Applying Proposition 2 and  $f(0) = \int \hat{f}(\xi) d\xi$ ,

$$(f * g)(0) = \int g(y)f(-y) dy = \int \mathcal{F}(f * g)(\xi) d\xi = \int \hat{f}(\xi)\hat{g}(\xi) d\xi \quad (\#)$$

In  $(\#)$  above, we want to take  $f = g$  and deal with the negative  $y$  part.

**Proposition 4 (Plancherel's Equality)**

For any  $f, g \in S$ ,  $\int g(y) \overline{f(y)} dy = \int \hat{g}(\xi) \overline{\hat{f}(\xi)} d\xi$ . Thus,  $\langle g, f \rangle_{L^2} = \langle \hat{g}, \hat{f} \rangle_{L^2}$ . In particular,  $\|f\|_{L^2} = \|\hat{f}\|_{L^2}$  in the case  $f = g$ .

**Proof.** Let  $f, g \in S$ . Set  $\tilde{f}(y) = \overline{f(-y)}$  for  $y \in \mathbb{R}^d$ . Note that  $\tilde{f} \in S$ . By (#),

$$\int g(y) \overline{f(y)} dy = \int g(y) \tilde{f}(-y) dy = \int \hat{g}(\xi) \mathcal{F}(\tilde{f})(\xi) d\xi$$

Now, we compute  $\mathcal{F}(\tilde{f})(\xi)$ .

$$\begin{aligned} \mathcal{F}(\tilde{f})(\xi) &= \int \overline{f(-y)} e^{-i2\pi y \cdot \xi} dy = \int \overline{f(x)} e^{i2\pi x \cdot \xi} dx \quad \text{where } x = -y, dx = dy |J| = dy \\ &= \overline{\int f(x) e^{-i2\pi x \cdot \xi} dx} = \overline{\hat{f}(\xi)} \end{aligned}$$

Hence,  $\int g(y) \overline{f(y)} dy = \int \hat{g}(\xi) \overline{\hat{f}(\xi)} d\xi \iff \langle g, f \rangle_{L^2} = \langle \hat{g}, \hat{f} \rangle_{L^2}$ .

**Remark.**  $\mathcal{F}$  and  $\mathcal{F}^{-1}$  can be extended to unitary maps on  $L^2$ . For  $f \in L^2$ , there exists  $f_n \in S$  with  $f_n \rightarrow f$  in  $L^2$  since  $C_c^\infty \subseteq S$  and  $C_c^\infty$  is dense in  $L^2$ . Then, by Proposition 4, if  $\|f_n - f_m\|_{L^2} \rightarrow 0$  as  $m, n \rightarrow \infty$ , then

$$\|f_n - f_m\|_{L^2} = \|\hat{f}_n - \hat{f}_m\|_{L^2} = \|\mathcal{F}^{-1}(f_n) - \mathcal{F}^{-1}(f_m)\|_{L^2}$$

We define  $\mathcal{F}f = \lim_{n \rightarrow \infty} \mathcal{F}(f_n)$  and  $\mathcal{F}^{-1}f = \lim_{n \rightarrow \infty} \mathcal{F}^{-1}(f_n)$ . This definition does not depend on the choice of the sequence  $f_n$ . Indeed, let  $g_n \in S$  such that  $g_n \rightarrow f$  in  $L^2$ . Then,  $\|g_n - f_n\|_{L^2} \rightarrow 0$  as  $n \rightarrow \infty$ . Therefore, by proposition 4,

$$\begin{aligned} \|g_n - f_n\|_{L^2} &= \|\hat{g}_n - \hat{f}_n\|_{L^2} = \|\mathcal{F}^{-1}g_n - \mathcal{F}^{-1}f_n\|_{L^2} \\ &\implies \hat{g}_n \rightarrow \mathcal{F}f \text{ and } \mathcal{F}^{-1}g_n \rightarrow \mathcal{F}^{-1}f_n \text{ in } L^2 \end{aligned}$$

Hence,  $\mathcal{F} : L^2 \rightarrow L^2$  is unitary with  $\mathcal{F}^* = \mathcal{F}^{-1}$ .

**Corollary 1**

$\mathcal{F}, \mathcal{F}^{-1} : S \rightarrow S$  can be extended to  $\mathcal{F}, \mathcal{F}^{-1} : L^2 \rightarrow L^2$  as

$$\mathcal{F}f = \hat{f} = \lim_{n \rightarrow \infty} \mathcal{F}f_n, \quad \mathcal{F}^{-1}f = \lim_{n \rightarrow \infty} \mathcal{F}^{-1}f_n \text{ in } L^2 \text{ with } f_n \in S, f_n \rightarrow f \text{ in } L^2.$$

We have  $\mathcal{F}\mathcal{F}^{-1}f = \mathcal{F}^{-1}\mathcal{F}f = f$  for any  $f \in L^2$ . Also,

$$\|\mathcal{F}f\|_{L^2} = \|\mathcal{F}^{-1}f\|_{L^2} = \|f\|_{L^2}, \quad f \in L^2.$$

### Proposition 6

Let  $f \in L^1 \cap L^2$ . Let  $\hat{f} = \lim_{n \rightarrow \infty} \hat{f}_n$ ,  $\mathcal{F}^{-1}f = \lim_{n \rightarrow \infty} \mathcal{F}^{-1}f_n$  in  $L^2$  with  $f_n \in S$  and  $f_n \rightarrow f$  in  $L^2$ . Then,  $\hat{f}(\xi) = \int f(x)e^{-i2\pi x \cdot \xi} dx$  with  $\xi \in \mathbb{R}^d$ .  $\mathcal{F}^{-1}f(x) = \int f(\xi)e^{i2\pi x \cdot \xi} d\xi$  with  $x \in \mathbb{R}^d$ .

**Proof.** By Proposition 3, for any  $h \in S$ ,

$$\int \hat{f}_n h = \int f_n \hat{h} \quad \text{and} \quad \int \mathcal{F}^{-1}(f_n) h = \int f_n \mathcal{F}^{-1}(h)$$

As  $n \rightarrow \infty$ ,

$$\begin{aligned} \int \hat{f}_n h &\rightarrow \int \hat{f} h, \quad \int f_n \hat{h} \rightarrow \int f \hat{h}, \quad \int \mathcal{F}^{-1}(f_n) h \rightarrow \int \mathcal{F}^{-1}(f) h, \quad \int f_n \mathcal{F}^{-1}(h) \rightarrow \int f \mathcal{F}^{-1}(h) \\ &\implies \int \hat{f} h = \int f \hat{h}, \quad \int \mathcal{F}^{-1}(f) h = \int f \mathcal{F}^{-1}(h) \end{aligned}$$

Now, for all  $h \in S$ ,

$$\begin{aligned} \int \hat{f}(y) h(y) dy &= \int f(x) \hat{h}(x) dx \\ &= \int f(x) \left( \int h(y) e^{-i2\pi x \cdot y} dy \right) dx \\ &= \int (f(x) e^{-i2\pi x \cdot y} dx) h(y) dy \quad (\text{By Fubini's Theorem}) \end{aligned}$$

Thus,

$$\hat{f}(y) = \int f(x) e^{-i2\pi x \cdot y} dx \quad \text{a.e.}$$

Fubini holds since  $\iint |f(x)| \cdot |h(y)| \cdot |e^{-i2\pi x \cdot y}| dx dy = \|f\|_{L^1} \cdot \|h\|_{L^1} < \infty$ . An analogous proof holds for  $\mathcal{F}^{-1}$ .

### Corollary 2

Let  $f \in L^2$ . Then,

$$\begin{aligned} \mathcal{F}f(\xi) &= \hat{f}(\xi) = \lim_{n \rightarrow \infty} \int_{|x| \leq n} f(x) e^{-i2\pi x \cdot \xi} dx \text{ in } L^2, \\ \mathcal{F}^{-1}f(x) &= \lim_{n \rightarrow \infty} \int_{|\xi| \leq n} f(\xi) e^{i2\pi x \cdot \xi} d\xi \text{ in } L^2. \end{aligned}$$

**Proof.** Since  $f \in L^2$ , we can approximate it by truncating it to the ball  $|x| \leq n$ . Define:

$$f_n(x) = f(x) \cdot \chi_{|x| \leq n},$$

where  $\chi_{|x| \leq n}$  is the indicator function of the ball of radius  $n$ . Then,  $f_n$  has compact support since it vanishes outside  $|x| \leq n$ . By Hölder's inequality, and  $f_n \in L^2$  since  $f \in L^2$ ,

$$\int |f_n| \leq \left( \int |f_n|^2 \right)^{1/2} \cdot \left( \int \chi_{|x| \leq n} \right)^{1/2} < \infty$$

Therefore,  $f_n \in L^1 \cap L^2$ .

$f_n \rightarrow f$  in  $L^2$  as  $n \rightarrow \infty$  by the dominated convergence theorem, since  $|f_n| \leq |f|$  and  $|f|^2$  is integrable. For  $f_n \in L^1 \cap L^2$ ,

$$\mathcal{F}f(\xi) = \hat{f}_n(\xi) = \int_{\mathbb{R}^d} f_n(x) e^{-i2\pi x \cdot \xi} dx = \int_{|x| \leq n} f(x) e^{-i2\pi x \cdot \xi} dx.$$

This is well-defined pointwise since  $f_n \in L^1$ . Since  $f_n \rightarrow f$  in  $L^2$ , and the Fourier transform  $\mathcal{F}$  is a unitary operator on  $L^2$ , we have:

$$\hat{f}_n \rightarrow \hat{f} \quad \text{in } L^2 \text{ as } n \rightarrow \infty.$$

This is because  $\|\hat{f}_n - \hat{f}\|_{L^2} = \|f_n - f\|_{L^2} \rightarrow 0$  by Plancherel's Equality. Now, define:

$$g_n(\xi) = f(\xi) \cdot \chi_{|\xi| \leq n}.$$

Then  $g_n \rightarrow f$  in  $L^2$ . By the continuity of  $\mathcal{F}^{-1}$  on  $L^2$ , we have  $\mathcal{F}^{-1}g_n \rightarrow \mathcal{F}^{-1}f$  in  $L^2$ , which gives the second formula:

$$\mathcal{F}^{-1}f(x) = \lim_{n \rightarrow \infty} \int_{|\xi| \leq n} f(\xi) e^{i2\pi x \cdot \xi} d\xi \quad (\text{in } L^2).$$

**Recall.**  $f \in S$ . Then,  $\mathcal{F}^{-1}\mathcal{F}(f)(x) = \mathcal{F}\mathcal{F}^{-1}(f)(x) = f(x)$ ,  $x \in \mathbb{R}^d$ .  $\mathcal{F}, \mathcal{F}^{-1} : S \rightarrow S$  are bijective.

## Plancherel's Equality

For  $f \in S$ ,  $\int |f(x)|^2 dx = \int |\hat{f}(\xi)|^2 d\xi$ , so  $\|f\|_{L^2} = \|\hat{f}\|_{L^2}$  where  $L^2 = L^2(\mathbb{R}^d)$ .

*Idea of the proof:* Use  $f(0) = \int \hat{f}(\xi) d\xi$  and the Fourier transform of the convolution: Let  $(f * g)(x) = \int f(x - y)g(y) dy$  and  $\mathcal{F}(f * g)(\xi) = \hat{f}(\xi)\hat{g}(\xi)$ . Applying Proposition 2 and  $f(0) = \int \hat{f}(\xi) d\xi$ ,

$$(f * g)(0) = \int g(y)f(-y) dy = \int \mathcal{F}(f * g)(\xi) d\xi = \int \hat{f}(\xi)\hat{g}(\xi) d\xi \quad (\#).$$

In  $(\#)$  above, we want to take  $f = g$  and deal with the negative  $y$  part.

### Proposition 4 (Plancherel's Equality)

For any  $f, g \in S$ ,  $\int g(y)\overline{f(y)} dy = \int \hat{g}(\xi)\overline{\hat{f}(\xi)} d\xi$ . Thus,  $\langle g, f \rangle_{L^2} = \langle \hat{g}, \hat{f} \rangle_{L^2}$ . In particular,  $\|f\|_{L^2} = \|\hat{f}\|_{L^2}$  in the case  $f = g$ .

**Proof.** Let  $f, g \in S$ . Set  $\tilde{f}(y) = \overline{f(-y)}$  for  $y \in \mathbb{R}^d$ . Note that  $\tilde{f} \in S$ . By  $(\#)$ ,

$$\int g(y)\overline{f(y)} dy = \int g(y)\tilde{f}(-y) dy = \int \hat{g}(\xi)\mathcal{F}(\tilde{f})(\xi) d\xi$$

Now, we compute  $\mathcal{F}(\tilde{f})(\xi)$ .

$$\begin{aligned}\mathcal{F}(\tilde{f})(\xi) &= \int \overline{f(-y)} e^{-i2\pi y \cdot \xi} dy = \int \overline{f(x)} e^{i2\pi x \cdot \xi} dx \quad \text{where } x = -y, dx = dy |J| = dy \\ &= \overline{\int f(x) e^{-i2\pi x \cdot \xi} dx} = \overline{\hat{f}(\xi)}\end{aligned}$$

Hence,  $\int g(y) \overline{f(y)} dy = \int \hat{g}(\xi) \overline{\hat{f}(\xi)} d\xi \iff \langle g, f \rangle_{L^2} = \langle \hat{g}, \hat{f} \rangle_{L^2}$ .

**Remark.**  $\mathcal{F}$  and  $\mathcal{F}^{-1}$  can be extended to unitary maps on  $L^2$ . For  $f \in L^2$ , there exists  $f_n \in S$  with  $f_n \rightarrow f$  in  $L^2$  since  $C_c^\infty \subseteq S$  and  $C_c^\infty$  is dense in  $L^2$ . Then, by Proposition 4, if  $|f_n - f_m|_{L^2} \rightarrow 0$  as  $m, n \rightarrow \infty$ , then

$$|f_n - f_m|_{L^2} = |\hat{f}_n - \hat{f}_m|_{L^2} = |\mathcal{F}^{-1}(f_n) - \mathcal{F}^{-1}(f_m)|_{L^2}$$

We define  $\mathcal{F}f = \lim_{n \rightarrow \infty} \mathcal{F}(f_n)$  and  $\mathcal{F}^{-1}f = \lim_{n \rightarrow \infty} \mathcal{F}^{-1}(f_n)$ . This definition does not depend on the choice of the sequence  $f_n$ . Indeed, let  $g_n \in S$  such that  $g_n \rightarrow f$  in  $L^2$ . Then,  $|g_n - f_n|_{L^2} \rightarrow 0$  as  $n \rightarrow \infty$ . Therefore, by Proposition 4,

$$\begin{aligned}|g_n - f_n|_{L^2} &= |\hat{g}_n - \hat{f}_n|_{L^2} = |\mathcal{F}^{-1}g_n - \mathcal{F}^{-1}f_n|_{L^2}, \\ \implies \hat{g}_n &\rightarrow \hat{f} \text{ and } \mathcal{F}^{-1}g_n \rightarrow \mathcal{F}^{-1}f_n \text{ in } L^2.\end{aligned}$$

Hence,  $\mathcal{F} : L^2 \rightarrow L^2$  is unitary with  $\mathcal{F}^* = \mathcal{F}^{-1}$ .

### Corollary 1

$\mathcal{F}, \mathcal{F}^{-1} : S \rightarrow S$  can be extended to  $\mathcal{F}, \mathcal{F}^{-1} : L^2 \rightarrow L^2$  as

$$\mathcal{F}f = \hat{f} = \lim_{n \rightarrow \infty} \mathcal{F}f_n, \quad \mathcal{F}^{-1}f = \lim_{n \rightarrow \infty} \mathcal{F}^{-1}f_n \text{ in } L^2 \text{ with } f_n \in S, f_n \rightarrow f \text{ in } L^2.$$

We have  $\mathcal{F}\mathcal{F}^{-1}f = \mathcal{F}^{-1}\mathcal{F}f = f$  for any  $f \in L^2$ . Also,

$$|\mathcal{F}f|_{L^2} = |\mathcal{F}^{-1}f|_{L^2} = |f|_{L^2}, \quad f \in L^2.$$

### Proposition 6

Let  $f \in L^1 \cap L^2$ . Let  $\hat{f} = \lim_{n \rightarrow \infty} \hat{f}_n, \mathcal{F}^{-1}f = \lim_{n \rightarrow \infty} \mathcal{F}^{-1}f_n$  in  $L^2$  with  $f_n \in S$  and  $f_n \rightarrow f$  in  $L^2$ . Then,  $\hat{f}(\xi) = \int f(x) e^{-i2\pi x \cdot \xi} dx$  with  $\xi \in \mathbb{R}^d$ .  $\mathcal{F}^{-1}f(x) = \int f(\xi) e^{i2\pi x \cdot \xi} d\xi$  with  $x \in \mathbb{R}^d$ .

**Proof.** By Proposition 3, for any  $h \in S$ ,

$$\int \hat{f}_n h = \int f_n \hat{h} \quad \text{and} \quad \int \mathcal{F}^{-1}(f_n) h = \int f_n \mathcal{F}^{-1}(h)$$

As  $n \rightarrow \infty$ ,

$$\int \hat{f}_n h \rightarrow \int \hat{f} h, \int f_n \hat{h} \rightarrow \int f \hat{h}, \int \mathcal{F}^{-1}(f_n) h \rightarrow \int \mathcal{F}^{-1}(f) h, \int f_n \mathcal{F}^{-1}(h) \rightarrow \int f \mathcal{F}^{-1}(h),$$

$$\implies \int \hat{f}h = \int f\hat{h}, \quad \int \mathcal{F}^{-1}(f)h = \int f\mathcal{F}^{-1}(h).$$

Now, for all  $h \in S$ ,

$$\begin{aligned} \int \hat{f}(y)h(y) dy &= \int f(x)\hat{h}(x) dx \\ &= \int f(x) \left( \int h(y)e^{-i2\pi x \cdot y} dy \right) dx \\ &= \int \left( \int f(x)e^{-i2\pi x \cdot y} dx \right) h(y) dy \quad (\text{By Fubini's Theorem}) \end{aligned}$$

Thus,

$$\hat{f}(y) = \int f(x)e^{-i2\pi x \cdot y} dx \quad \text{a.e.}$$

Fubini holds since  $\iint |f(x)| \cdot |h(y)| \cdot |e^{-i2\pi x \cdot y}| dx dy = \|f\|_{L^1} \cdot \|h\|_{L^1} < \infty$ . An analogous proof holds for  $\mathcal{F}^{-1}$ .

## Corollary 2

Let  $f \in L^2$ . Then,

$$\begin{aligned} \mathcal{F}f(\xi) &= \hat{f}(\xi) = \lim_{n \rightarrow \infty} \int_{|x| \leq n} f(x)e^{-i2\pi x \cdot \xi} dx \text{ in } L^2, \\ \mathcal{F}^{-1}f(x) &= \lim_{n \rightarrow \infty} \int_{|\xi| \leq n} f(\xi)e^{i2\pi x \cdot \xi} d\xi \text{ in } L^2. \end{aligned}$$

**Proof.** Let  $f \in L^2$ . Set  $f_n = \chi_{|x| \leq n} f$ . Then  $f_n \in L^1 \cap L^2$  (by Hölder,  $\int |f_n| \leq \|f_n\|_{L^2} \|\chi_{|x| \leq n}\|_{L^2} < \infty$ ) and  $f_n \rightarrow f$  in  $L^2$  by the dominated convergence theorem, since  $|f_n| \leq |f|$  and  $|f|^2$  is integrable. By Proposition 6, since  $f_n \in L^1 \cap L^2$ ,

$$\hat{f}_n(\xi) = \int_{\mathbb{R}^d} f_n(x)e^{-i2\pi x \cdot \xi} dx = \int_{|x| \leq n} f(x)e^{-i2\pi x \cdot \xi} dx.$$

Since  $\mathcal{F}$  is unitary on  $L^2$ ,  $\|\hat{f}_n - \hat{f}\|_{L^2} = \|f_n - f\|_{L^2} \rightarrow 0$ , so  $\hat{f}_n \rightarrow \hat{f}$  in  $L^2$ . Hence

$$\hat{f}(\xi) = \lim_{n \rightarrow \infty} \int_{|x| \leq n} f(x)e^{-i2\pi x \cdot \xi} dx \quad \text{in } L^2.$$

An analogous argument with  $g_n = \chi_{|\xi| \leq n} f$  proves the formula for  $\mathcal{F}^{-1}$ .

## Some Applications of the Fourier Transform

### 1. Heisenberg Principle

Let  $\psi \in S(\mathbb{R})$  be a wave function,  $\int_{\mathbb{R}} |\psi(x)|^2 dx = 1$ . Then, in quantum mechanics,  $|\psi(x)|^2$  is the pdf of a particle's location and

$$\int |\hat{\psi}(\xi)|^2 d\xi = \int |\psi(x)|^2 dx = 1.$$



$|\hat{\psi}(\xi)|^2$  is the pdf of a particle's momentum. If  $x_0, \xi_0$  are the measured location and momentum, then their mean square errors are

$$(\Delta x)^2 = \int_{\mathbb{R}} |x - x_0|^2 |\psi(x)|^2 dx, \quad (\Delta \xi)^2 = \int_{\mathbb{R}} |\xi - \xi_0|^2 |\hat{\psi}(\xi)|^2 d\xi.$$

**Theorem 2 (Heisenberg Uncertainty Principle).** Let  $\psi \in S(\mathbb{R})$  with  $\int_{\mathbb{R}} |\psi(x)|^2 dx = 1$ . Then

$$(\Delta x)^2 (\Delta \xi)^2 \geq \frac{1}{16\pi^2}.$$

**Proof.** First take  $x_0 = \xi_0 = 0$ , so that  $(\Delta x)^2 = \int x^2 |\psi(x)|^2 dx$  and  $(\Delta \xi)^2 = \int \xi^2 |\hat{\psi}(\xi)|^2 d\xi$ . Since  $\psi \in S(\mathbb{R})$ , it decays faster than any polynomial, so  $x|\psi(x)|^2 \rightarrow 0$  as  $|x| \rightarrow \infty$ . Integrating by parts,

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} |\psi(x)|^2 dx = \left[ x |\psi(x)|^2 \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} x (\psi'(x) \overline{\psi(x)} + \psi(x) \overline{\psi'(x)}) dx \\ &= -2 \operatorname{Re} \int_{-\infty}^{\infty} x \psi'(x) \overline{\psi(x)} dx. \end{aligned}$$

Hence, by the triangle inequality and then Cauchy–Schwarz,

$$1 \leq 2 \int_{-\infty}^{\infty} |x \psi(x)| |\psi'(x)| dx \leq 2 \left( \int x^2 |\psi(x)|^2 dx \right)^{1/2} \left( \int |\psi'(x)|^2 dx \right)^{1/2} = 2 (\Delta x) \left( \int |\psi'|^2 dx \right)^{1/2}.$$

By Plancherel's equality and Property 1,  $\widehat{\psi'}(\xi) = i2\pi\xi \hat{\psi}(\xi)$ , so

$$\int |\psi'(x)|^2 dx = \int |\widehat{\psi'}(\xi)|^2 d\xi = \int (2\pi|\xi|)^2 |\hat{\psi}(\xi)|^2 d\xi = 4\pi^2 (\Delta \xi)^2.$$

Therefore  $1 \leq 2(\Delta x) \cdot 2\pi(\Delta \xi) = 4\pi(\Delta x)(\Delta \xi)$ , which gives

$$(\Delta x)^2 (\Delta \xi)^2 \geq \frac{1}{16\pi^2}.$$

For general  $x_0, \xi_0$ , apply the result to  $\varphi(x) = e^{-i2\pi\xi_0 x} \psi(x + x_0)$ . Then  $\int |\varphi|^2 = 1$ ,  $|\hat{\varphi}(\xi)|^2 = |\hat{\psi}(\xi + \xi_0)|^2$ , and a change of variables gives  $\int x^2 |\varphi(x)|^2 dx = (\Delta x)^2$  and  $\int \xi^2 |\hat{\varphi}(\xi)|^2 d\xi = (\Delta \xi)^2$ . Applying the case above to  $\varphi$  yields the general inequality.

## 2. Helmholtz Equation in $S = S(\mathbb{R}^d)$

Consider

$$u(x) - \Delta u(x) = f(x), \quad x \in \mathbb{R}^d, \quad f \in S, \quad (\text{H})$$

where  $\Delta u(x) = \frac{\partial^2}{\partial x_1^2} u(x) + \cdots + \frac{\partial^2}{\partial x_d^2} u(x)$ . We found (Example 1, Property 1)

$$\widehat{\Delta u}(\xi) = -4\pi^2 |\xi|^2 \hat{u}(\xi).$$

### Proposition 7

For any  $f \in S$ , there exists a unique  $u \in S$  solving (H); i.e.,  $I - \Delta : S \rightarrow S$  is bijective.

**Proof.** Let  $f \in S$ . Note that  $\hat{f} \in S$  and, since  $1 + 4\pi^2|\xi|^2$  is a polynomial that is bounded below by 1 with  $\frac{1}{1+4\pi^2|\xi|^2} \in C^\infty$  having all derivatives bounded, the product  $\frac{1}{1+4\pi^2|\xi|^2} \hat{f}(\xi) \in S$ .

(1) *Uniqueness.* Let  $u \in S$  with  $u - \Delta u = f$ . Taking Fourier transforms,

$$\widehat{u - \Delta u}(\xi) = \hat{f}(\xi) \implies (1 + 4\pi^2|\xi|^2) \hat{u}(\xi) = \hat{f}(\xi) \implies \hat{u}(\xi) = \frac{\hat{f}(\xi)}{1 + 4\pi^2|\xi|^2}.$$

Thus  $\hat{u}$  is uniquely determined, and since  $\mathcal{F} : S \rightarrow S$  is a bijection,  $u$  is unique.

(2) *Existence.* Let  $f \in S$  and set  $g(\xi) = \frac{\hat{f}(\xi)}{1 + 4\pi^2|\xi|^2} \in S$ . Then  $u(x) = (\mathcal{F}^{-1}g)(x) \in S$  solves (H): indeed  $\hat{u} = g$ , so

$$\widehat{u - \Delta u}(\xi) = (1 + 4\pi^2|\xi|^2) \hat{u}(\xi) = (1 + 4\pi^2|\xi|^2) g(\xi) = \hat{f}(\xi),$$

and injectivity of  $\mathcal{F}$  gives  $u - \Delta u = f$ .

## Tempered Distributions

**Notation.** Let  $S' = S'(\mathbb{R}^d)$  be the space of all linear continuous maps  $T : S \rightarrow \mathbb{C}$ .

**Recall.** A linear map  $T : S \rightarrow \mathbb{C}$  is continuous if and only if there exist  $m \geq 1$ ,  $C \geq 0$ ,  $\alpha_1, \alpha_2, \dots, \alpha_m \in \mathbb{N}_0^d$ , and  $N_1, N_2, \dots, N_m \in \mathbb{N}_0$  such that

$$|T(f)| \leq C \sum_{k=1}^m |f|_{N_k, \alpha_k}, \quad f \in S.$$

Equivalently, a linear map  $T : S \rightarrow \mathbb{C}$  is continuous if and only if there exist  $N \in \mathbb{N}_0$  and  $C \geq 0$  such that  $|T(f)| \leq C|f|_{(N)}$ .

**Definition.**  $T \in S'$  is called a *tempered distribution*, and  $S' = S'(\mathbb{R}^d)$  is called the space of tempered distributions.

### Example 1 ( $\mathcal{O}_M$ )

Let  $\mathcal{O}_M$  be the space of all  $f \in C^\infty$  such that for every  $\alpha \in \mathbb{N}_0^d$  there exist  $C = C(\alpha) \geq 0$  and  $N = N(\alpha) \in \mathbb{N}_0$  with

$$|\partial^\alpha f(x)| \leq C(1 + |x|)^N, \quad x \in \mathbb{R}^d.$$

(a) Every  $g \in \mathcal{O}_M$  defines  $T_g \in S'$  by  $T_g(f) = \int_{\mathbb{R}^d} g(x)f(x) dx$ ,  $f \in S$ .

**Proof.** Suppose  $|g(x)| \leq C(0)(1 + |x|)^N$  for  $x \in \mathbb{R}^d$  (the bound for  $\alpha = 0$ ). Then

$$\begin{aligned} \int |g(x)| \cdot |f(x)| dx &\leq C(0) \int (1 + |x|)^N |f(x)| dx \\ &= C(0) \int (1 + |x|)^{N+d+1} |f(x)| \cdot \frac{dx}{(1 + |x|)^{d+1}} \\ &\leq C(0) |f|_{N+d+1,0} \int \frac{dx}{(1 + |x|)^{d+1}}. \end{aligned}$$

Hence

$$|T_g(f)| \leq \left| \int g(x)f(x) dx \right| \leq C |f|_{N+d+1,0}, \quad C = C(0) \int \frac{dx}{(1 + |x|)^{d+1}} < \infty,$$

so  $T_g$  is continuous, i.e.  $T_g \in S'$ .

(b) The map  $\mathcal{O}_M \ni g \mapsto T_g \in S'$  is injective.

**Proof.** Let  $g_1, g_2 \in \mathcal{O}_M$  with  $T_{g_1}(f) = T_{g_2}(f)$  for all  $f \in S$ . Then  $\int g_1(x)f(x) dx = \int g_2(x)f(x) dx$  for all  $f \in S$ . Since  $g_1, g_2$  are locally integrable and  $\int g_1 f = \int g_2 f$  for all  $f \in C_c^\infty$ , it follows that  $g_1 = g_2$  a.e.

(c) Let  $\alpha \in \mathbb{N}_0^d$ . Then  $\partial^\alpha g \in \mathcal{O}_M$  for  $g \in \mathcal{O}_M$ , and since  $f \in S$  decays rapidly, integration by parts gives

$$\langle \partial^\alpha g, f \rangle = \int_{\mathbb{R}^d} \partial^\alpha g \cdot f dx = (-1)^{|\alpha|} \int g \partial^\alpha f dx = (-1)^{|\alpha|} \langle g, \partial^\alpha f \rangle = (-1)^{|\alpha|} T_g(\partial^\alpha f), \quad f \in S.$$

**Notation.**  $C_c^\infty \subseteq S \subseteq \mathcal{O}_M \subseteq S'$ . We also write  $T_g = g$  and  $T_g(f) = \langle g, f \rangle$  for  $f \in S$ .

**Definition.** Let  $T_n, T \in S'$ . We say  $T_n \rightarrow T$  in  $S'$  if  $T_n(f) \rightarrow T(f)$  for every  $f \in S$ .

## Example 2 (Polynomial Growth)

We say a measurable function  $g : \mathbb{R}^d \rightarrow \mathbb{C}$  is of *polynomial growth* if

$$|g(x)| \leq C(1 + |x|)^N, \quad x \in \mathbb{R}^d, \text{ for some } C > 0, N \geq 1.$$

**Remark.** If  $g$  is of polynomial growth, then  $g$  defines  $T_g \in S'$  by  $T_g(f) = \int g f$ ,  $f \in S$  (equivalently  $\langle g, f \rangle = \int g f$ ), and  $|T_g(f)| \leq C |f|_{N+d+1,0}$ ,  $f \in S$ . Again we write  $T_g = g$ .

For  $\alpha \in \mathbb{N}_0^d$  we define the distributional derivative  $\partial^\alpha g \in S'$  by

$$\langle \partial^\alpha g, f \rangle = (-1)^{|\alpha|} \langle g, \partial^\alpha f \rangle = (-1)^{|\alpha|} T_g(\partial^\alpha f), \quad f \in S, \quad (\#)$$

Formally this reads  $\langle \partial^\alpha g, f \rangle = \int \partial^\alpha g(x) f(x) dx = (-1)^{|\alpha|} \int g(x) \partial^\alpha f(x) dx$ . As an operator,

$$\partial^\alpha T_g = (-1)^{|\alpha|} T_g \circ \partial^\alpha, \quad S \xrightarrow{\partial^\alpha} S \xrightarrow{T_g} \mathbb{C}.$$

More generally,  $(\#)$  defines  $\partial^\alpha T$  for any  $T \in S'$  by  $(\partial^\alpha T)(f) = (-1)^{|\alpha|} T(\partial^\alpha f)$ ,  $f \in S$ .

**Definition.**  $T \in S'$  is *regular* if  $T = T_g$  where  $g$  is of polynomial growth.

**E.g.** Let

$$g(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad (\text{the ramp function}).$$

(a) Find  $\frac{d}{dx}g = g'$ . For  $f \in S$ ,

$$\begin{aligned} \left\langle \frac{d}{dx}g, f \right\rangle &= - \int_{-\infty}^{\infty} g(x) f'(x) dx = - \int_0^{\infty} x f'(x) dx \\ &= - \left[ x f(x) \right]_0^{\infty} + \int_0^{\infty} f(x) dx = \int_0^{\infty} f(x) dx = \int_{-\infty}^{\infty} H(x) f(x) dx, \end{aligned}$$

where the boundary term vanishes since  $f \in S$ . Hence  $g' = H$ , the Heaviside function,

$$H(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}, \quad H = g' = \frac{d}{dx}g.$$

(b) Find  $\frac{dH}{dx}$ . For  $f \in S$ ,

$$\begin{aligned} \left\langle \frac{dH}{dx}, f \right\rangle &= - \int_{-\infty}^{\infty} H(x) f'(x) dx \\ &= - \int_0^{\infty} f'(x) dx \\ &= - \left[ f(x) \right]_0^{\infty} = f(0) \\ &= \int_{-\infty}^{\infty} f(x) \delta_0(dx) = \int_{-\infty}^{\infty} f(x) dH(x). \end{aligned}$$

Thus  $\frac{dH}{dx} = \delta_0$ , the Dirac measure, where (informally)

$$\delta_0(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases} \quad (\text{Dirac measure}), \quad \langle \delta_0, f \rangle = f(0).$$

Formally,  $\int_{-\infty}^{\infty} f(x) H'(x) dx = \int_{-\infty}^{\infty} f(x) dH(x)$ .

### Example 3 ( $L^p$ )

Let  $g \in L^p$  with  $p \geq 1$ . Then  $g$  defines  $T_g \in S'$  by  $T_g(f) = \int g(x) f(x) dx$ ,  $f \in S$ .

**Proof.** By Hölder's inequality, with  $\frac{1}{p} + \frac{1}{q} = 1$ ,

$$\int |g(x)| \cdot |f(x)| dx \leq |g|_{L^p} |f|_{L^q}.$$

Now, for  $f \in S$ , writing  $m = \lceil \frac{d+1}{q} \rceil$ ,

$$|f|_{L^q}^q = \int |f(x)|^q dx = \int \left( |f(x)|(1+|x|)^{\frac{d+1}{q}} \right)^q \cdot \frac{dx}{(1+|x|)^{d+1}} \leq |f|_{m,0}^q \int \frac{dx}{(1+|x|)^{d+1}},$$

where  $\int \frac{dx}{(1+|x|)^{d+1}} = C < \infty$ . Then  $|f|_{L^q} \leq C^{1/q} |f|_{m,0}$ , so

$$|T_g(f)| \leq |g|_{L^p} |f|_{L^q} \leq C^{1/q} |g|_{L^p} |f|_{m,0} < \infty,$$

showing  $T_g \in S'$ . The map  $g \in L^p \mapsto T_g \in S'$  is injective (similar proof to  $\mathcal{O}_M$  for injectivity).

## Tempered Distributions (continued)

We define  $L^p \subseteq S'$  and often write  $g = T_g$ , where

$$T_g(f) = \langle g, f \rangle = \int g(x)f(x) dx, \quad f \in S.$$

We define  $\partial^\alpha g$  by

$$\langle \partial^\alpha g, f \rangle = (-1)^{|\alpha|} \langle g, \partial^\alpha f \rangle, \quad f \in S.$$

In particular  $L^2 \subseteq S'$ . Recall that if  $g \in L^2$ , then  $\mathcal{F}g = \hat{g}$  and  $\mathcal{F}^{-1}g \in L^2$ . By Proposition 3 of Chapter 7,

$$\int \hat{g} f = \int g \hat{f}, \quad \int \mathcal{F}^{-1}g f = \int g \mathcal{F}^{-1}f, \quad f \in S,$$

and this even holds for  $f \in L^2$ :

$$\langle \hat{g}, f \rangle = \langle g, \hat{f} \rangle, \quad \langle \mathcal{F}^{-1}g, f \rangle = \langle g, \mathcal{F}^{-1}f \rangle, \quad f \in S.$$

For  $g \in L^p$  with  $p \geq 1$ , we define  $\mathcal{F}g = \hat{g}$ ,  $\mathcal{F}^{-1}g \in S'$  by

$$\langle \mathcal{F}g, f \rangle = \langle \hat{g}, f \rangle = \langle g, \mathcal{F}f \rangle = \langle g, \hat{f} \rangle, \quad \langle \mathcal{F}^{-1}g, f \rangle = \langle g, \mathcal{F}^{-1}f \rangle, \quad f \in S.$$

**Note.**  $S \xrightarrow{\mathcal{F}} S \xrightarrow{T_g} \mathbb{C}$ , i.e.  $\mathcal{F}g = T_g \circ \mathcal{F}$ .

### Example 4 ( $\mathcal{M}_{\text{mod}}$ )

Let  $\mathcal{M}_{\text{mod}}$  (moderate) be the space of signed Borel measures  $\mu$  on  $\mathbb{R}^d$  of *moderate growth*:

$$\int (1+|x|)^{-N} d|\mu| < \infty \quad \text{for some integer } N \geq 1. \quad (\#)$$

**E.g.** Lebesgue measure on  $\mathbb{R}^d$ :  $N = d + 1$  works (indeed any  $N = d + \epsilon$ ,  $\epsilon > 0$ ).

Every  $\mu \in \mathcal{M}_{\text{mod}}$  defines  $T_\mu \in S'$  by  $T_\mu(f) = \int f d\mu$ ,  $f \in S$ .

**Proof (continuity).**

$$\int |f| d|\mu| = \int |f(x)|(1+|x|)^N \cdot \frac{1}{(1+|x|)^N} d|\mu| \leq |f|_{N,0} \int \frac{d|\mu|}{(1+|x|)^N} < \infty \quad \text{by } (\#),$$

for  $f \in S$ . Hence  $|T_\mu(f)| \leq C \|f\|_{N,0}$ , so  $T_\mu \in S'$ .

The map  $\mu \in \mathcal{M}_{\text{mod}} \mapsto T_\mu \in S'$  is injective; we write  $\mathcal{M}_{\text{mod}} \subseteq S'$ ,  $T_\mu(f) = \langle \mu, f \rangle$  for  $f \in S$ , and  $\mu = T_\mu$ . We define

$$\langle \partial^\alpha \mu, f \rangle = (-1)^{|\alpha|} \langle \mu, \partial^\alpha f \rangle, \quad \langle \mathcal{F} \mu, f \rangle = \langle \mu, \mathcal{F} f \rangle, \quad \langle \mathcal{F}^{-1} \mu, f \rangle = \langle \mu, \mathcal{F}^{-1} f \rangle, \quad f \in S.$$

Let  $\mu$  be a finite signed Borel measure on  $\mathbb{R}^d$ . Then, by Fubini,

$$\begin{aligned} \langle \mathcal{F}^{-1} \mu, f \rangle &= \langle \mu, \mathcal{F}^{-1} f \rangle = \int \mathcal{F}^{-1} f \, d\mu = \int \left( \int_{\mathbb{R}^d} f(\xi) e^{i2\pi x \cdot \xi} \, d\xi \right) d\mu(x) \\ &= \int_{\mathbb{R}^d} f(\xi) \left( \int e^{i2\pi x \cdot \xi} \, d\mu(x) \right) d\xi = \int_{\mathbb{R}^d} f(\xi) \phi(\xi) \, d\xi, \end{aligned}$$

where

$$\phi(\xi) = \int e^{i2\pi x \cdot \xi} \, d\mu(x) = E[e^{i2\pi x \cdot \xi}], \quad \xi \in \mathbb{R}^d,$$

(the characteristic function of  $\mu$ ). Thus  $\mathcal{F}^{-1} \mu = \phi$  in  $S'$ , where  $\phi = T_\phi$  is defined by  $T_\phi(f) = \int f \phi$ ,  $f \in S$ .

## Main Operations on $S' = S'(\mathbb{R}^d)$

**Differentiation.** Given  $T \in S'$  and  $\alpha \in \mathbb{N}_0^d$ , we define  $\partial^\alpha T \in S'$  by

$$\partial^\alpha T(f) = (-1)^{|\alpha|} T(\partial^\alpha f), \quad f \in S, \quad \text{i.e.} \quad \langle \partial^\alpha T, f \rangle = (-1)^{|\alpha|} \langle T, \partial^\alpha f \rangle, \quad f \in S,$$

where, for  $R \in S'$ ,  $R(f) = \langle R, f \rangle$ .

**Fourier Transform.** Given  $T \in S'$ , we define  $\mathcal{F}T = \hat{T}$ ,  $\mathcal{F}^{-1}T \in S'$  by  $\hat{T}(f) = T(\hat{f})$ , i.e.

$$\mathcal{F}T(f) = T(\mathcal{F}f), \quad \mathcal{F}^{-1}T(f) = T(\mathcal{F}^{-1}f), \quad f \in S.$$

**Notation.**  $\langle \hat{T}, \hat{f} \rangle = \langle T, f \rangle$ ,  $\langle \mathcal{F}^{-1}T, f \rangle = \langle T, \mathcal{F}^{-1}f \rangle$ .

**Multiplication by  $F \in \mathcal{O}_M$ .** Given  $T \in S'$  and  $F \in \mathcal{O}_M$ , we define  $FT \in S'$  by

$$FT(f) = T(Ff), \quad f \in S, \quad \text{i.e.} \quad \langle FT, f \rangle = \langle T, Ff \rangle, \quad f \in S,$$

which is well defined since  $f \in S \mapsto Ff \in S$  is continuous.

### Proposition 1

$\mathcal{F}, \mathcal{F}^{-1} : S' \rightarrow S'$  are bijections (recall  $S \subseteq L^2 \subseteq S'$ ).

**Proof.** Let  $f \in S$ . Then

$$\mathcal{F}^{-1} \mathcal{F}T(f) = \mathcal{F}T(\mathcal{F}^{-1}f) = T(\mathcal{F} \mathcal{F}^{-1}f) = T(f), \quad \mathcal{F} \mathcal{F}^{-1}T(f) = \mathcal{F}^{-1}T(\mathcal{F}f) = T(\mathcal{F}^{-1} \mathcal{F}f) = T(f).$$

Thus  $\mathcal{F}^{-1} \mathcal{F}T = T = \mathcal{F} \mathcal{F}^{-1}T$ .

### Example

Let  $T \in S'$ ,  $\alpha \in \mathbb{N}_0^d$ . Show that  $\widehat{\partial^\alpha T} = (i2\pi\xi)^\alpha \hat{T}$ .

We know  $(i2\pi\xi)^\alpha \in \mathcal{O}_M$ . Let  $f \in S$ . Then

$$\begin{aligned}\widehat{\partial^\alpha T}(f) &= \partial^\alpha T(\hat{f}) = (-1)^{|\alpha|} T(\partial^\alpha \hat{f}) = (-1)^{|\alpha|} T(\hat{g}) \quad \text{with } g(x) = (-i2\pi x)^\alpha f(x) \in S \\ &= (-1)^{|\alpha|} T(\widehat{(-i2\pi \cdot)^\alpha f}) = (-1)^{|\alpha|} \hat{T}((-i2\pi \cdot)^\alpha f) \quad \text{with } (-i2\pi \cdot)^\alpha \in \mathcal{O}_M \\ &= (-1)^{|\alpha|} (-i2\pi \cdot)^\alpha \hat{T}(f) = (i2\pi \cdot)^\alpha \hat{T}(f),\end{aligned}$$

using Property 3,  $\partial^\alpha \hat{f} = \hat{g}$  with  $g = (-i2\pi \cdot)^\alpha f$ . Hence  $\widehat{\partial^\alpha T} = (i2\pi\xi)^\alpha \hat{T}$ .

**Recall.** Let  $g(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases}$ ,  $g \in S'$ . We found  $\frac{d^2}{dx^2}g = \delta_0$ , and  $\delta_0 \in S'$  with  $\int_{\mathbb{R}^d} f(x) d\delta_0(x) = f(0)$ . Also recall that  $g : \mathbb{R}^d \rightarrow \mathbb{C}$  is of *polynomial growth* if  $|g(x)| \leq C(1 + |x|)^N$ ,  $x \in \mathbb{R}^d$ , for some  $C > 0$  and integer  $N \geq 1$ .

### Theorem 1 (Structure Theorem)

For each  $T \in S'$ , there exist a continuous function  $g$  of polynomial growth and  $\alpha \in \mathbb{N}_0^d$  such that  $T = \partial^\alpha g$ , i.e.

$$T(f) = \langle T, f \rangle = \langle \partial^\alpha g, f \rangle = (-1)^{|\alpha|} \langle g, \partial^\alpha f \rangle = (-1)^{|\alpha|} \int_{\mathbb{R}^d} g(x) \partial^\alpha f(x) dx.$$

**Proof.** We prove it for  $d = 1$ . In particular, we prove the following first.

#### Lemma 1

Let  $h : \mathbb{R} \rightarrow \mathbb{C}$  be of polynomial growth. Then there exists a continuous function  $H$  of polynomial growth so that

$$\langle h, f^{(n)} \rangle = \langle H, f^{(n+1)} \rangle, \quad f \in S, n \in \mathbb{N}_0, \quad \text{i.e.} \quad \int_{-\infty}^{\infty} h(x) f^{(n)}(x) dx = \int_{-\infty}^{\infty} H(x) f^{(n+1)}(x) dx.$$

**Proof.** Set  $G(x) = \int_0^x h(t) dt$ ,  $x \in \mathbb{R}$ . Suppose  $|h(x)| \leq C_1(1 + |x|)^N$ . Then, using  $|x| \leq \frac{|x|^2 + 1}{2}$ ,

$$|h(x)| \leq C_2(1 + |x|^2)^N, \quad \text{and also} \quad |h(x)| \leq C_3(1 + |x|^{2N}), \quad x \in \mathbb{R},$$

(since  $|x|^k \leq \frac{|x|^{2N}}{C_4} + C_5$ ). Then  $G$  is a continuous function of polynomial growth: for  $x \geq 0$ ,

$$|G(x)| \leq \int_0^x |h(t)| dt \leq C_6 \int_0^x (1 + t^{2N}) dt = C_6 x + C_6 \frac{x^{2N+1}}{2N+1},$$

and for  $x < 0$ ,  $|G(x)| \leq \int_x^0 |h(t)| dt \leq C_6|x| + C_6 \frac{|x|^{2N+1}}{2N+1}$ ; hence  $|G(x)| \leq C_7(1 + |x|)^{2N+1}$ .

$$\begin{aligned} \langle G, f^{(n+1)} \rangle &= \int_{-\infty}^{\infty} G(x) f^{(n+1)}(x) dx \\ &= \left[ G(x) f^{(n)}(x) \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f^{(n)}(x) h(x) dx \\ &= 0 - \langle h, f^{(n)} \rangle \quad (f^{(n)} \in S, h \in S'). \end{aligned}$$

Take  $H = -G$ , which finishes the proof of the lemma.

Now we return to Theorem 1. Let  $T \in S'$ . There exist  $C > 0$ ,  $N \in \mathbb{N}_0$  such that

$$|T(f)| \leq C|f|_{(N)} \leq C \sum_{k=0}^N \sup_x (1 + |x|)^N |f^{(k)}(x)| \leq C_1 \sum_{k=0}^N \sup_x (1 + |x|^2)^N |f^{(k)}(x)|, \quad f \in S. \quad (1)$$

Let  $\varphi(x) = (1 + |x|^2)^{-N}$ . Note  $\varphi \in \mathcal{O}_M$  and  $|\varphi^{(\ell)}(x)| \leq C_\ell \varphi(x)$  for all  $\ell \in \mathbb{N}_0$ ,  $x \in \mathbb{R}$ . Consider  $\varphi T(f) = T(\varphi f)$ ,  $f \in S$ .

**Claim 1.**  $|\varphi T(f)| \leq C_2 \sum_{k=0}^N \sup_x |f^{(k)}(x)|$ ,  $f \in S$ .

**Proof.** By (1),  $|\varphi T(f)| = |T(\varphi f)| \leq C_1 \sum_{k=0}^N \sup_x (1 + |x|^2)^N |(\varphi f)^{(k)}(x)|$ . By Leibniz,

$$|(\varphi f)^{(k)}(x)| \leq \sum_{j=0}^k \frac{k!}{j!(k-j)!} |f^{(j)}(x)| |\varphi^{(k-j)}(x)| \leq \sum_{j=0}^k C'_{k,j} |f^{(j)}(x)| \varphi(x), \quad C'_{k,j} = \binom{k}{j} C_{k-j}.$$

Since  $(1 + |x|^2)^N \varphi(x) = 1$ ,

$$|\varphi T(f)| \leq C_1 \sum_{k=0}^N \sum_{j=0}^k C'_{k,j} \sup_x |f^{(j)}(x)| \leq C_2 \sum_{k=0}^N \sup_x |f^{(k)}(x)|.$$

**Claim 2.** There exist complex (finite Borel) measures  $\mu_k$ ,  $k = 0, 1, \dots, N$ , on  $\mathbb{R}$  such that

$$\varphi T(f) = \sum_{k=0}^N \int_{\mathbb{R}} f^{(k)}(x) d\mu_k, \quad f \in S.$$

**Proof.** Let  $C_0(\mathbb{R})$  be the Banach space of continuous functions on  $\mathbb{R}$  vanishing at  $\pm\infty$ , with norm  $|g|_u = \sup_x |g(x)|$ . Let  $E = \underbrace{C_0(\mathbb{R}) \times \dots \times C_0(\mathbb{R})}_{N+1}$  with norm  $|(g_0, \dots, g_N)|_u = |g_0|_u + \dots + |g_N|_u$ ;  $E$  is a Banach space. Let

$$M = \{(f^{(0)}, f^{(1)}, \dots, f^{(N)}) : f \in S\} \subseteq E,$$



and consider  $\Lambda : M \rightarrow \mathbb{C}$  defined by  $\Lambda(f^{(0)}, f^{(1)}, \dots, f^{(N)}) = \varphi T(f)$ . By Claim 1,

$$|\Lambda(f^{(0)}, \dots, f^{(N)})| = |\varphi T(f)| \leq C_2 |(f^{(0)}, \dots, f^{(N)})|_u,$$

so  $\Lambda$  is bounded on  $M$ . By a corollary of the Hahn–Banach theorem,  $\Lambda$  extends to  $\Lambda \in E^*$  with  $\|\Lambda\|_{E^*} \leq C_2$ . Now  $|\Lambda(g_0, 0, \dots, 0)| \leq C_2 |g_0|_u$  for  $g_0 \in C_0(\mathbb{R})$ , so by the Riesz representation theorem there exists a finite complex Borel measure  $\mu_0$  on  $\mathbb{R}$  with  $\Lambda(g_0, 0, \dots, 0) = \int_{\mathbb{R}} g_0 d\mu_0$ . Continuing slot by slot, we obtain  $\mu_1, \dots, \mu_N$  with

$$\Lambda(g_0, g_1, \dots, g_N) = \sum_{k=0}^N \int_{\mathbb{R}} g_k(x) d\mu_k, \quad (g_0, \dots, g_N) \in E.$$

(for example,  $\Lambda(g_0, g_1) = \Lambda(g_0, 0) + \Lambda(0, g_1)$ ). Hence  $\varphi T(f) = \Lambda(f^{(0)}, \dots, f^{(N)}) = \sum_{k=0}^N \int_{\mathbb{R}} f^{(k)} d\mu_k$ .

**Claim 3.** There exist bounded measurable functions  $F_k$ ,  $k = 0, 1, \dots, N$ , such that

$$\varphi T(f) = \sum_{k=0}^N \int_{\mathbb{R}} f^{(k+1)}(x) F_k(x) dx, \quad f \in S.$$

**Proof.** Let  $G_k(x) = \mu_k((-\infty, x])$ ,  $x \in \mathbb{R}$ ; then  $G_k$  is bounded (as  $\mu_k$  is finite) and of bounded variation. Then

$$\begin{aligned} \varphi T(f) &= \sum_{k=0}^N \int_{-\infty}^{\infty} f^{(k)}(x) d\mu_k(x) = \sum_{k=0}^N \int_{-\infty}^{\infty} f^{(k)}(x) dG_k(x) \\ &= \sum_{k=0}^N \left[ f^{(k)} G_k \right]_{-\infty}^{\infty} - \sum_{k=0}^N \int_{-\infty}^{\infty} f^{(k+1)}(x) G_k(x) dx. \end{aligned}$$

The boundary terms vanish ( $f^{(k)} \in S$ ,  $G_k$  bounded). Set  $F_k = -G_k$ .

**Finishing the proof.** Since  $\frac{1}{\varphi} = (1 + |x|^2)^N \in \mathcal{O}_M$  and  $\frac{1}{\varphi} f \in S$ ,

$$T(f) = \frac{1}{\varphi} \cdot \varphi T(f) = \varphi T\left(\frac{1}{\varphi} f\right) = \sum_{k=0}^N \int_{\mathbb{R}} F_k(x) \left(\frac{1}{\varphi} f\right)^{(k+1)}(x) dx.$$

By Leibniz,  $\left(\frac{1}{\varphi} f\right)^{(k+1)} = \sum_{\ell=0}^{k+1} \binom{k+1}{\ell} f^{(\ell)} \left(\frac{1}{\varphi}\right)^{(k+1-\ell)}$ , so

$$T(f) = \sum_{k=0}^N \sum_{\ell=0}^{k+1} \int_{\mathbb{R}} f^{(\ell)}(x) H_{k,\ell}(x) dx, \quad H_{k,\ell} = \binom{k+1}{\ell} F_k \left(\frac{1}{\varphi}\right)^{(k+1-\ell)},$$

where each  $H_{k,\ell}$  is measurable of polynomial growth (bounded  $\times$  polynomial growth). Applying Lemma 1 repeatedly to raise every term to order  $N+2$ ,

$$T(f) = \sum_{k=0}^N \sum_{\ell=0}^{k+1} \int_{\mathbb{R}} f^{(N+2)}(x) \widetilde{H}_{k,\ell}(x) dx,$$

where the  $\widetilde{H}_{k,\ell}$  are continuous of polynomial growth. Set

$$g(x) = (-1)^{N+2} \sum_{k=0}^N \sum_{\ell=0}^{k+1} \widetilde{H}_{k,\ell}(x), \quad x \in \mathbb{R},$$

a continuous function of polynomial growth. Then  $T(f) = (-1)^{N+2} \int g(x) f^{(N+2)}(x) dx = \langle \partial^{N+2} g, f \rangle$ , i.e.  $T = \partial^{N+2} g$  (so  $\alpha = N+2$ ). This concludes the proof of Theorem 1.

## Some Applications of Tempered Distributions

### Helmholtz Equation in $S' = S'(\mathbb{R}^d)$

Consider

$$u - \Delta u = g \quad \text{in } S', \quad (\text{H})$$

Given  $g \in S'$ , find  $u \in S'$  solving (H).

### Proposition 2

For any  $g \in S'$ , there exists a unique  $u \in S'$  solving (H); i.e.  $I - \Delta : S' \rightarrow S'$  is bijective. Moreover

$$u = \mathcal{F}^{-1} \left( \frac{1}{1 + 4\pi^2 |\xi|^2} \hat{g} \right), \quad \text{where } \frac{1}{1 + 4\pi^2 |\xi|^2} \in \mathcal{O}_M.$$

**Proof.** Note that both  $1 + 4\pi^2 |\xi|^2$  and  $\frac{1}{1 + 4\pi^2 |\xi|^2}$  are in  $\mathcal{O}_M$ . Using  $\widehat{\partial^\alpha T} = (i2\pi\xi)^\alpha \hat{T}$  (so  $\widehat{\Delta u} = -4\pi^2 |\xi|^2 \hat{u}$ ),

$$\begin{aligned} u - \Delta u = g \text{ in } S' &\iff \widehat{u - \Delta u} = \hat{g} \text{ in } S' \iff (1 + 4\pi^2 |\xi|^2) \hat{u} = \hat{g} \text{ in } S' \\ &\iff \hat{u} = \frac{1}{1 + 4\pi^2 |\xi|^2} \hat{g} \text{ in } S' \iff u = \mathcal{F}^{-1} \left( \frac{1}{1 + 4\pi^2 |\xi|^2} \hat{g} \right) \text{ in } S'. \end{aligned}$$

This determines  $u$  uniquely and provides a solution, so  $I - \Delta : S' \rightarrow S'$  is bijective.

**(H) with  $g \in L^2$**

**Definition.**  $H_2^2 = \{u \in S' : u, \partial^\alpha u \in L^2 \text{ for all } |\alpha| \leq 2\}$  with norm

$$|u|_{2,2} = \left( \sum_{|\alpha| \leq 2} |\partial^\alpha u|_{L^2}^2 \right)^{1/2}.$$

In fact  $H_2^2$  is a Hilbert space with inner product  $\langle u, v \rangle = \sum_{|\alpha| \leq 2} \int_{\mathbb{R}^d} \partial^\alpha u \overline{\partial^\alpha v} dx$ .

**Note.**  $u \in H_2^2$  means  $u \in L^2$  and there exists  $\partial^\alpha u \in L^2$  so that  $\int \partial^\alpha u f dx = (-1)^{|\alpha|} \int u \partial^\alpha f dx$ ,  $f \in S$ , for all  $|\alpha| \leq 2$ .

### Proposition 3

For all  $g \in L^2$ , there exists a unique  $u \in H_2^2$  solving (H). Moreover,  $u - \Delta u = g$  a.e., and  $|u|_{2,2} \leq C|g|_{L^2}$  with  $C$  independent of  $g$ . Also  $I - \Delta : H_2^2 \rightarrow L^2$  is bijective.

**Proof.** Let  $g \in L^2$  and  $u = \mathcal{F}^{-1}\left(\frac{1}{1+4\pi^2|\xi|^2}\hat{g}\right) \in L^2$ . Then for all  $|\alpha| \leq 2$ ,

$$\widehat{\partial^\alpha u} = (i2\pi\xi)^\alpha \hat{u} = \frac{(i2\pi\xi)^\alpha}{1+4\pi^2|\xi|^2} \hat{g} \implies |\widehat{\partial^\alpha u}(\xi)| \leq |\hat{g}(\xi)|,$$

since for  $|\alpha| \leq 2$ ,  $|(i2\pi\xi)^\alpha| \leq (2\pi|\xi|)^{|\alpha|} \leq 1 + 4\pi^2|\xi|^2$ . Hence  $\partial^\alpha u \in L^2$  and, by Plancherel,

$$|\partial^\alpha u|_{L^2} = |\widehat{\partial^\alpha u}|_{L^2} \leq |\hat{g}|_{L^2} = |g|_{L^2}.$$

Summing over  $|\alpha| \leq 2$  gives  $|u|_{2,2} \leq C|g|_{L^2}$ . Uniqueness follows from Proposition 2 (uniqueness in  $S'$ ). Hence  $I - \Delta : H_2^2 \rightarrow L^2$  is bijective.

**Solution of (H) as a limit with  $g \in L^2$ .** Let  $g \in L^2$ . There exist  $g_n \in S$  with  $g_n \rightarrow g$  in  $L^2$ . For each  $g_n \in S$  there is a unique  $u_n \in S$  solving (H) (since  $I - \Delta : S \rightarrow S$  is bijective,  $S \subseteq L^2$ , and  $S$  is dense in  $L^2$ ). Then  $(u_n - u_m) - \Delta(u_n - u_m) = g_n - g_m$ , and by Proposition 3, for all  $|\alpha| \leq 2$ ,

$$|\partial^\alpha u_n - \partial^\alpha u_m|_{L^2} \leq C|g_n - g_m|_{L^2} \rightarrow 0 \quad \text{as } n, m \rightarrow \infty.$$

Hence there exist  $u_\alpha \in L^2$  with  $\partial^\alpha u_n \rightarrow u_\alpha$  in  $L^2$  for  $|\alpha| \leq 2$ ; write  $u = u_0$ . For any  $f \in S$  and  $|\alpha| \leq 2$ ,  $\int \partial^\alpha u_n f = (-1)^{|\alpha|} \int u_n \partial^\alpha f$ , and passing to the limit,

$$\int u_\alpha f = (-1)^{|\alpha|} \int u \partial^\alpha f, \quad f \in S,$$

which means  $\partial^\alpha u = u_\alpha$  in  $S'$ . Thus  $u \in H_2^2$  and  $u - \Delta u = g$  a.e.

**(H) with  $g \in H_2^s$**

**Definition.** Let  $s \in \mathbb{N}_0$ . Define

$$H_2^s = \left\{ u \in S' : u, \partial^\alpha u \in L^2, |\alpha| \leq s \right\}, \quad |u|_{s,2} = \left( \sum_{|\alpha| \leq s} |\partial^\alpha u|_{L^2}^2 \right)^{1/2}.$$

**Note.**  $H_2^s \subseteq L^2$ .

### Proposition 4

For any  $g \in H_2^s$ , there exists a unique  $u \in H_2^{s+2}$  solving (H); i.e.  $I - \Delta : H_2^{s+2} \rightarrow H_2^s$  is bijective.

**Proof.**  $u - \Delta u = g$  implies  $\partial^\alpha u - \Delta \partial^\alpha u = \partial^\alpha g$  in  $S'$ . Since  $\partial^\alpha g \in L^2$  for all  $|\alpha| \leq s$ , by Proposition 3 we have, for all  $|\beta| \leq 2$ ,

$$|\partial^{\beta+\alpha} u|_{L^2} = |\partial^\beta(\partial^\alpha u)|_{L^2} \leq C|\partial^\alpha g|_{L^2}.$$

Every  $\gamma$  with  $|\gamma| \leq s+2$  can be written  $\gamma = \beta + \alpha$  with  $|\alpha| \leq s$ ,  $|\beta| \leq 2$ , so

$$|\partial^\gamma u|_{L^2} \leq C|g|_{s,2} \quad \text{for all } |\gamma| \leq s+2.$$

Hence  $u \in H_2^{s+2}$  solves (H), and uniqueness follows from Proposition 2. Therefore  $I - \Delta : H_2^{s+2} \rightarrow H_2^s$  is bijective.

## Weak Convergence

Let  $E$  be a normed vector space and  $E^* = \mathcal{L}(E, \mathbb{R})$  be its dual. Let  $x \in E$  and let  $x_n \in E$  be a sequence “approximating”  $x$ . In what sense do we mean “approximating”? By the Hahn–Banach theorem,

$$|x_n - x|_E = \sup_{|\ell|_{E^*} \leq 1} |\ell(x_n - x)| \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

if and only if for any  $R > 0$ ,  $\sup_{|\ell|_{E^*} \leq R} |\ell(x_n) - \ell(x)| \rightarrow 0$  as  $n \rightarrow \infty$ . That is, norm convergence asks for  $\ell(x_n) \rightarrow \ell(x)$  *uniformly* over bounded sets of functionals. Weak convergence relaxes this to pointwise.

**Definition.** We say  $x_n \rightarrow x$  *weakly*, written  $x_n \xrightarrow{w} x$ , if

$$\ell(x_n) \rightarrow \ell(x) \quad \text{as } n \rightarrow \infty \quad \text{for all } \ell \in E^*, \quad \text{i.e.} \quad \langle \ell, x_n \rangle \rightarrow \langle \ell, x \rangle \quad \text{as } n \rightarrow \infty \quad \text{for all } \ell \in E^*.$$

### Remark 1.

- (a) Let  $E = L^p$ ,  $p \in [1, \infty)$ . Then  $(L^p)^* = L^q$  with  $\frac{1}{p} + \frac{1}{q} = 1$ . Therefore  $f_n \xrightarrow{w} f$  in  $L^p$  if and only if  $\int f_n g \rightarrow \int f g$  for all  $g \in L^q$ .
- (b) Let  $X$  be a compact metric space and  $E = C(X)$  with the  $|f|_u$  (sup) norm. Then  $[C(X)]^* = M(X)$ , the space of finite signed Borel measures, and  $f_n \xrightarrow{w} f$  if and only if  $\int f_n d\mu \rightarrow \int f d\mu$  for all  $\mu \in M(X)$ .
- (c) Let  $E = H$  be a Hilbert space. Then  $H^* = H$  and  $x_n \xrightarrow{w} x$  if and only if  $\langle z, x_n \rangle \rightarrow \langle z, x \rangle$  as  $n \rightarrow \infty$  for all  $z \in H$ . (Recall Problem 3 of HW4:  $x_n \rightarrow x$  in  $H$  if and only if  $x_n \xrightarrow{w} x$  and  $\limsup_n |x_n|_H \leq |x|_H$ .)

### Proposition 1

Let  $E$  be finite dimensional (e.g.  $E = \mathbb{R}^d$ ). Then

$$x_n \rightarrow x \quad \text{if and only if} \quad x_n \xrightarrow{w} x.$$

**Proof.** ( $\Rightarrow$ ) In any normed space, if  $x_n \rightarrow x$  then for each  $\ell \in E^*$ ,  $|\ell(x_n) - \ell(x)| \leq |\ell|_{E^*} |x_n - x|_E \rightarrow 0$ , so  $x_n \xrightarrow{w} x$ .

( $\Leftarrow$ ) Suppose  $\dim E = d$  and  $x_n \xrightarrow{w} x$ . Fix a basis  $v_1, \dots, v_d$  of  $E$  with coordinate functionals  $\ell_1, \dots, \ell_d \in E^*$  ( $\ell_i(v_j) = \delta_{ij}$ ), so  $y = \sum_{i=1}^d \ell_i(y) v_i$  for every  $y \in E$ . Weak convergence gives

$\ell_i(x_n) \rightarrow \ell_i(x)$  for each  $i$ . Since all norms on a finite-dimensional space are equivalent,

$$|x_n - x|_E \leq C \sum_{i=1}^d |\ell_i(x_n) - \ell_i(x)| \rightarrow 0,$$

so  $x_n \rightarrow x$  in norm.

**E.g.** For  $x^{(n)}, x \in \mathbb{R}^d$  with  $\{x^{(n)}\}$  a sequence,  $x^{(n)} \rightarrow x$  iff  $x_i^{(n)} \rightarrow x_i$  for every  $i = 1, 2, \dots, d$ .

**Remark 2.** In general,  $x_n \xrightarrow{w} x$  does *not* imply  $x_n \rightarrow x$  in norm. For example, let  $H$  be an infinite-dimensional Hilbert space with a CONS  $\{e_n : n \geq 1\}$ . Then for  $n \neq m$ ,

$$|e_n - e_m|_H^2 = |e_n|^2 - 2\langle e_n, e_m \rangle + |e_m|^2 = 2,$$

so  $\{e_n\}$  has no norm-convergent subsequence. But  $e_n \xrightarrow{w} 0$ : for every  $x \in H$ , by Parseval  $\sum_{n=1}^{\infty} |\langle x, e_n \rangle|^2 = |x|^2 < \infty$ , hence  $|\langle x, e_n \rangle|^2 \rightarrow 0$ , i.e.  $\langle x, e_n \rangle \rightarrow 0 = \langle x, 0 \rangle$  as  $n \rightarrow \infty$ .

Recall the canonical embedding  $E \subseteq (E^*)^* = E^{**}$ : every  $x \in E$  defines a functional  $T_x \in (E^*)^*$  by  $T_x(\ell) = \ell(x)$ ,  $\ell \in E^*$ . By the Hahn–Banach theorem,

$$\|T_x\| = \sup_{|\ell|_{E^*} \leq 1} |T_x(\ell)| = \sup_{|\ell|_{E^*} \leq 1} |\ell(x)| = |x|_E,$$

so  $x \mapsto T_x$  is an isometry.

**Definition.** A sequence  $\{x_n\}$  in  $E$  is *weakly Cauchy* if  $\{\ell(x_n)\}$  is Cauchy for all  $\ell \in E^*$ .

**Note.**  $x_n \xrightarrow{w} x \implies \{x_n\}$  is weakly Cauchy.

## Proposition 2

Let  $E$  be a normed vector space. Then:

- (a)  $x_n \xrightarrow{w} x \implies |x|_E \leq \liminf_n |x_n|_E$ .
- (b) If  $\{x_n\}$  is weakly Cauchy, then  $\{x_n\}$  is bounded:  $\sup_n |x_n|_E < \infty$ .
- (c) Suppose  $E$  is a reflexive Banach space. Then every weakly Cauchy sequence in  $E$  has a weak limit in  $E$ .

**Note on reflexive spaces.**  $E \subseteq (E^*)^* = E^{**}$ . The space is *reflexive* when  $E = E^{**}$ : for all  $T \in (E^*)^*$  there exists  $x \in E$  such that  $T(\ell) = \ell(x)$  for all  $\ell \in E^*$ .

**Proof.** (a) Let  $x_n \xrightarrow{w} x$ . By a corollary of the Hahn–Banach theorem, there exists  $\ell \in E^*$  with  $|\ell|_{E^*} = 1$  and  $\ell(x) = |x|_E$ . Then

$$|x|_E = \ell(x) = \lim_n \ell(x_n), \quad |\ell(x_n)| \leq |\ell|_{E^*} |x_n|_E = |x_n|_E \text{ for any } n.$$

Hence  $|x|_E = \lim_n \ell(x_n) = \liminf_n \ell(x_n) \leq \liminf_n |x_n|_E$ .

(b) Let  $\{x_n\}$  be weakly Cauchy. Then  $\sup_n |\ell(x_n)| < \infty$  for all  $\ell \in E^*$ , because  $\{\ell(x_n)\}$  is Cauchy in  $\mathbb{R}$ , hence convergent and bounded. Consider  $T_{x_n} \in (E^*)^*$  defined by  $T_{x_n}(\ell) =$

$\ell(x_n)$ ,  $\ell \in E^*$ . Then  $\sup_n |T_{x_n}(\ell)| = \sup_n |\ell(x_n)| < \infty$  for all  $\ell \in E^*$ . Since  $E^*$  is Banach, by the Banach–Steinhaus theorem (uniform boundedness),

$$\sup_n \|T_{x_n}\| = \sup_n \sup_{|\ell|_{E^*} \leq 1} |T_{x_n}(\ell)| = \sup_n \sup_{|\ell|_{E^*} \leq 1} |\ell(x_n)| = \sup_n |x_n|_E < \infty.$$

(c) Let  $\{x_n\}$  be a weakly Cauchy sequence in a reflexive Banach space  $E$ . By (b),  $M := \sup_n |x_n|_E < \infty$ . Set  $T(\ell) = \lim_n \ell(x_n)$ ; the limit exists since  $\{x_n\}$  is weakly Cauchy, and  $T : E^* \rightarrow \mathbb{R}$  is linear. Now  $|\ell(x_n)| \rightarrow |T(\ell)|$  and

$$|\ell(x_n)| \leq |\ell|_{E^*} |x_n|_E \leq M |\ell|_{E^*} \implies |T(\ell)| \leq M |\ell|_{E^*}, \quad \ell \in E^*,$$

so  $T \in (E^*)^*$ . By reflexivity  $(E^*)^* = E$ , so there exists  $x \in E$  with  $T(\ell) = \ell(x)$  for all  $\ell \in E^*$ . Then  $\ell(x) = \lim_n \ell(x_n)$  for all  $\ell \in E^*$ , i.e.  $x_n \xrightarrow{w} x$ .

**Note.** “ $\lim_n x_n = x$ ” means (a)  $\lim_n x_n$  exists and (b) it equals  $x$ . For (a), it helps to know there is a convergent subsequence  $x_{n_k}$  — equivalently, that  $\{x_n : n \geq 1\}$  is relatively compact.

**Fact 1.** If  $E$  is infinite dimensional, the weak topology on  $E$  is not metrizable; in particular, weak convergence cannot be defined by any metric.

**Proof.** Suppose, for contradiction, that the weak topology is induced by a metric. Then 0 has a countable weak-neighborhood basis  $\{V_k\}_{k \geq 1}$ . Every weak neighborhood of 0 contains a basic neighborhood

$$W(\ell_1, \dots, \ell_m; \epsilon) = \{x \in E : |\ell_i(x)| < \epsilon, i = 1, \dots, m\}, \quad \ell_i \in E^*, \epsilon > 0,$$

so each  $V_k$  contains such a  $W_k$  built from a finite set  $F_k \subseteq E^*$ . Let  $F = \bigcup_k F_k$ , a countable subset of  $E^*$ .

Given any  $\ell \in E^*$ , the set  $\{x : |\ell(x)| < 1\}$  is a weak neighborhood of 0, hence contains some  $V_k \supseteq W_k = W(\phi_1, \dots, \phi_m; \epsilon)$  with  $\phi_j \in F_k$ . In particular  $\bigcap_j \ker \phi_j \subseteq \{x : |\ell(x)| < 1\}$ , and by scaling  $\bigcap_j \ker \phi_j \subseteq \ker \ell$ . By the elementary lemma (if  $\bigcap_j \ker \phi_j \subseteq \ker \ell$  then  $\ell \in \text{span}\{\phi_1, \dots, \phi_m\}$ ), we get  $\ell \in \text{span}(F_k) \subseteq \text{span}(F)$ .

Thus  $E^* = \text{span}(F)$  has a countable spanning set, i.e. countable Hamel dimension. But  $E^*$  is a Banach space, and by the Baire category theorem an infinite-dimensional Banach space cannot have countable Hamel dimension. Since  $E$  is infinite dimensional, so is  $E^*$  — a contradiction. Hence the weak topology is not metrizable.